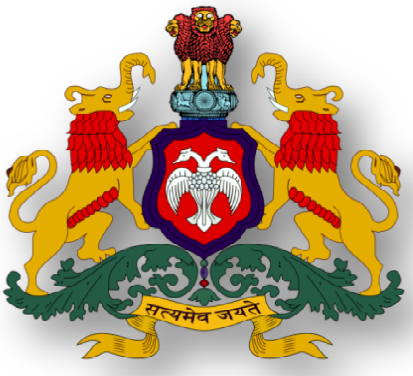


GOVERNMENT OF KARNATAKA

DEPARTMENT OF SCHOOL EDUCATION (PRE-UNIVERSITY)

18th Cross, Malleshwaram Bengaluru- 560012

Web site: https://www.pue.karnataka.gov.in/

&

KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD

6th Cross, Malleshwaram Bengaluru - 560003

Web site: https://kseab.karnataka.gov.in/

2025-26

REVISED QUESTION BANK

II PUC BIOLOGY (36)

DISCLAIMER

The question bank is prepared for the benefit of students and teachers (Lecturers).The committee worked for the preparation of question bank made all efforts to make it comprehensive and foolproof. However, if any mistakes, errors found, please mail: **iipuquestionbankkseab@gmail.com.** There is no guarantee that questions from this question bank will only appear in the examination conducted by the KSEAB Board.

COPYRIGHTS



The copyrights of the question bank lies with the Director, Department of School Education (Pre- University). The question bank is prepared for academic purpose only. No part of the question bank is allowed to be used for commercial gains. No part of this publication may be reproduced, stored in retrieval system or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior permission of the publisher.

2025 - 26

II PUC BIOLOGY QUESTION BANK DEVELOPMENT COMMITTEE

1	Dr. DEVARAJU. R GOVERNMENT PU COLLEGE (AS0161) KR Puram, Bangalore South (Dist)	COORDINATOR
2	Dr. SAMPATH KUMARA K.K GOVERNMENT PU COLLEGE (Boys) PB Road, PJ Extension, Davanagere (Dist)	REVIEWER
3	Dr. DIVAKAR GOVERNMENT PU COLLEGE BH Road, Gubbi, Tumkuru (Dist)	MEMBER
4	MRS. SMITHA JOSHMA FERNANDES GOVERNMENT PU COLLEGE Sanoor, Karkala, Udupi (Dist)	MEMBER
5	Dr. NAGABHUSHAN GOVERNMENT PU COLLEGE (Boys) Krishnarajpet Mandya (Dist)	MEMBER
6	MRS. VIMALA. R GOVERNMENT PU COLLEGE Yelahanka, Bangalore North (Dist)	MEMBER

II PUC CHAPTERWISE MARKS ALLOTMENT

TIME: 3 HOURS **BIOLOGY(36)** **MAX MARKS: 70**

	CHAPTERS	No of Teaching Hours (Periods)	Marks Allotted	Page Numbers
1	Sexual reproduction in flowering plants	12	11	01 - 09
2	Human reproduction	11	10	10 - 17
3	Reproductive health	7	6	18 - 26
4	Principles of Inheritance	15	13	27 - 35
5	Molecular basis of inheritance	15	13	36 - 44
6	Evolution	8	7	45 - 50
7	Human health and disease	13	11	51 - 60
8	Microbes in Human welfare	8	7	61 - 66
9	Biotechnology: Principles and Processes	8	7	67 - 73
10	Biotechnology and its applications	7	6	74 - 79
11	Organism and populations	6	6	80 - 84
12	Ecosystem	5	4	85 - 89
13	Biodiversity and conservation	5	4	90 - 95
	Total	**120**	**105**	

1	**2024 Board Exam Question Papers (Three Papers)**	96 - 108
2	**2025 Board Exam Question Papers (Three Papers)**	109 - 122
3	**2025 Model Question Papers (Three Papers)**	123 - 140
3	**2025 Practice Papers (Two Papers)**	141 - 150

CHAPTER 1
SEXUAL REPRODUCTION IN FLOWERING PLANTS

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. The function of tapetum in microsporangium is
   a) It nourishes the developing pollen grains
   b) It helps in dehiscence of anther to release the pollen
   c) It helps in protection
   d) It helps in division **[Easy]**

2. Select the correct sequence of anther wall layers:
   a) Epidermis → Middle layers → Endothecium → Tapetum
   b) Epidermis → Endothecium → Middle layers → Tapetum
   c) Epidermis → Tapetum → Middle layers → Endothecium
   d) Tapetum → Middle layers → Epidermis → Endothecium **[Average]**

3. Select the correct sequence of events in microsporogenesis:
   a) Sporogenous tissue → Microspore mother cell → Microspore tetrad →Microspores
   b) Microspores → Microspore mother cell → Microspore tetrad→ Sporogenous tissue
   c) Sporogenous tissue → Microspore tetrad →Microspores → Microspore mother cell
   d) Microspores → Sporogenous tissue → Microspore tetrad → Microspore mother cell **[Average]**

4. The most resistant organic material present on exine of pollen grain is
   a) Tapetum b) Pectin c) Sporopollenin d) Cellulose **[Easy]**

5. Which of the following has proved helpful in preserving pollen as fossils?
   a) Pollen kit b) Cellulosic intine c) Oil content d) Sporopollenin **[Easy]**

6. Select the incorrect statement with respect to vegetative cell of pollen grain.
   a) It is bigger than generative cell b) It has abundant food reserve
   c) It has a large irregularly shaped nucleus d) It floats in the cytoplasm of generative cell **[Easy]**

7. Which one of the following statements is not true?
   a) Pollen grains of many species cause severe allergies.
   b) Stored pollen in liquid nitrogen can be used in the crop breeding programmes.
   c) Tapetum helps in the dehiscence of anther.
   d) Exine of pollen grains is made up of sporopollenin **[Average]**

8. Pollen grains can be stored for several years in liquid nitrogen having a temperature of

a) -120°C b) -80°C c) -160°C d) -196°C **[Easy]**

9. The functional megaspore in an angiosperm develops into an

a) Endosperm b) embryo sac c) embryo d) ovule **[Easy]**

10. In angiosperms, microsporogenesis and megasporogenesis

a) Involve meiosis b) Occur in ovule

c) Occur in anther d) Form gametes without further divisions **[Easy]**

11. Function of filiform apparatus is to

a) Recognize the suitable pollen at stigma b) stimulate division of generative cell

c) Produce nectar d) Gide the entry of pollen tube **[Easy]**

12. From the statements given below choose the option that is true for a typical female gametophyte of a flowering plant.

A) It is 8-nucleate and 7-celled at maturity B) It is free-nuclear during the development

B) It is situated inside the integument but outside the nucellus

C) It has an egg apparatus situated at the chalazal end

a) A) and D) b) B) and C) c) A) and B) d) B) and D) **[Difficult]**

13. Identify the correct sequence of arrangement of cells in the mature embryo sac micropylar end.

a) Synergids → Egg Cell → Central cell →Antipodals

b) Antipodals → Central Cell → Egg cell → Synergids

c) Synergids → Central Cell → Egg cell → Antipodals

e) Antipodals → Egg cell → Central Cell → Synergids **[Average]**

14. Which of the following is not a correct statement?

a) Placenta is located outside the ovarian cavity.

b) Central cell develops into PEC after fertilization.

c) Orchids contain more than one ovule in an ovary.

d) Egg apparatus consists of one egg cell and two synergid cells. **[Average]**

15. Even in the absence of pollinating agents, seed setting is assured in

a) *Cucurbita* b) Papaya c) *Commelina* d) Maize **[Easy]**

16. Transfer of pollen grains from the anther to stigma of another flower of the same plant is known as

a) Autogamy b) Xenogamy c) Cleistogamy d) Geitonogamy **[Easy]**

17. Which pollination brings genetically different types of pollen grains to the stigma?

a) Geitonogamy b) Xenogamy c) Autogamy d) Cleistogamy **[Easy]**

18. Dioecious plants are

a) Autogamous only b) Both autogamous and geitonogamous

c) Both geitonogamous and xenogamous d) Xenogamous only **[Average]**

19. **Statement I:** In *Viola*, both autogamy and geitonogamy is prevented.

Statement II: Geitonogamy is similar to autogamy since pollen grains comes from the same plant.

a) Both statement I and statement II are correct b) Both statement I and statement II are incorrect

c) Statement I is incorrect but statement II is correct d) Statement I is correct but statement II is incorrect **[Difficult]**

20. Which is not a characteristic feature of wind pollinated flowers?

a) Feathery stigma b) Presence of single ovule in ovary

c) Pollen grains are non-sticky d) Flowers are very colourful **[Easy]**

21. Wind pollination is common in

a) Legumes b) Lilies c) Grasses d) Orchids **[Easy]**

22. Double fertilization is

a) Fusion of two male gametes of a pollen tube with two different eggs

b) Fusion of one male gamete with two polar nuclei

c) Fusion of two male gametes with one egg

d) Syngamy and triple fusion **[Average]**

23. In a fertilized embryo sac, the haploid, diploid and triploid structures are represented by

a) Synergid, zygote and primary endosperm nucleus

b) Synergid, antipodal and polar nuclei

c) Antipodal, synergid and primary endosperm nucleus

d) Synergid, polar nuclei and zygote **[Average]**

24. Haploid condition is not observed in which of the following cells of embryo sac

a) Synergids and Egg b) Zygote and PEN

c) Antipodal and Egg d) Antipodal and Synergids **[Easy]**

25. Endosperm persist in the mature seed in

a) Groundnut b) Castor c) Pea d) Beans **[Easy]**

26. Coconut water that from a tender coconut is

(a) Innermost layers of the seed coat (b) Degenerated nucellus

(c) Immature embryo (d) Free nuclear endosperm **[Easy]**

27. In a monocot embryo, shoot apex and leaf primordia are enclosed in a hollow foliar structure called

a) Pericarp b) Coleoptile c) Endosperm d) Coleorrhiza **[Easy]**

28. Perisperm differs from endosperm in

a) Being a diploid tissue b) Being a haploid tissue c) Having no reserve food

d) Its formation by fusion of secondary nucleus with several sperms **[Average]**

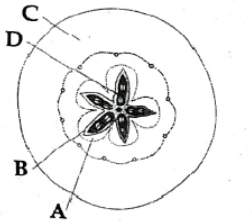
29. Non-albuminous seed is produced in

a) Maize b) Castor c) Wheat d) Pea **[Easy]**

30. Which one of the following fruits is parthenocarpic?

a) Jackfruit b) Banana c) Brinjal d) Apple **[Easy]**

31. Which part of the fruit, labelled in the given figure makes it a false fruit?



a) D-Seed b) A-Mesocarp c) B-endocarp d) C-Thalamus **[Average]**

32. Occurrence of more than one embryo in a seed is seen in

a) Grasses b) Asteraceae c) *Citrus* d) Both grasses and asteraceae **[Easy]**

33. **Statement I:** Apomixis is the production of seeds from unfertilized ovules.

Statement II: Embryos produced from apomictic seeds are not generally identical to the parent plant.

a) Statement I is true, statement II is false b) Statement I is false, statement II is true

c) Statement I and statement II both are true d) Statement I and statement II both are false **[Difficult]**

34. **Statement I:** Formation of fruit without fertilization is called apomixis.

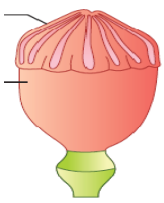
Statement II: In some species of Asteraceae and grasses seeds are formed without fertilization.

a) Both Statement I and Statement II are correct b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Difficult]**

35. The following picture is an example for



a) Multicarpellary, apocarpous gynoecium b) Multicarpellary, syncarpous gynoecium

c) Monocarpellary gynoecium d) Apocarpous gynoecium **[Average]**

36. The correct sequence of embryogemy in dicots is

a) Proembryo→ Globular embryo → Heart shaped embryo → Mature embryo

b) Globular embryo → Proembryo → Heart shaped embryo → Mature embryo

c) Proembryo→ → Heart shaped embryo → Globular embryo → Mature embryo

d) Heart shaped embryo → Proembryo → Globular embryo → Mature embryo

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket.

1 Mark Each

(Embryogeny, *Amorphophallus*, Self-incompatibility, Micropyle, Perisperm, Polyembryony, *Zostera*, Tapetum, Strawberry, Microsporogenesis, Lupine, Pollen grain, Scutellum, Emasculation, Castor)

1. The process of formation of microspores from a pollen mother cell through meiosis is called _____ **[Easy]**
2. _____ is the innermost wall layer of microsporangium. **[Easy]**
3. Embryo sac is to ovule as _____ is to an anther. **[Easy]**
4. The opening of the angiosperm ovule where the integument is absent is known as _____. **[Easy]**
5. _____ is an example for water pollinated plant. **[Easy]**
6. An example for plant that gives floral reward by providing safe place to lay insect eggs is _____. **[Easy]**
7. The genetic mechanism which prevents self-pollen from fertilizing the ovules by inhibiting pollen germination or pollen tube growth in the pistil is known as _____. **[Easy]**
8. The removal of anthers from the flower bud before the anther dehiscence in bisexual flower is referred as _____. **[Easy]**
9. The early stages of embryo development are called _____. **[Easy]**
10. An example for albuminous seed is _____. **[Easy]**
11. The presence of residual persistent nucellus in some seeds is called_____. **[Easy]**
12. In the grass family the cotyledon of embryo is called **[Easy]**
13. _____ is an example for false fruit. **[Easy]**
14. An example for oldest viable seed is _____. **[Easy]**
15. If more than one embryo occurs in a seed then it is known as _____. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Mention four wall layers of microsporangium. **[Easy]**
2. Write a note on tapetum. **[Easy]**

3. Mention two differences between microsporogenesis and megasporogenesis. **[Easy]**

4. Complete the tabular column given below with respect to the male gametophyte of angiosperms.

Cells of the male gametophyte		
Shape of nucleus of the cells		

[Difficult]

5. Write the significance of sporopollenin. **[Easy]**

6. What is filiform apparatus? What is its significance? **[Easy]**

7. Differentiate between chasmogamous and cleistogamous flowers. **[Difficult]**

8. Cleistogamous flowers are invariably autogamous. Justify. **[Difficult]**

9. Cleistogamous flowers produce assured seed set even in the absence of pollinators. Justify. **[Difficult]**

10. What are cleistogamous flowers? Name a plant showing this. **[Easy]**

11. Differentiate between geitonogamy and xenogamy. **[Difficult]**

12. List the characteristic features of wind pollinated flowers. **[Easy]**

13. Mention any two floral rewards offered by insect pollinated flowers to their pollinators. **[Easy]**

14. "The flowering plants have evolved strategies to prevent self-pollination". Justify by giving two reasons. **[Difficult]**

15. Define: a) Emasculation b) Bagging **[Easy]**

16. Define a) Syngamy b) Triple fusion. **[Easy]**

17. Name the four stages in dicotyledonous embryo development. **[Easy]**

18. Draw a neat labeled diagram of typical dicotyledonous embryo. **[Average]**

19. Distinguish between albuminous and non–albuminous seeds. **[Difficult]**

20. What is perisperm? Give an example. **[Easy]**

21. Differentiate between false fruit and true fruit. **[Difficult]**

22. Why apple is called a false fruit? Which part(s) of the flower forms the fruit? **[Average]**

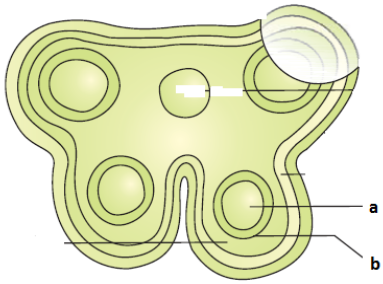
23. What is parthenocarpy? Give an example. **[Easy]**

24. Name the phenomenon of formation of seed without fertilization in angiosperms. Give an example of plant which exhibits this phenomenon. **[Easy]**

25. Define polyembryony. Mention the plant which exhibits polyembryony. **[Easy]**

26. How are parthenocarpic fruits produced by some plants and apomictic seeds by some others? **[Difficult]**

27. In the T.S. of an anther given below, identify "*a*" and "*b*" and mention their function. **[Average]**



PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Draw a labelled diagram of transverse section of a young anther. **[Average]**
2. Describe the structure of microsporangium. **[Average]**
3. Write any two commercial uses of pollen grains and mention the viability period of pollen grains of Solanaceae members. **[Easy]**
4. Draw a neat labeled diagram of a typical anatropous ovule. **[Average]**
5. Describe the structure of megasporangium. **[Average]**
6. Explain the structure of a mature female gametophyte in flowering plants. **[Average]**
7. Draw a neat labeled diagram of a mature embryo sac. **[Average]**
8. List the characters of insect pollinated flowers. **[Easy]**
9. What is artificial hybridization? Name the techniques that are employed to achieve this? **[Easy]**
10. What is triple fusion? Where does it take place in the megasporangium? Name the nuclei involved in triple fusion. **[Easy]**
11. Draw a labeled diagram of the fertilized female gametophyte and mention the ploidy of any one of the products of double fertilization. **[Difficult]**
12. Name the product of fertilization that forms the kernel of coconut. How does the kernel differfrom coconut water? **[Average]**
13. Draw a L.S. of grass embryo diagram and label the following parts:
    a) Scutellum b) Coleoptile c) Shoot apex d) Radicle e) Root cap f) Coleorrhiza **[Easy]**
14. Draw a neat labeled diagram of a monocot seed. **[Average]**
15. 'Fertilization is not an obligatory event for fruit production in certain plants'. Justify the statement. **[Difficult]**
16. Explain any three advantages the seeds offer to angiosperms. **[Average]**
17. Why are some seeds referred to as apomictic seeds? Mention its advantages to a farmer who uses them. **[Average]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Explain the structure of pollen grain. **[Average]**

2. Schematically represent the development of a female gametophyte in angiosperms. **[Difficult]**

3. Answer the following:

   a) Explain different types of pollination based on the source of pollen. 3 **[Average]**

   b) List the characteristic features of water pollinated flowers. 2 **[Easy]**

4. How does pollination occur in *Vallisneria* and *Zostera*. **[Average]**

5. Pollen-Pistil interaction is a dynamic process involving pollen recognition followed by promotion or inhibition. Explain. **[Average]**

6. Flowering plants have developed many devices to discourage self-pollination and encourage Cross pollination. Comment. **[Difficult]**

7. Answer the following:

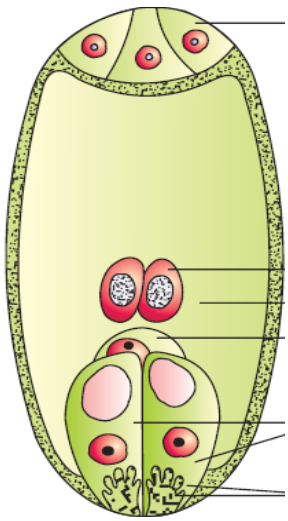
   a) Double fertilization is a unique event taking place in flowering plants. Briefly explain. 3

   b) Arrange the following terms in the correct developmental sequence:

   Pollen grain, Sporogenous tissue, Microspore tetrad, Pollen mother cell, Male gametes. 2 **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Picture of a mature angiosperm embryo sac is given below. Answer the questions that follow.



   a) Which cells/nuclei of the embryo sac produce zygote and primary endosperm nucleus? 2

   b) What is the ploidy of antipodal cells and primary endosperm nucleus? 2

   c) Why the endosperm development precedes embryo development? 1 **[Easy]**

2. Picture related to pollination is given below. Answer the questions that follow.



a) Will this pollination confirm fertilization? Justify your answer with suitable reasons. 2

b) What are the floral rewards provided by the plants to the insects to revisit? 1

c) Mention the characteristics of flower in this pollination other than floral rewards. 2 **[Average]**

3. Answer the following:

a) Plan an experiment and prepare a flow chart of the steps that you would follow to ensure that the seeds are formed only from the desired sets of pollen grains. 3

b) Name the type of experiment that you carried out. 1

c) Write the importance of such experiments. 1 **[Difficult]**

4. Double fertilization is the unique feature of angiosperms and the products of this double fertilization are zygote and PEN. In context of this when a hexaploid plant is pollinated by a tetraploid plant find out the ploidy of zygote and PEN through a schematic illustration. **[Difficult]**

5. Why is the process of fertilization in angiosperms termed as double fertilization? Draw a diagram of an angiospermic embryo sac where fertilization is just completed and label the following parts:

a) The part that develops into an embryo

b) The part that develops into an endosperm

c) The degenerating cells at the chalazal end. **[Average]**

6. Answer the following:

a) Double fertilization is reported in plants of both, castor and groundnut. However, the mature seed of groundnut is non-albuminous and castor is a luminous. Explain the post-fertilization events that are responsible for it. 3

b) Banana is a parthenocarpic fruit whereas oranges show polyembryony. How are they different from each other with respect to seeds? 2 **[Difficult]**

****

CHAPTER 2
HUMAN REPRODUCTION

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. Scrotum helps in maintaining the temperature of testes which is **A** lower than the normal body temperature. Each of testes has about **B** compartments and measures **C** cm in length.
   Select the option that fills the blanks correctly.

	A	B	C	
a)	1-3°C	250	4-6	
b)	2-2.5°C	250	4-5	
c)	3-3.5°C	300	2-3	
d)	2-2.5°C	250	2-3	**[Easy]**

2. Spot the odd one out from the following structures with reference to the male reproductive system.
   a) Rete testis b) Epididymis c) Vasa efferentia d) Isthmus **[Average]**
3. Which of the following is not a male accessory gland?
   a) Seminal vesicle b) Ampulla c) Prostate d) Bulbourethral gland **[Easy]**
4. The vas deferens receives duct from the seminal vesicle and opens into urethra as
   a) epididymis b) ejaculatory duct c) efferent ductule d) ureter **[Easy]**
5. The function of myometrium layer present in the uterine wall is to
   a) Undergo cyclical changes during menstrual cycle.
   b) Exhibit strong uterine contraction during parturition.
   c) Give protection to the uterus.
   d) Help in the implantation process. **[Easy]**
6. Sertoli cells are found in
   a) Ovaries and secrete progesterone
   b) Adrenal cortex and secrete adrenaline
   c) Seminiferous tubules and provide nutrition to germ cells
   d) Pancreas and secrete cholecystokinin **[Easy]**
7. The cells that secrete testicular hormones (androgens) are _____.
   a) Sertoli celles b) Leydig cells c) Germ cells d) Spermatogonia **[Easy]**
8. Seminal plasma in human males is rich in
   a) Fructose and Calcium b) Glucose and Calcium

c) DNA and Testosterone d) Ribose and Potassium **[Easy]**

9. **Statement I:** The process of release of sperms from the seminiferous tubule is called spermatogenesis.
   Statement II: The spermatids are transformed into spermatozoa by the process called spermiogenesis. Choose the correct answer from the options given below:
   a) Both statement I and statement II are correct
   b) Both statement I and statement II are incorrect
   c) Statement I is correct but statement II is incorrect
   d) Statement I is incorrect but statement II is correct **[Easy]**
10. In spermatogenesis, if FSH hormone is not secreted from the anterior pituitary gland, which stage is affected
    a) Formation of primary spermatocyte b) Formation of spermatozoa
    c) Formation of spermatid d) Formation of secondary spermatocyte **[Average]**
11. To produce 2000 sperm and 400 ova, how many spermatogonia and oogonia are required?
    a) 500 spermatogonia and 200 oogonia b) 500 spermatogonia and 400 oogonia
    c) 250 spermatogonia and 250 oogonia d) 250 spermatogonia and 400 oogonia **[Easy]**
12. Diploid number of chromosomes in a primary spermatocyte is 46. What is the total number of chromatids in each of its secondary spermatocyte?
    a) 46 b) 23 c) 92 d) 69 **[Average]**
13. The secondary spermatocytes undergo the second meiotic division to produce four equal, haploid
    a) Spermatogonium b) Spermatids c) Primary spermatocyte d) Secondary spermatocyte **[Easy]**
14. The process of transformation of spermatids into sperm is
    a) Spermiation b) Spermatogenesis c) Spermiogenesis d) Insemination **[Easy]**
15. The release of sperms from the seminiferous tubules is called _____.
    a) Spermiation b) Spermatogenesis c) Spermiogenesis d) Spermioteliosis **[Easy]**
16. What is the correct sequence of sperm formation?
    a) Spermatogonia → Spermatozoa →Spermatocytes → Spermatids
    b) Spermatogonia → Spermatocytes → Spermatids → Spermatozoa
    c) Spermatids → Spermatocytes → Spermatogonia → Spermatozoa
    d) Spermatogonia, → Spermatocytes → Spermatozoa → Spermatids **[Easy]**
17. In female oogenesis is initiated during
    a) Ovulation b) Puberty c) Childhood d) Embryonic development stage **[Easy]**

18. Which among the following has 23 chromosomes?

a) Spermatogonia b) Zygote c) Secondary oocyte d) Oogonia **[Average]**

19. Gonadotropin Releasing Hormone (GnRH) is a hypothalamic hormone which stimulates pituitary gland to secrete _____. **[Easy]**

a) LH and FSH b) FSH and Oxytocin c) LH and Oxytocin d) Progestorone

20. Identify the wrong pair with respect to menstrual cycle:

a) Formation of corpus luteum-15th to 28th day b) Secretion of progesterone -15th to 28th day

c) Regeneration of endometrium-5th to 13th day d) Disintegration of endometrium- 5th to 13th day **[Average]**

21. The corpus luteum secretes large amounts of progesterone, which is essential for maintenance of

a) Perimetrium b) Myometrium c) Endometrium d) Cervical canal **[Easy]**

22. Progesterone hormone is secreted by

a) Testis b) Pituitary gland c) Corpus luteum d) Graafian follicle **[Easy]**

23. Which of the following statement is true?

a) LH surge induces menstruation.

b) Regression of corpus luteum increases progesterone.

c) Diploid egg is formed after second meiotic division.

d) Embryo differentiates from the inner cell mass of blastocyst. **[Easy]**

24. Match the items given in column I with those in column II and select the correct option given below.

Column I	**Column II**
A. Proliferative phase	(i) Breakdown of endometrial lining
B. Secretory phase	(ii) Follicular phase
C. Menstruation	(iii) Luteal phase

	A	**B**	**C**
a)	(iii)	(ii)	(i)
b)	(i)	(iii)	(ii)
c)	(ii)	(iii)	(i)
d)	(iii)	(i)	(ii)

[Average]

25. When does the second meiotic division occur in the secondary oocyte?

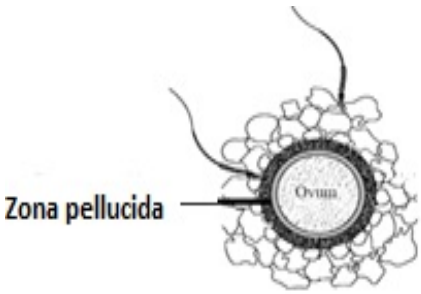
a) At the time of birth b) At the time of fertilization

c) At the time of puberty d) At the time of ovulation **[Easy]**

26. The secondary oocyte after ovulation is covered by a non-cellular layer of

a) Cumulus oophorus b) Corona radiata c) Zona pellucida d) Cortical layer **[Average]**

27. Choose the correct option from the table given below for the formation and dissolution of the labeled part in the given diagram.



	Formed by	Dissolved
a)	Primary oocyte	before fertilisation
b)	Primary oocyte	after fertilisation
c)	Secondary oocyte	before fertilisation
d)	Secondary oocyte	after fertilisation

[Average]

28. Which of the following hormones is produced only during pregnancy?

a) hCG b) LH c) Estrogen d) FSH **[Easy]**

29. The _____ hormone is secreted by the ovary in the later phase of pregnancy.

a) Androgens b) Estrogens c) Relaxin d) Progestogens **[Easy]**

30. Which of the following hormones is not secreted by human placenta?

a) hCG b) Estrogens c) Progesterone d) LH **[Easy]**

31. Morula is a developmental stage

a) Between the zygote and blastocyst b) Between the blastocyst and gastrula

c) After the implantation d) Between implantation and parturition **[Easy]**

32. Which one of the following is not the function of placenta?

a) Facilitates removal of carbon dioxide and waste material from embryo

b) Secretes oxytocin during parturition

c) Facilitates supply of oxygen and nutrients to embryo

d) Secretes estrogen **[Average]**

33. During pregnancy the foetus develops limbs and digits

a) After three weeks b) After eight weeks

c) After six weeks d) After nine weeks **[Easy]**

34. The first movements of the fetus and appearance of hair on its head are usually observed during which month of pregnancy?

a) Fourth month b) Fifth month c) Sixth month d) Third month **[Easy]**

35. During embryonic development embryo's heart formed at the end of

a) First trimester b) Second trimester c) First month d) Third month **[Easy]**

36. Signals for parturition originate from

a) Both placenta as well as fully developed fetus b) Oxytocin released from maternal pituitary

c) Placenta only d) Fully developed foetus only **[Easy]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket.

1 Mark Each

(Oxytocin, Menarche, Placenta, Alveoli, Fimbriae, Ovulation, Antrum, Acrosome, Endometrium, Colostrum, Stem, Fertilization, Competent cells, Parturition, Blastocyst, Luteinising hormone)

1. The interstitial space in seminiferous tubules consists of immunologically _____. **[Easy]**
2. The edges of the infundibulum possess finger-like projections called _____. **[Easy]**
3. The cells of _____ secrete milk in the mammary glands. **[Easy]**
4. The anterior portion of the sperm head is covered by a cap like structure called _____. **[Easy]**
5. Tertiary follicle is characterized by a fluid filled cavity called _____. **[Easy]**
6. The first menstruation begins at puberty is called _____. **[Easy]**
7. The glandular layer of uterus that undergoes cyclic changes during menstrual cycle is __ **[Easy]**
8. The process of release of ovum from a mature follicle is called _____. **[Easy]**
9. Ovulation is induced by a hormone called _____. **[Easy]**
10. Several hormones like hCG, hPL, estrogen and progesterone are produced by _____. **[Easy]**
11. The fusion of male and female gametes is called _____. **[Easy]**
12. The _____ cells of the inner mass of embryo have the potency to give rise to all the tissues and organs. **[Easy]**
13. Zygote divides to form _____ which is implanted in the uterus. **[Easy]**
14. The hormone which helps in parturition is _____. **[Easy]**
15. The process of delivery of the foetus is called _____. **[Easy]**
16. _____ contains several antibodies which is absolutely essential to develop resistance for the new-born babies. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

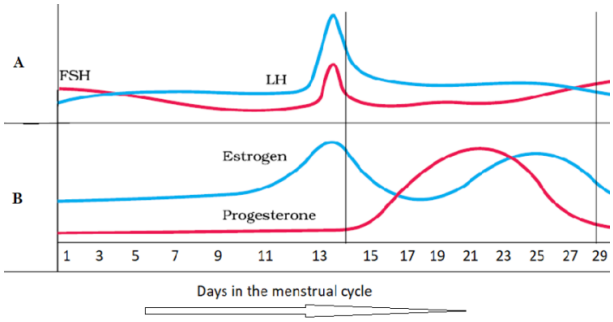
1. Why are the human testes located outside the abdominal cavity? Name the pouch in which they are present. **[Average]**
2. List the accessory ducts of human male reproductive system. **[Easy]**
3. Draw a neat labelled diagram of sectional view of seminiferous tubules. **[Average]**

4. Mention the two functions of sertoli cells. **[Easy]**
5. What are Leydig cells? Mention their function. **[Easy]**
6. Name any two accessory ducts of the human female reproductive system. **[Easy]**
7. Name any two accessory glands associated with the human male reproductive system. **[Easy]**
8. What is the number of chromosomes in the following cells of a human female?
   a) Primary oocyte b) Secondary oocyte **[Easy]**
9. Mention any four parts of the external genitalia in human females. **[Easy]**
10. Mention the roles of LH and FSH during spermatogenesis. **[Easy]**
11. Name the cells which synthesize the androgen. Mention the hormone which stimulates the cells to produce androgen. **[Easy]**
12. Write the function each for the following:
    a) Seminal vesicle b) Acrosome of human sperm **[Average]**
13. "Acrosome and middle piece of the sperm are very essential parts of the sperm without which fertilization does not occur". Justify the statement. **[Easy]**
14. Differentiate spermiogenesis from spermiation. **[Easy]**
15. How is 'oogenesis' markedly different from 'spermatogenesis' with respect to the growth till puberty in the humans? **[Average]**
16. List the events of menstrual cycle. **[Easy]**
17. Differentiate between Menarche and Menopause. **[Easy]**
18. Define ovulation. Mention the hormone that induces ovulation. **[Easy]**
19. Mention two events that occur during the luteal phase of menstrual cycle. **[Easy]**
20. Mention the fate of corpus luteum and its effect on the uterus in absence of fertilisation of the ovum in a human female. **[Average]**
21. Define fertilization. In which part of the fallopian tube of the female reproductive system fertilization occurs? **[Easy]**
22. Define: a) Cleavage b) Implantation **[Easy]**
23. List any four hormones secreted by placenta. **[Easy]**
24. Why is oxytocin necessary for parturition? **[Average]**
25. What is parturition? Name the hormone released by the pituitary which induces parturition. **[Easy]**
26. "Unless foetal ejection reflex is produced, normal parturition does not occur". Substantiate the statement. **[Easy]**
27. What is colostrum? Mention its benefits. **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Draw a neat labelled diagram of human sperm. **[Average]**
2. Write the schematic representation of spermatogenesis. **[Easy]**
3. Trace the path of spermatozoa from the testes up to the ejaculatory duct. **[Average]**
4. Explain the organisation of glandular tissue in a breast of human females. **[Average]**
5. Schematically represent oogenesis. **[Easy]**
6. Mention the events that occur during the follicular phase of menstrual cycle. **[Easy]**
7. What is placenta? List the functions of human placenta. **[Easy]**
8. Parturition is induced by complex neuroendocrine mechanism. Comment. **[Easy]**
9. Explain the changes that occur in ovary and uterus during luteal phase of menstrual cycle. **[Average]**
10. In the figure given below, parts A and B show the level of hormones which influence the menstrual cycle. Study the figure and answer the questions that follow:



a) Name the organs/glands which secrete the hormones represented in labelled parts A and B. 1

b) State the role of hormones secreted from part B on the uterus of human female during menstrual cycle. 2 **[Average]**

12. Briefly explain the process of fertilization in humans. **[Average]**
13. In humans, sex of the baby is determined by the father and not the mother. Justify. **[Easy]**
14. When and where do chorionic villi appear in humans? State their function. **[Average]**
15. Study the following flow chart. Name the hormones involved at each stage. Explain their functions.

Hypothalamus → Pituitary →Testes → Sperms **[Average]**

PART- D

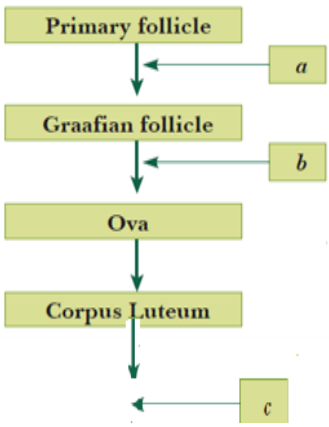
V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Draw a neat labeled diagram of the sectional view human male reproductive system. **[Average]**
2. Draw a neat labeled diagram of the sectional view of human female reproductive system. **[Average]**
3. Draw a neat labeled diagram of the sectional view of mammary gland. **[Average]**
4. Differentiate between spermatogenesis and oogenesis. **[Easy]**

5. Trace the development of oocytes till ovulation in human female reproductive system (in menstrual cycle). **[Easy]**
6. What is menstrual cycle? Describe the various events that occur during menstrual cycle. **[Average]**
7. Briefly explain the embryonic development in humans. **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Draw a sectional view of seminiferous tubule of a human and label the following cells in the seminiferous tubule:
   a) Cells that divide by mitosis to increase their number.
   b) Cells that undergo Meiosis I.
   c) Cells that undergo Meiosis II.
   d) Cells that help in the process of spermiogenesis.
   e) Cells produced due to spermiogenesis. **[Easy]**
2. Answer the following:
   a) Given below is a flow chart showing ovarian changes during menstrual cycle.
   Fill in the spaces with the hormonal factor(s) responsible for the events shown. 3 **[Average]**



   b) Women experience two major events in their life time, one at menarche and the second at menopause. Mention the characteristics of both the events. 2 **[Average]**
3. Answer the following:
   a) Why does corpus luteum secrete large amount of progesterone during luteal/secretory phase of the menstrual cycle? 3 **[Easy]**
   b) Medically it is advised to all young mothers that breast feeding is the best for their newborn babies. Do you agree? Give reasons in support of your answer. 2 **[Easy]**

****

CHAPTER 3
REPRODUCTIVE HEALTH

PART- A

I. Select the correct alternative from the choices given: **1Mark Each**

1. According to WHO, reproductive health means a total well being in all aspects of reproduction like _____

   a) Physical, Emotional, Behavioural, Social

   b) Physical, Emotional, Social, Life style

   c) Physical, Social, Behavioural, Life style

   d) Physical, Emotional, Behavioural, Life style **[Easy]**

2. Abbreviate 'Reproductive and Child Health Care' programme.

   a) RACHC b) RCHC c) RCH d) RC **[Easy]**

3. What are the major tasks under reproductive and child health care programs?

   a) Create awareness about reproductive health among people.

   b) Educate people about reproductive process

   c) Establish reproductive health care in hospitals

   d) Reproductive and child health. **[Easy]**

4. **Statement I:** Introduction of sex education in schools should be encouraged.

   Statement II: This will encourage children to believe in myths about sex related aspects.

   a) Both Statement I and Statement II are correct

   b) Both Statement I and Statement II are incorrect

   c) Statement I is correct and Statement II is incorrect

   d) Statement I is incorrect and Statement II is correct **[Average]**

5. Decreased MMR and IMR and increased people in reproducible age in a population will result in

   a) Increase in growth rate b) Decrease in growth rate

   c) No significant change in growth rate d) Stable growth rate **[Easy]**

6. Which of the following is an indicator of a reproductively unhealthy society?

   a) Increase in infant mortality rate

   b) Decline in maternal mortality rate

   c) Widespread awareness of contraceptive methods

   d) Detection and cure of sexually transmitted diseases **[Easy]**

7. **Statement I:** Amniocentesis is legally banned for sex determination.

   Statement II: Amniocentesis was being misused for aborting normal female foetus

   a) Both Statement I and Statement II are correct

   b) Both Statement I and Statement II are incorrect

   c) Statement I is correct and Statement II is incorrect

   d) Statement I is incorrect and Statement II is correct **[Average]**

8. Amniocentesis is a medical procedure where some of the amniotic fluid of the developing foetus is drawn to analyse the foetal cells and dissolved substances. It was banned in India in order to stop female foeticides. Which of the following highlight the importance of amniocentesis despite the ban?

   a) Identification of a decline in sex ratio

   b) Prevention of sexually transmitted diseases

   c) Determination of biological sex of the unborn baby

   d) Detection of genetic disorders **[Average]**

9. Which of the following cannot be detected in a developing foetus by amniocentesis?

   a) Downs' syndrome b) Haemophilia c) Sex of the foetus d) Hepatitis **[Easy]**

10. CDRI stands for_____

    a) Common Drug Research Institute b) Chandigarh Drug Research Institute

    c) Calcutta Drug Research Institute d) Central Drug Research Institute **[Easy]**

11. What is true for natural methods of contraception?

    a) They increase phagocytosis of sperms

    b) They act as barriers to prevent fertilisation

    c) They avoid chances of ovum and sperms meeting.

    d) They are terminal methods **[Easy]**

12. **Statement I:** Lactational amenorrhea is absence of menstruation during intense lactation

    Statement II: This is a highly reliable method of contraception.

    a) Both Statement I and Statement II are correct

    b) Both Statement I and Statement II are incorrect

    c) Statement I is correct and Statement II is incorrect

    d) Statement I is incorrect and Statement II is correct **[Average]**

13. **Statement I:** IUDs are ideal contraceptives for the females who want to delay pregnancy or space children.

    Statement II: IUDs are most widely accepted methods contraception in India.

    a) Both Statement I and Statement II are correct

b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Average]**

14. **Statement I:** Use of condom is a safeguard against AIDS and sexual diseases besides checking pregnancy.

Statement II: Certain contraceptives are planted under the skin of the upper arm to prevent pregnancy.

a) Both Statement I and Statement II are correct

b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Average]**

15. Choose the incorrect statement with regard to diaphragm:

a) They are reusable b) They block the entry of sperms

c) They are placed to cover the cervix d) They prevent ovulation **[Easy]**

16. Which of the following is not a copper-releasing IUD?

a) LNG 20 b) CuT c) Cu7 d) Multiload 375 **[Easy]**

17. What makes the uterus unsuitable for implantation and the cervix hostile to sperms?

a) Cu7 b) Lippe's loop c) Multiload 375 d) Progestasert **[Easy]**

18. Select the correct statement with respect to IUDs.

a) Once inserted need not be replaced. b) Inserted by user itself

c) Increase phagocytosis of sperms in the uterus d) Suppress gametogenesis **[Average]**

19. **Statement I:** IUDs can cause excess menstrual bleeding and pain.

Statement II: IUDs can perforate uterus.

a) Both Statement I and Statement II are correct

b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Average]**

20. Oral contraceptive pills mainly contain

a) Estrogen and luteinising hormone b) Progesterone, estrogen and luteinising hormone

c) Progesterone and estrogen d) Progesterone and luteinising hormone **[Easy]**

21. Select the incorrect statement regarding vasectomy?

a) No sperms in seminal fluid b) No sperm in epididymis

c) Vas deferentia is cut and tied d) No spermatogenesis **[Average]**

22. A widely accepted method of contraception in India is _____

a) IUDs b) Diaphragms c) Tubectomy d) Cervical caps **[Easy]**

23. Which of the following contraceptives involve a role of hormones?

a) Pills, Emergency contraceptives, Implants b) Pills, IUDs, Vaults

c) Pills, Condoms, CuT d) Pills, Implants, Diaphragms **[Easy]**

24. Which of the following does not give the defined action of a contraceptive?

a) Hormonal method prevents or retards the entry of sperms, prevent ovulation and fertilisation

b) Vasectomy prevents spermatogenesis c) Barrier method prevent fertilisation

d) Intra uterine devices increase phagocytosis of sperms, suppress sperm motility and fertilising capacity of sperms **[Difficult]**

25. **Statement I:** All contraceptives have the added advantage of protection against sexually transmitted diseases.

Statement II: Natural birth control methods have negligible side effects.

a) Both Statement I and Statement II are correct

b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Average]**

26. Which of the following are not completely curable?

a) Hepatitis B, Chlamydiasis, HIV infection c) Hepatitis B, Trichomoniasis, HIV infection

b) Trichomoniasis, Chlamydiasis, HIV infection d) Hepatitis B, Genital herpes, HIV infection

[Easy]

27. The incidences of sexually transmitted infections are reported to be very high among persons in the age group of _____ years.

a) 05-14 b) 15-24 c) 25-34 d) 35-44 **[Easy]**

28. A specialised procedure to form a embryo in the laboratory is ____

a) Gamete Intra Fallopian Transfer b) Intra Cytoplasmic Sperm Injection

c) Artificial Insemination d) Intra Uterine Insemination **[Average]**

29. **Statement I:** GIFT is an in-vitro fertilisation technique.

Statement II : In GIFT, gametes are fertilised outside the female body

a) Both Statement I and Statement II are correct b) Both Statement I and Statement II are incorrect

c) Statement I is correct and Statement II is incorrect

d) Statement I is incorrect and Statement II is correct **[Average]**

30. *In vivo* fertilisation occurs in

a) ZIFT b) GIFT c) ICSI d) IVF **[Easy]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Family planning, Periodic abstinence, Coitus interrupts, Infertility, ICSI, surgical intervention, venereal, Saheli, 72, 12, Amniocentesis, Condom)

1. The programs to get total reproductive health as a social goal at the national level are called____ **[Easy]**
2. A pre-natal diagnostic technique which was misused to illegally determine the sex of the developing foetus is _____ **[Easy]**
3. The collective term given to the diseases which are transmitted through sexual intercourse is ____ diseases **[Easy]**
4. A natural method of contraception in which the couples avoid coitus during the period of menstrual cycle when ovulation could be expected is _____ **[Easy]**
5. _____ is a natural method of birth control based on the withdrawal of penis before ejaculation. **[Easy]**
6. The contraceptive that protects the user from contracting STIs and AIDS is _____ **[Easy]**
7. _____ methods of contraception blocks gamete transport and thereby prevent conception. **[Average]**
8. _____ is an oral contraceptive for females, developed by scientists at the Central Drug Research Institute, Lucknow. **[Easy]**
9. Administration of progestogens or progestogen-estrogen combinations or IUDs are effective as emergency contraceptives within _____ hours of coitus **[Easy]**
10. MTPs are considered relatively safe upto _____ weeks of pregnancy **[Easy]**
11. Inability of the couple to produce children inspite of unprotected sexual co-habitation is _____ **[Easy]**
12. An assisted reproductive technology in which a sperm is directly injected into the ovum to ensure conception is _____ **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Mention the major tasks under RCH programmes. **[Easy]**
2. Introduction of sex education in schools should be encouraged. Why? **[Average]**
3. Which days of menstrual cycle are identified as “fertile period” and why? **[Easy]**
4. Mention two benefits of condoms which are used as contraceptive devices. **[Easy]**
5. What are IUDs? Give any two examples. **[Easy]**

6. Parents of one year old child wanted to space their second child. Their doctor suggested Cu-T. Explain its contraceptive actions. **[Difficult]**
7. Mention any two events that are inhibited by the intake of oral contraceptive pills to prevent pregnancy in humans. **[Easy]**
8. Why do women use 'Saheli', the oral contraceptive pill as a well accepted method? **[Easy]**
9. What are emergency contraceptives? Write their hormonal combination. **[Easy]**
10. List one contraceptive method each in males and females with poor reversibility. **[Easy]**
11. Differentiate between vasectomyand tubectomy. **[Average]**
12. List the possible ill-effects of various contraceptive methods. **[Easy]**
13. Mention four early symptoms of STD that occur in the genital region. **[Easy]**
14. What are assisted reproductive technologies? Mention any two assisted reproductive technologies (ARTs). **[Easy]**
15. Suggest any two ARTs that can help the couple to have a child where the problem is with the female partner. **[Average]**
16. Explain the technique that can help a healthy married woman who is unable to produce viable ova but can provide a suitable environment for fertilization and further development and wants to bear a child. **[Average]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Mention three strategies to promote reproductive health. **[Easy]**
2. How is amniocentesis performed? Why it is legally banned in our country? **[Average]**
3. Mention the parameters that indicate improved reproductive health of the society. **[Easy]**
4. List the measures taken by the government to check the population growth rate. **[Easy]**
5. What are contraceptives? Mention the qualities of an ideal contraceptive. **[Easy]**
6. Explain natural methods of contraception. **[Average]**
7. Explain barrier methods of contraception. **[Average]**
8. List the types of intra uterine devices with one example for each. **[Easy]**
9. Categorize the following into different kinds of IUDs such as non-medicated, copper releasing or hormone releasing ones:
   a) Progestasert b) Lippes loop c) CuT d) LNG-20 e) Cu7 f) Multiload 375 **[Average]**
10. Name any two copper releasing intra-uterine devices (IUDs). List two reasons that make them effective contraceptives. **[Easy]**

11. The picture shows the contraceptive device used by women.



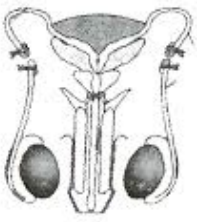
a). Identify the contraceptive device shown in the picture. 1

b). Mention how it prevents fertilization and conception. 2

[Difficult]

12. At the time of independence, the population of India was 350 million which exploded to over 1 billion by May 2000. List any three reasons for this rise in population and any two steps taken by the government to check this population explosion. **[Difficult**

13. Shown below is the surgical method of contraception in human males.

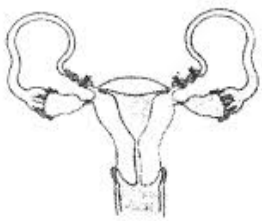


a) Give the technical term for the process shown above. 1

b) What is the principle behind it as a contraceptive method? 1

c) How is the process carried out? 1 **[Difficult]**

14. Shown below is the surgical method of contraception in human females.



a) Give the technical term for the process shown above. 1

b) How is the process carried out? 1

c) What is the principle behind it as a method of contraception? 1 **[Difficult]**

15. Name the hormonal composition of oral contraception used by human females. Explain how does it act as contraceptive? **[Easy]**

16. List any six reproductive tract infections or sexually transmitted diseases. **[Easy]**

17. List three principles through which one could be free of STDs. **[Easy]**

18. Define venereal disease. Mention the modes of transmission of HIV / Hepatitis B virus. **[Easy]**
19. Suggest any three ARTs that can help the couple to have a child where the problem is with the male partner. **[Average]**
20. Explain the steps involved in *In vitro* fertilization popularly known as 'test tube baby' **[Average]**
21. A healthy couple came to know that both of them are unable to produce functional gametes and should look for an ART (Assisted Reproductive Technique). Name the ARTthat can be suggested to help them bear a child and explain the procedure involved in it. **[Average]**
22. A large number of couples all over the world are infertile. Such infertile couples could be assisted to have children through certain special techniques, commonly called 'Assisted Reproductive Technologies' (ART). Give the technical term for each of the methods described below.
    a) Transfer of ovum collected from a donor into the fallopian tube of another female, who cannot produce functional ova, but can provide suitable environment for fertilization and embryo development.
    b) Semen collected from the husband or a healthy donor is artificially introduced into the uterus of a woman.
    c) An embryo is formed in the laboratory by directly injecting the sperm into the ovum. **[Difficult]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Explain five measures that can be taken to build up a socially responsible and reproductively healthy society? **[Average]**
2. A wide range of contraceptive methods are presently available. Broadly group them into different categories. **[Average]**
3. a) What is Medical Termination of Pregnancy? 1
   b) Mention two reasons for a woman to undergo medical termination of pregnancy. 2
   c) Mention any two medical grounds on which pregnancy can be terminated. 2 **[Easy]**
4. Medical termination of pregnancy is termination of pregnancy before the foetus become viable. To reduce the incidence of illegal abortion and consequent maternal mortality, MTP Amendment Act, 2017 was enacted by the government of India. About 40-50 million MTPs are done in a year all over the world.
   a) When was MTP legalized by government of India? 1
   b) When is MTP relatively safe? 1
   c) What is the significance of MTP? 3 **[Average]**
5. What are IUDs? Mention any two types with an example for each. How do IUDs help in preventing conception? **[Easy]**

6. Intrauterine devices are most widely accepted methods of contraception.
7. These are used by females and are inserted by doctor or nurses in the uterus through vagina.
8. However these devices are not recommended for those who eventually intend to conceive.

a) How does CuT prevent conception? 2

b) Which IUDs makes uterus unsuitable for implantation? 1

c) Name two hormone releasing IUDs. 2 **[Average]**

9. Explain how oral contraceptives and sterilization help in contraception. **[Average]**
10. What are sexually transmitted diseases? Mention four early symptoms of STD that occur in the genital region. Mention any four complications of STDs if they are not detected early and treated properly. **[Easy]**
11. What are assisted reproductive technologies? How infertility is treated by assisted reproductive technologies? a) IVF-ET b) GIFT c) ZIFT **[Easy]**
12. a) What is infertility? Mention any two causes for infertility. 3

b) How is infertility treated by ICSI, AI? 2 **[Easy]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Based on the picture related to contraceptives answer the question given below given below.

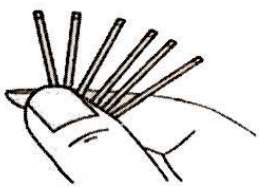


a) Identify the male and the female contraceptive devices, A and B. 2

b) How do they prevent conception? 1

c) Mention two advantages (other than contraception), why they are preferred. 2 **[Difficult]**

2. Explain natural method and sterilization method of contraception. **[Average]**
3. The picture shows the contraceptive device used by women.
4. a) Identify the contraceptive device shown in the picture. 1



b) What does it contain? 1

c) How is it used? 1

d) How does it function as a contraceptive? 2 **[Difficult]**

****

CHAPTER 4
PRINCIPLES OF INHERITANCE AND VARIATION

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. A plant having the genotype AaBb will produce _____ kinds of gametes
   a) 5 b) 4 c) 3 d) 2 **[Average]**
2. The colour based contrasting traits in seven contrasting pairs, studied by Mendel in pea plant were
   a) 1 b) 2 c) 3 d) 4 **[Average]**
3. Which of the following characters was not chosen by Mendel?
   a) Pod shape b) Pod colour c) Position of flower d) Position of pod **[Average]**
4. Which of the following is not a pair of opposing characteristics Mendel studied?
   a) Axial and terminal flowers b) Full and constricted pods
   c) Green and yellow pods d) Pink and white flowers. **[Average]**
5. A cross between two tall plants resulted in tall and dwarf plants. What would be the genotypes of both the parents?
   a) TT and Tt b) Tt and Tt c) TT and TT d) tt and tt **[Average]**
6. In a monohybrid cross between two heterozygous individuals, percentage of pure homozygous individuals obtained in F_1 generation will be
   a) 25 % b) 50 % c) 75 % d) 100 % **[Average]**
7. What is the probability of production of dwarf offspring in a cross between two heterozygous tall pea plants?
   a) Zero b) 50 % c) 25 % d) 100 % **[Average]**
8. In the F_2 generation of a Mendelian dihybrid cross the number of phenotypes and genotypes are
   a) Phenotypes – 4: Genotypes – 16 b) Phenotypes – 9: Genotypes – 4
   c) Phenotypes – 4: Genotypes – 8 d) Phenotypes – 4: Genotypes – 9 **[Average]**
9. Which of the following crosses will give tall and dwarf pea plants in same proportions?
   a) TT × tt b) Tt × tt c) TT × Tt d) tt × tt **[Average]**
10. Mendel's work was rediscovered by three scientists in the year
    a) 1865 b) 1900 c) 1910 d) 1920 **[Easy]**
11. Which three scientists independently rediscovered Mendel's work?
    a) Avery, McLeod, McCarty b) Sutton, Morgan and Bridges
    c) Bateson, Punnett and Bridges d) De Vries, Correns and Tschemark **[Easy]**

12. In *Antirrhinum* (Snapdragon), phenotypic ratio in F_2 generation for the inheritance of flower colour would be

    a) 3 : 1 b) 1 : 2 : 1 c) 1 : 1 d) 2 : 1 **[Easy]**

13. What is the possible blood group of a offspring when both parents have AB blood group?

    a) AB only b) A, B and AB c) A, B, AB and O d) A and B only **[Difficult]**

14. Genes located very close to one another on same chromosome tend to be transmitted together and are called as

    a) Allelomorphs b) Identical genes c) Linked genes d) Recessive genes **[Easy]**

15. Chromosome maps or genetic maps were first prepared by

    a) Sutton and Boveri b) Bateson and Punnett c) Morgan d) Sturtevant **[Easy]**

16. Which of the following is incorrect regarding ZW – ZZ type of sex determination?

    a) It occurs in birds

    b) Females are homogametic and males are heterogametic.

    c) Females are heterogametic and males are homogametic.

    d) Total number of chromosomes is same in both male and female. **[Average]**

17. A couple has three daughters. What is the possibility of them having a son next time?

    a) 0 % b) 25 % c) 50 % d) 100 % **[Average]**

18. Number of autosomes present in a normal human female cell is

    a) 21 b) 22 pairs c) 23 d) 23 pairs **[Easy]**

19. Haplo-diploid sex determination is found in

    a) Grasshoppers b) Birds c) Humans d) Honeybees **[Easy]**

20. In XO type of sex determination

    a) Females produce two different types of gametes

    b) Males produce two different types of gametes

    c) Females produce gametes with Y chromosome

    d) Males produce gametes with Y chromosome. **[Average]**

21. Which of the following techniques may be used to identify an inheritance pattern within a family?

    a) Pedigree analysis b) Chromosome analysis

    c) Nuclear analysis d) Cytoplasm analysis **[Easy]**

22. When is it possible for a woman to be colorblind?

    a) The father has normal vision and the mother is a carrier

    b) The father has normal vision and the mother is colorblind

    c) The father is colorblind and the mother has a normal vision

d) The father is colorblind and the mother is a carrier. **[Average]**

23. A woman with normal vision, whose brother is colour-blind, marries a man with normal vision. They have four children: two sons and two daughters. One son is colour-blind, while the other son and both daughters have normal vision. The pattern of inheritance for the disease is:

a) Sex-linked recessive b) Sex-linked dominant

c) Autosomal dominant d) Autosomal recessive **[Difficult]**

24. In sickle cell anaemia,Glutamic acid is replaced by

a) Valine b) Histidine c) Glutamate d) Aspartate **[Average]**

25. How many possible sickle cell genotypes are there in the population?

a) One b) Two c) Three d) Four **[Easy]**

26. If both parents are unaffected carriers for Thalassemia, which is an autosomal recessive disorder, what are the chances of pregnancy resulting in an affected child?

a) 25 % b) 100 % c) 0% d) 50 % **[Average]**

27. Which of the following assertions about alpha-thalassemia is incorrect?

a) Involves the HBA1 and HBA2 genes

b) Is associated with the mutation or deletion of genes on chromosome 11

c) Is associated with the mutation or deletion of genes on chromosome 16

d) Leads to a reduction in the synthesis of globin **[Average]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Dominant, Consanguineous, Genotype, Sutton, Linkage, Incomplete dominance, Thomas Hunt Morgan, Aneuploidy, Phenylketonuria, Factors, Monohybrid cross, Test cross, Dihybrid cross, Thalassemia, Codominance, Alleles)

1. An individual's collection of genes is called _____ **[Easy]**
2. Genes which code for a pair of contrasting traits are known as _____ **[Easy]**
3. According to Mendel, characters are controlled by discrete units called _____ **[Easy]**
4. The _____ characters appear in the F_1 in heterozygous condition in pea plants. **[Average]**
5. Mendel formulated the law of dominance on the basis of _____ **[Easy]**
6. Law of independent assortment can be explained with the help of _____ **[Easy]**
7. To determine the genotype of a tall plant of F_1 generation, Mendel crossed this plant with a dwarf plant. This cross represents a _____ **[Average]**
8. The inheritance of flower colour in dog flower is an example for _____ **[Easy]**
9. AB blood group is an example for _____ **[Easy]**

10. _____ united the knowledge of chromosomal segregation with Mendelian principles. **[Easy]**
11. Experimental verification of the chromosomal theory of inheritance was given by _____ **[Easy]**
12. The physical association of two genes is _____ **[Easy]**
13. The diseases caused by mutation in the gene that codes for the enzyme phenylalanine hydroxylase is _____ **[Easy]**
14.  denotes _____ mating **[Average]**
15. An autosome linked recessive blood disease is _____ **[Average]**
16. Failure of segregation of chromatids during cell division resulting in the gain or loss of chromosomes is called _____ **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Differentiate between homozygous and heterozygous conditions. **[Average]**
2. Differentiate between dominant and recessive alleles. **[Average]**
3. What is test cross? Write its significance. **[Average]**
4. State the law of segregation and law of independent assortment. **[Easy]**
5. What is incomplete dominance? Give an example. **[Easy]**
6. A pea plant with purple flowers was crossed with plant having white flowers. The progeny produced only purple flowers. On selfing, these plants produced 482 plants with purple flowers and 162 plants with white flowers. Explain the mechanism that accounts for these results? **[Difficult]**
7. Name the respective pattern of inheritance where F_1 phenotype
   a) Does not resemble either of the two parents and is in between the two
   b) Resembles only one of the two parents. **[Average]**
8. What is codominance? Give an example. **[Easy]**
9. Distinguish between incomplete dominance and codominance. **[Average]**
10. What are multiple alleles? Give an example. **[Easy]**
11. A child has blood group **O**. If the father has blood group **A** and mother blood group **B**, work out the genotypes of the parents and the possible genotypes of the other offspring. **[Average]**
12. What is polygenic inheritance? Give an example. **[Easy]**
13. ABO blood grouping in human population exhibits four possible phenotypes from six different genotypes. Explain different mechanisms of inheritance involved in exhibiting the possibility of four phenotypes and six genotypes. **[Average]**
14. Define pleiotropy. Give an example. **[Easy]**
15. Differentiate between linkage and recombination. **[Average]**

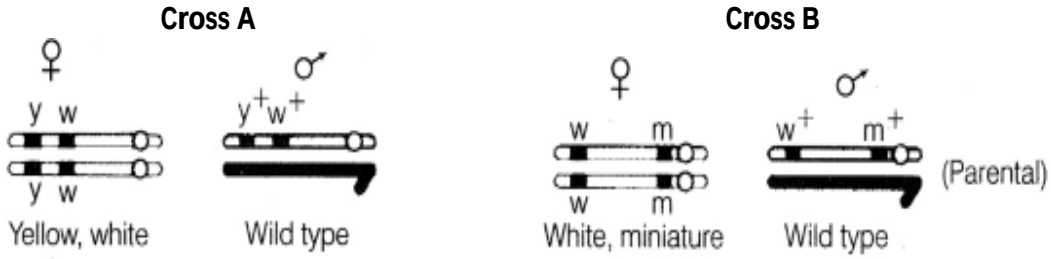
16. List two features of *Drosophila melanogaster* which make it suitable for genetic experiments. **[Easy]**
17. Differentiate male heterogamety from female heterogamety. **[Average]**
18. Define mutation. What are mutations causing agents called? **[Easy]**
19. What are mutagens? Give example. **[Easy]**
20. What is point mutation? Give an example. **[Easy]**
21. What are Mendelian disorders? Give an example. **[Easy]**
22. Schematically represent a pedigree chart of myotonic dystrophy. **[Difficult]**
23. Schematically represent a pedigree chart of sickle cell anaemia. **[Difficult]**
24. Mention the sixth codon of the beta globin chain in normal haemoglobin and the haemoglobin of person with sickle celled anaemia. **[Average]**
25. Distinguish between α (Alpha) thalassemia and β (Beta) thalassemia. **[Average]**
26. What is aneuploidy? Give an example. **[Easy]**
27. What is polyploidy? Give an example. **[Easy]**
28. Distinguish between aneuploidy and polyploidy. **[Average]**
29. Write the genotype of
    a) An individual who is carrier of sickle-cell anaemia gene but apparently unaffected 1
    b) An individual affected with the of sickle-cell anaemia disease. 1 **[Average]**
30. Mention a Mendelian disorder and its types where there is reduced synthesis of haemoglobin. **[Average]**

PART – C

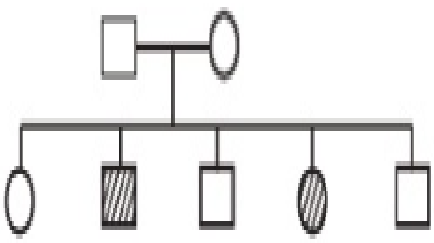
IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Schematically represent the inheritance of stem height in pea plant. **[Average]**
2. Schematically represent the inheritance of seed shape and seed colour in pea plants. **[Average]**
3. Schematically represent incomplete dominance in *Antirrhinum*. **[Average]**
4. When a tall pea plant was selfed, it produced one-fourth of its progeny as dwarf. Explain with the help of a cross. **[Average]**
5. A tall Pea plant with yellow seeds (heterozygous for both) is crossed with a dwarf Pea plant with green seeds. Using a Punnett square work out the cross to show the phenotypes and the genotypes of F_1 generation. **[Average]**
6. A woman with blood group O married a man with AB group. Show the possible blood groups of the progeny. List the alleles involved in this inheritance. **[Average]**
7. Four children with four different blood groups are born to parents where the mother has blood group ‘A’ and the father has blood group ‘B’. Work out the cross to show the genotypes of the parents and all four children. **[Average]**

8. "Mendel's work was not recognized during his time." Why? **[Average]**
9. Mention the similarities between the behaviours of chromosomes and genes according to chromosomal theory of inheritance. **[Average]**
10. During his studies on genes in Drosophila that were sex-linked T.H. Morgan found F_2 population phenotypic ratios deviated from expected 9:3:3:1. Explain the conclusion he arrived at. **[Average]**
11. T H Mogan used *Drosophila melanogaster* for genetical experiments. Why? **[Average]**
12. Study the figures given below and answer the question.
    Identify in which of the crosses, the strength of linkage between the genes is higher. Give reasons in support of your answer. **[Average]**



13. Explain how is sex determined in human beings. **[Average]**
14. Explain sex determination in *Drosophila melanogaster*. **[Average]**
15. Explain sex determination in birds. **[Average]**
16. Explain sex determination mechanism in grasshopper. **[Average]**
17. Explain sex determination mechanism in honey bees. **[Average]**
18. What is pedigree analysis? Mention its significance. **[Average]**
19. The following pedigree shows a particular trait which is absent in the parents but found in the subsequent generation irrespective of the sexes. Analyze the pedigree and draw a conclusion. **[Difficult]**



20. Mention the cause and symptoms of phenylketonuria. **[Average]**
21. Mention the cause and symptoms of colourblndness. **[Average]**
22. Mention the cause, karyotype and symptoms of Down's syndrome. **[Average]**
23. Mention the cause, karyotype and symptoms of Klinefelters' syndrome. **[Average]**

24. Mention the cause, karyotype and symptoms of Turners' syndrome. **[Average]**

25. Identify the genetic disorders given below:

    a) Sterile females, Chromosomal complement 45 with XO. 1

    b) Short statured, furrowed tongue, characteristic palm crease. 1

    c) Sterile males, Gynaecomastia, Karyotype of 47 1 **[Average]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. State law of segregation. Schematically represent it with reference to the inheritance of height in pea plants. **[Average]**
2. Schematically represent inheritance of colour and shape of seed in pea plant. **[Average]**
3. What is incomplete dominance? Explain it with reference to flower colour in *Snapdragon*. **[Average]**
4. A colour-blind father has a daughter with normal vision. The daughter marries a man with a normal vision. What is the probability of her children to be colourblind? Explain with the help of a pedigree chart. **[Difficult]**
5. Write a note on sickle cell anemia. **[Average]**
6. Write a note on haemophilia. **[Average]**
7. Explain thalassemia as an example for Mendelian disorder. **[Average]**
8. Give an example for each of the following traits:

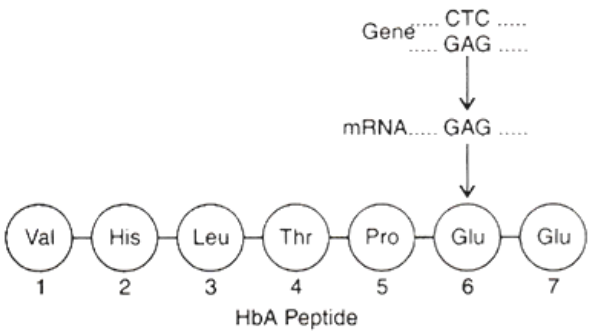
    a) Sex-linked recessive disorder b) Autosome linked recessive disease

    c) An inborn error of metabolism d) A single gene that exhibit multiple phenotypic expression

    e) A trait generally controlled by three or more genes **[Easy]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Given below is the representation of amino acid composition of the relevant translated portion of B chain of haemoglobin, related to the shape of human red blood cells.



a) Is this representation indicating a normal human or a sufferer from certain related genetic disease? Give reason in support of your answer. 2

b) What difference would be notice in the phenotype of the normal and the sufferer related to this gene? 1

c) Who are likely to suffer more from the defect related to the gene represented the males, the females or both males and females equally? Why? 2 .

[Difficult]

2. Observe the representation of a dihybrid cross and answer the questions that follow.

RRYY (Round and yellow) X rryy (Wrinkled and green)

↓

RrYy (Selfing of F_1)

↓

♀ \ ♂	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

a) Write the phenotypic ratio. 1

b) Identify homozygous dominant genotype in progeny. 1

c) Identify recessive genotype in progeny. 1

d) Number of parental type progeny. 1

e) Number of recombinant type progeny. 1 **[Difficult]**

3. A cross between a normal couples resulted in a son who was haemophilic and a normal daughter. The daughter was married to a normal man. To their surprise, the grandson was also haemophilic.

a) Represent the cross in the form of a pedigree chart. Give the genotypes of the daughter and her husband. 3

b) Write the conclusion you draw of the inheritance pattern of this disease. 2 **[Average]**

4. Identify the genetic disorders given below:

a) Sex-linked recessive disorder occurring in about 8% of males and only about 0.4% of females leading to defect in cone of eye. 1

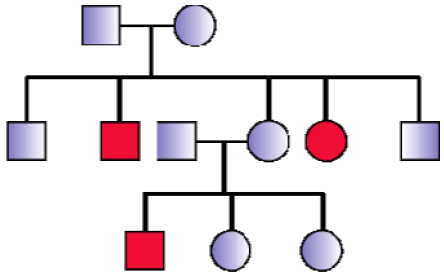
b) Sex-linked recessive disorder affecting a single protein involved in the clotting of blood. 1

c) Inborn error of metabolism, autosomal recessive trait, accumulation of phenylalanine. 1

d) Autosome linked recessive blood disease, substitution of amino acid in beta globin chain of haemoglobin. 1

e) Autosome linked recessive blood disease, reduced rate of synthesis of globin chains of haemoglobin. 1**[Average]**

5. Refer to the representation of a pedigree chart of a particular trait and answer the questions that follow.



a) Which genetic disorder does the pedigree represent?

b) Mention the cause of the above trait

c) Write any three features of the trait.

****

CHAPTER 5
MOLECULAR BASIS OF INHERITANCE

PART-A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. The two nucleotides of DNA are linked through 3 and 5 carbon of pentose sugar by
   a) N- Glycosidic bond b) Phosphodiester bond
   c) H bond d) Phosphoester bond **[Easy]**
2. In biochemical characterization of transforming principle, which of the following enzymes affect transformation
   a) Protease b) DNase c) Amylase d) Lipase **[Easy]**
3. The experiment that proves the semiconservative principle of DNA replication
   a) Griffith experiment
   b) Hershey and Chase experiment
   c) O.Avery, MacLeod & M.McCarty experiment
   d) M.Meselson & F.Stahl experiment **[Easy]**
4. Average rate of polymerization during DNA replication is
   a) 200 bp/sec b) 200 bp/min c) 2000 bp/sec d) 2000 bp/min **[Easy]**
5. RNA polymerase I transcribes
   a) rRNA b) mRNA c) tRNA d) hnRNA **[Easy]**
6. GGU, GGC, GGA and GGG all code for glycine. Which salient feature of the genetic code does it prove?
   a) Unambigous b) Degenerate codons c) Universal code d) Initiator code **[Difficult]**
7. Aminoacylation of tRNA occurs during
   a) Transcription b) Translation c) Regulation of operon d) DNA fingerprinting **[Easy]**
8. In a single stranded DNA, if the ratio of A:G is 60:40 then the ratio of T:C is
   a) 60:40 b) 40:60 c) 20:80 d) None of the above **[Difficult]**
9. The components of deoxyguanosine is
   a) Guanine + Ribose b) Guanine + Deoxyribose
   c) Guanine + Deoxyribose + Phosphate d) Guanine + Ribose + Phosphate **[Average]**
10. The coding sequences present in processed RNA is called
    a) Genes b) Exons c) Introns d) Cistron **[Average]**

11. A mature (processed) mRNA transcribed from a gene is 636 nucleotides long. Assuming it it codes continuously without UTR and stop codons for a polypeptide, how many amino acids does it encode?

    a) 212 b) 211 c) 210 d) 636 **[Difficult]**

12. If a mRNA 3'-AUGUUUUUUUUUUAA-5' undergoes translation, how many types of amino acids are present

    a) 4 types b) 3 types c) 2 types d) 1 type **[Difficult]**

13. **Statement 1:** Deoxyribonucleotide triphosphate (dNTPs) acts as substrate during DNA replication.
    Statement II: Deoxyribonucleotide triphosphate (dNTPs) also acts as the source of energy in the process of DNA replication.
    Choose the correct statement

    a) Both statement I and statement II are correct

    b) Both statement I and statement II are incorrect

    c) Statement I is correct but statement II is incorrect

    d) Statement I is incorrect but statement II is correct **[Average]**

14. In pea plants, each character is governed by a gene which is

    a) Monocistron b) Polycistron c) Operon d) Intron **[Difficult]**

15. Which of these are correct?

    I. UTR (Untranslated regions are located on DNA)

    II.UTR is present both at 5' before the start codon and also at 3' after the stop codon

    III.UTR provides stability to mRNA IV. UTR is required for efficient translation

    a) I, II, IV b) I, III, IV c) I, II, III d) II, III, IV **[Difficult]**

16. During transcription the site of DNA molecule at which RNA polymerase binds is called

    a) Promoter b) Regulator c) Enhancer d) Inducer **[Average]**

17. Methyl Guanosine Triphosphate is added at the 5' of hnRNA in a process of

    a) Splicing b) Capping c) Tailing d) Coding **[Average]**

18. If the template of a DNA strand is 3'ATGCAATCT5' its complementary DNA strand would be

    a) 3'TACGTTAGA 5' b) 3'ATGCAATCT5'

    c) 5'TACGTTAGA 3' d) 5'ATGCAATCT3' **[Average]**

19. Expand YAC

    a) Yeast artificial centromere b) Yeast artificial chromosome

    c) Yeast automated chromosome d) Yeast annotated chromosome **[Easy]**

20. Expand BAC

a) Bacterial artificial centromere b) Bacterial artificial chromosome

c) Bacterial automated chromosome d) Bacterial annotated chromosome **[Easy]**

21. Match the following:

A. Alec Jeffreys i. Lac operon

B. F. Sanger ii. Transforming principle

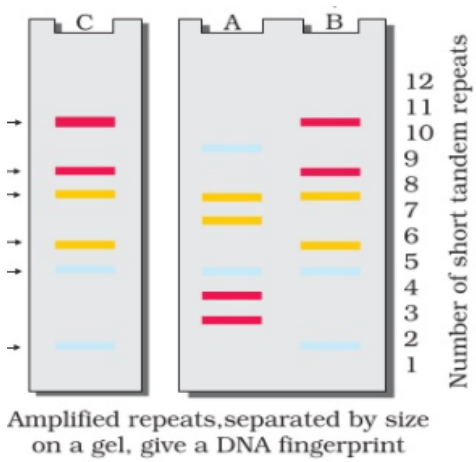
C. Jacob & Monad iii. Annotated DNA sequencing

D. Griffith iv. DNA finger print

a) A-iv, B-ii, C-i, D-iii b) A-iv, B-iii, C-i, D-ii

c) A-iii, B-iv, C-ii, D-I d) A-iii, B-i, C-ii, D-iii **[Average]**

22. If A & B are the DNA of two mothers and C is DNA of child. On observation of the banding pattern of the DNA finger printing, it can be concluded that



Amplified repeats,separated by size on a gel, give a DNA fingerprint

a) Child C belongs to mother A b) Child C belongs to mother B

c) Child C does not belongs to A or B d) No conclusions can be drawn. **[Difficult]**

23. **Statement I:** In Bio-chemical analysis of the transforming principle only DNAase could not transform R strain to S strain.

Statement II: This experiment proved that proteins and RNAase not the transforming principle.

a) Both statement I and statement II are correct

b) Both statement I and statement II are incorrect

c) Statement I is correct but statement II is incorrect

d) Statement I is incorrect but statement II is correct **[Average]**

24. In a DNA strand if there are 100 nucleotides and 30 are adenine the number of thymine, guanine, cytosines respectively are:

a) 30, 20, 20 b) 30, 20, 30 c) 20, 30, 20 d) 20, 20, 30 **[Difficult]**

25. The chemical method developed by Har Gobind Khorana was
    a) Synthesizing RNA molecules with defined combination of basis
    b) Synthesizing RNA without a template.
    c) Categorization of RNA
    d) Cell-free Protein synthesis **[Average]**
26. In a structural gene consisting of 300 base pairs, if 4 nucleotides are added and 1 is deleted at the end point it leads to:
    a) Point mutation b) Frame shift mutation.
    c) Unaltered change. d) 101 aminoacids. **[Average]**
27. Human insulin is produced in *E.coli* cells by rDNA technology. This proves that genetic code is/a
    a) Universal. b) Triplet codon. c) Degenerate. d) Initiator codon **[Average]**
28. **Statement I:** In Bio-chemical analysis of the transforming principle only DNAase could not transform R strain to S strain.
    Statement II: This experiment proved that proteins and RNAase not the transforming principle.
    a) Both statements I and statement II are correct
    b) Both statement I and statement II are incorrect
    c) Statement I is correct but statement II is incorrect
    d) Statement I is incorrect but statement II is correct **[Average]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Gene, Lysine, Automated DNA sequencer, DNA, Glycosidic, RNA polymerase I, 5'→ 3', Severo Ochoa, Bioinformatics, Inducer)

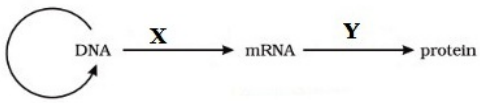
1. A nitrogenous base is linked to the OH of carbon of pentose sugar by_____ bond **[Easy]**
2. Histones are rich in basic amino acid residues like _____. **[Easy]**
3. The polarity of the coding strand of a transcription unit is _____. **[Easy]**
4. Template independent enzymatic synthesis of RNA was achieved by using the enzyme ____ **[Easy]**
5. A _____ is defined as a structural unit of inheritance. **[Easy]**
6. DNA sequencing is done by using _____ developed by Frederick Sanger. **[Easy]**
7. Lactose regulates switching on and off of the lac operon hence it is termed as_____ **[Average]**
8. _____ is branch of Biology that deals with store of DNA sequences in data bases and data analysis tools. **[Easy]**
9. 28S, 18S and 5.8S are all transcribed by the enzyme_____ **[Easy]**
10. The radioactive P was used to label _____ of the viral particle in Hershey and Chase experiment. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. What is the structural unit of a nucleic acid called? Name the chemical components of the unit. **[Easy]**
2. Mention the four nitrogenous bases present in DNA. **[Easy]**
3. Mention the four nitrogenous bases present in RNA. **[Easy]**
4. List the nucleosides of DNA. **[Easy]**
5. Mention the nucleosides of RNA. **[Easy]**
6. If a double stranded DNA has 20% of cytosine, calculate the percent of adenine. **[Difficult]**
7. Explain the central dogma of molecular biology with a schematic representation. **[Average]**
8. Draw a labelled diagram of nucleosome. **[Average]**
9. Define the terms transcription and translation. **[Easy]**
10. How many base pairs of DNA are present in a typical nucleosome? Mention the types of proteins present in a nucleosome. **[Average]**
11. Write the conclusion that can be drawn from Griffith's experiment. **[Average]**
12. Mention any two criteria that a molecule must fulfil to act as genetic material. **[Average]**
13. DNA is a better genetic material than RNA. Justify with two reasons. **[Difficult]**
14. Mention the functions of DNA polymerase and DNA ligase during replication. **[Average]**
15. Draw a labelled diagram of replication fork. **[Average]**
16. "Both the strands of DNA molecule do not act as templates during transcription". Justify with two reasons. **[Difficult]**
17. Identify the following:
    a) Pentose sugar present in DNA
    b) Pyrimidine present only in DNA **[Average]**
18. If 5'- ATGC - 3' is a template DNA strand, write its complementary strand with polarity. **[Average]**
19. If the total length of DNA in an egg is 2.1×10^6 bp, what is the length of the DNA in zygote? **[Average]**
20. In a ds DNA of 400 bp, if there are 150 Adenine, find the number of the other bases. **[Difficult]**
21. If a DNA 34nm in length has 30 Guanine, calculate the number of other nucleotides. **[Difficult]**
22. Applying Chargaff's rule, if cytosine content of a DNA sample is 100, calculate the length of DNA. **[Difficult]**
23. Differentiate between euchromatin and heterochromatin. **[Average]**
24. In Meselson and Stahl experiment, in 40 min we obtain one light and one hybrid band. What would be the proportion of the band after 60 min? **[Difficult]**

25. Draw a schematic representation of transcription unit. **[Average]**
26. If Template strand of DNA has following sequence:
    3' T A C T G T G A G C T A C T C A T T 5' Write the sequence of mRNA. **[Average]**
27. Write any two differences between prokaryotic and eukaryotic transcription unit. **[Average]**
28. Differentiate monocistronic transcriptional unit from polycistronic transcriptional. **[Average]**
29. Differentiate between introns and exons. **[Average]**
30. Differentiate template strand from coding strand in DNA. **[Average]**
31. Write the coding strand and the sequence of base on mRNA synthesized by
    DNA template with the sequence 3'-ATGCATAGCAT-5' **[Average]**
32. Write a note on the role of sigma factor and the rho factor in transcription in prokaryotes. **[Easy]**
33. What is heterogeneous nuclear RNA (hnRNA)? How is it converted into functional mRNA? **[Average]**
34. Name the two amino acids richly present in histones that impart positive charge. **[Average]**
35. If 60% of a DNA is heterochromatinised, what percent of DNA is genetically active? **[Average]**
36. Label X and Y with respect to central dogma: **[Average]**



37. According to Griffith experiment why did the mice die when heat killed S-strain and
    live R-strain was injected into mice. **[Average]**
38. Mention any four properties of a genetic material. **[Easy]**
39. In Human cell DNA replication: Whose $2n = 6.6 \times 10^9$ bp takes place at the rate of polymerization,
    2000 bp/sec. How long it would take to complete the process. **[Difficult]**
40. DNA is more stable than RNA. Substantiate. **[Average]**
41. **Match the following**:

A. DNA ligase	1. Synthesis of rRNA
B. DNA polymerase	2. Synthesis of tRNA
C. RNA polymerase I	3. Joining of Okazaki fragments
D. RNA polymerase III	4. Synthesis of DNA strand

[Average]

42. In the context of DNA replication, what is the polarity of leading strand and lagging strands? **[Average]**
43. 5' – 3' strands of DNA are not copied during transcription. Give reason. **[Average]**
44. List the essential roles of ribosome during translation. **[Average]**
45. Genetic code is said to triplet and universal. Why? **[Average]**
46. AUG has dual function. Justify. **[Average]**
47. There are no tRNAs for stop codons. Substantiate. **[Average]**

48. tRNA is regarded as adapter molecule. Substantiate. **[Average]**
49. Mention the initiator and terminator codons. **[Easy]**
50. Name the initiator codon and the amino acid coded by it. **[Easy]**
51. If AUG codes for methionine and UAC codes for Tryptophan, write its anticodon on tRNA. **[Average]**
52. Differentiate point mutation and frame-shift mutation. **[Average]**
53. Mention the four levels in regulation of gene expression in eukaryotes. **[Easy]**
54. Mention the role of repressor and inducer in regulation of gene expression. **[Average]**
55. Mention the functions of β-galactosidase and permease in lactose metabolism in *E. coli*. **[Easy]**
56. How does the Lac-operon function in the presence of inducer (lactose) in the medium? **[Average]**
57. How does the Lac-operon function in the absence of inducer (lactose) in the medium? **[Average]**
58. Mention any two goals of Human Genome Project (HGP). **[Easy]**
59. Mention two applications of Human Genome Project. **[Easy]**
60. Mention two salient features of human genome. **[Easy]**
61. Which chromosome in human genome has
    a) Highest number of genes
    b) Least number of genes **[Easy]**
62. Mention the two approaches with respect to the methodology of sequencing DNA in Human Genome Project. **[Easy]**
63. Mention two common vectors used for the amplification of DNA fragments during the sequencing of human genome. **[Easy]**
64. Name the two commonly used host cell and the vectors in HGP. **[Easy]**
65. Which is the largest known human gene? Mention the number of genes in it. **[Easy]**
66. Mention any two differences with reference to transcription in prokaryotes and eukaryotes. **[Average]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. What are the components of Thymine monophosphate (TMP)? **[Average]**
2. What are the components of Uridine monophosphate (UMP)? **[Average]**
3. Mention the locationsof N - glycosidic bond, Hydrogen bond & phosphoester bond present in DNA? **[Average]**
4. Identify the following :
    a) Pyrimidine present both in DNA and RNA
    b) Purines present in DNA and RNA
    c) Chemical bond present between 2 nucleotides. **[Average]**

5. Mention the three codons that do not code for any amino acid. **[Easy]**
6. Lac operon is a polycistronic gene. Give reason **[Average]**
7. Mention the three structural genes and the proteins they code for in *E.coli.* **[Average]**
8. RNA polymerase exhibit division of labour. Substantiate **[Difficult]**
9. The hnRNA undergoes processing to become stable mRNA. List out the processes involved. **[Average]**
10. Mention one function of each mRNA, rRNA and tRNA. **[Easy]**
11. Mention the functions of RNA polymerase I, RNA polymerase II and RNA polymerase II in eukaryotes. **[Easy]**
12. With reference to transcription, define: a) Splicing b) Capping c) Tailing **[Easy]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. What are the functions of
   a) *i gene* b) *lac z gene* c) *lac y gene* d) *lac a gene* e) inducer. **[Easy]**
2. a) Mention the three steps involved in transcription in prokaryotes. 3
   b) What is role of **σ** factor and **ρ** factor in transcription of prokaryotes? 2 **[Easy]**
3. Describe the double helix model of DNA proposed by Watson & Crick. **[Average]**
4. Explain Griffith's experiment to explain the 'transforming principle'. **[Average]**
5. Explain the steps involved in DNA replication. **[Average]**
6. Explain the experiment that gave the final proof for DNA as the genetic material. **[Difficult]**
7. Explain the experiment to prove that DNA replicates semiconservatively. **[Average]**
8. Enumerate the characteristics of Genetic code. **[Easy]**
9. Explain the process of transcription in prokaryotes. **[Average]**
10. Explain the regulation of operon that is involved in utilization of lactose in *E.coli.* **[Average]**
11. Mention the goals of HGP. **[Easy]**
12. Explain the steps involved in translation process. **[Difficult]**
13. Mention the salient features of human genome. **[Easy]**
14. Mention the steps involved in DNA finger printing technique developed by Alec Jeffrey. **[Easy]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. The following sequence of nucleotides 3' AUG UAC UUU AGG GAG UAA 5' in mRNA codes for a sequence of 5 amino acids. Generate the salient features of genetic code. **[Difficult]**
2. In context to lac operon, when Jacob & Monad cultured *E.coli* cell in a media devoid of lactose, *E.coli* cells formed colonies and when lactose was added, the *E.coli* cells metabolized lactose by synthesizing enzymes. Explain how lac operon switches off & switches on. **[Average]**

3. Regarding Hershey and Chase experiment

   a) Mention the three steps involved. 3

   b) Radioactive Sulphur & radioactive Phosphorus get incorporated in which parts of viral particle. 2 **[Average]**

4. Name the process involved in the following:

   a) One parental DNA strand is conserved by daughter cell.

   b) Separation of DNA fragment.

   c) Removal of introns in hnRNA.

   d) Attachment of Rho factor to RNA polymerase during transcription.

   e) Polymerization of amino acids to form a polypeptide on mRNA. **[Average]**

5. A policeman from the crime department approached the forensic science department to help identify the culprit. DNA samples were collected from the crime site and of the two suspects.

   a) Explain the process involved.

   b) If the culprit had monozygotic twin, can this method give results. **[Average]**

****

CHAPTER 6
EVOLUTION

PART - A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. Theory of spontaneous generation was experimentally disproved by

   a) Louis Pasteur b) Haldane & Oparin c) S.L. Miller d) Charles Darwin **[Easy]**

2. The thorns of *Bougainvillea* and Tendrils of *Cucurbita* represents

   i. Homology ii. Convergent evolution iii. Divergent evolution iv. Analogy

   a) i & ii are correct. b) i & iii are correct c) ii & iv are correct. d) ii & iii are correct. **[Average]**

3. **Statement 1:** Man has domesticated dogs & has created different breeds by artificial breeding.

   Statement 2: Intensive breeding of dogs signifies divergent evolution.

   a) Both statements 1 & 2 are correct.

   b) Statement 1 is correct & 2 is incorrect.

   c) Statement 1 is incorrect & 2 is correct.

   d) Both the statements are incorrect. **[Average]**

4. **Statement 1:** Antibiotic or drug resistance is due to evolution by anthropogenic action.

   Statement 2: Many antibiotic or drugs resistant organisms arise due to over usage of antibiotics.

   a) Both statements 1 & 2 are correct. b) Statement 1 is correct & 2 is incorrect.

   c) Statement 1 is incorrect & 2 is correct. d) Both the statements are incorrect. **[Average]**

5. Charles Darwin in Galapagos Island found that finches had different forms of beaks.

   This is process of evolution represents

   a) Natural selection b) Adaptive radiation

   c) Darwinian Fitness d) Struggle for existence. **[Easy]**

6. A brown billed duck is newly added to a farm of white billed ducks. This indicates

   a) Gene flow b) Genetic drift c) Mutation d) Natural selection **[Average]**

7. Change in frequency of genes and alleles is due to

   a) Mutation b) Gene flow c) Both a & b d) Neither a or b **[Easy]**

8. The first amphibians that lived both on land and water evolved from

   a) Fishes b) Turtles c) Lobefins d) Frog **[Easy]**

9. Arrange in correct sequence of their occurrence on geological time scale.

   i. Amphibians ii. Fishes iii. Reptiles iv. Mammals

   a) i ii iii iv b) ii iii i iv c) ii i iii iv d) iii ii iv I **[Difficult]**

10. The biggest land reptiles that existed during Mesozoic era was

a) *Ichthyosaurs* b) *Tyrannosaurs* c) Sea cows d) Crocodiles **[Easy]**

11. Arrange the following in the order of evolution.

a. Neanderthal man b. *Australopithecus* c. *Ramapithecus*

d. *Homo habilis* e. *Homo erectus*

a) a → b → c → d → e b) e → d → c → b → a

b) b → c → d → e → a d) c → b → d → e → a **[Average]**

12. Koala, Tasmanian wolf, Kangaroo & Marsupial rat, all of these represent

a) Darwin flinches b) Australian Marsupials

c) Placental mammals d) Non placental mammals **[Easy]**

13. **Statement 1:** Hugo De Vries believed mutation leads to spontaneous evolution.

Statement 2: Mutations are random and direction less & cause saltation.

a) Both statements 1 & 2 are correct.

b) Statement 1 is correct & 2 is incorrect.

c) Statement 1 is incorrect & 2 is correct.

d) Both the statements are incorrect. **[Average]**

14. Elongation of neck in giraffes, due to inheritance of acquired characters was given by

a) Lamarck b) Thomas Malthus c) Charles Darwin d) Hardy-Weinberg **[Easy]**

15. According to Oparin, which one of the following was not present in the primitive atmosphere of the earth?

a) Water vapour b) Oxygen c) Methane d) Hydrogen **[Easy]**

16. Single step large mutation leading to speciation is also called

a) Natural selection b) Branching descent c) Founder's effect d) Saltation **[Easy]**

17. Which of the following theories attempts to explain origin of universe?

a) Panspermia theory b) Big Bang theory

c) Chemical evolution theory d) Darwin's theory **[Easy]**

18. In Miller's experiment, the environment of primitive earth was created & temperature was

a) 400°C b) 450°C c) 800°C d) 850°C **[Easy]**

19. The organic compounds formed in Miller's experiment is/are

i. Amino acids ii. Nitrogen bases iii. Pigment iv. Fats

a) i only b) i & ii only c) i, ii & iii only d) i, ii, ii & iv. **[Easy]**

20. The paleontological evidence for evolution is obtained from

a) Fossils b) Embryos c) Homologous organs d) Analogous organs **[Easy]**

21. Fossils are hard remains of life forms and are found in

    a) Igneous rocks b) Metamorphic rocks c) Sedimentary rocks d) Any type of rocks **[Easy]**

22. **Statement 1:** The study of fossils in various sedimentary layers indicates the geological period in which they exit.

    Statement 2: Ages of the fossil can be determined by radioactive dating.

    a) Both statements 1 & 2 are correct.

    b) Statement 1 is correct & 2 is incorrect.

    c) Statement 1 is incorrect & 2 is correct.

    d) Both the statements are incorrect. **[Easy]**

23. Embryological evidence for evolution was disapproved by

    a) Ernst Heckel b) Karl Ernst Von Baer c) Charles Darwin d) Alferd Wallace **[Easy]**

24. Epochs, periods and eras depict

    a) Geological history b) Biological history

    c) Physiological history d) Embryological history **[Easy]**

25. Arrange the following plants as appeared on different geological periods

    a) Conifers, Gnetales, Dicotyledons, Monocotyledons

    b) Gnetales,Dicotyledons,Monocotyledons,Conifers

    c) Conifers,Monocotyledons,Dicotyledons,Gnetales

    d) Conifers, Gnetales,Monocotyledons,Dicotyledons **[Difficult]**

26. Potato is respectively homologous & analogous to

    a) Sweet potato & tendrils of *Cucurbita*

    b) Tendrils of *Cucurbita* & sweet potato

    c) Thorns of *Bougainvillea* & tendrils of *Cucurbita*

    d) Sweet potato & wings of butterfly **[Average]**

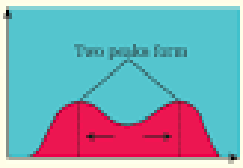
27. **Statement 1:** Pouched mammals of Australia survived because of lack of competition.

    Statement 2: Continental drift has both positive & negative effect on population.

    a) Both statement 1 & 2 are correct. b) Statement 1 is correct & 2 is incorrect.

    c) Statement 1 is incorrect & 2 is correct. d) Both the statements are incorrect. **[Average]**

28. Observe the following diagrammatic representation of operation of natural selection on different traits. Here, Natural selection depicts



a) Stabilisation b) Directional c) Disruptive d) Cannot be depicted **[Average]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Conifers, Panspermia, *Homo habilis*, Divergent evolution, Chemical, *Ichthyosaurs*, Founder effect, Analogy, Embryological, Neanderthal)

1. Convergent evolution is indicated by _____ **[Easy]**
2. Adaptive radiation of marsupials of Australia indicate _____ **[Easy]**
3. If the allele frequency is very different in the new sample of population they become different species. This is called _____ **[Easy]**
4. _____ were group of plants that pre-dominated earth during Mesozoic era. **[Easy]**
5. _____ was a land reptile that went back into water to evolve into fish-like reptiles. **[Easy]**
6. _____ was believed to come from outer space that created life on earth. **[Easy]**
7. Inorganic chemicals formed organic molecules which lead to first form of life. This was supported by _____ evolution. **[Easy]**
8. Karl Ernst Von Baer disapproved _____ evidence for evolution. **[Easy]**
9. _____ stage in human evolution used hides to protect their body and buried their dead. **[Easy]**
10. The first human - like being was _____. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Name any two theories that explain the origin of life. **[Easy]**
2. Write a short note on Big bang theory. **[Easy]**
3. How could the experiment of Louis Pasteur dismiss theory of spontaneous generation of life? **[Average]**
4. How does Miller's experiment support the theory of chemical evolution? **[Average]**
5. List any four areas which provide evidences for evolution. **[Easy]**
6. What are fossils? How do they support evolution? **[Average]**
7. What are homologous organs? Give an example. **[Easy]**
8. What are analogous organs? Give an example. **[Easy]**
9. What is convergent evolution? Give an example. **[Easy]**
10. What is divergent evolution? Give an example. **[Easy]**
11. Distinguish between divergent and convergent evolution. **[Average]**
12. Comment on the similarity between the thorn of *Bougainvillea* and tendril of *Cucurbita* with reference to evolution. **[Average]**

13. Analyze the evolutionary relationship between forelimbs of whales and human. **[Average]**
14. Differentiate between homology and analogy. **[Average]**
15. What is convergent evolution? Give an example of convergent evolution in plants and animals. **[Average]**
16. Explain with the help of an example the type of evolution which is based on analogy. **[Average]**
17. State the evolutionary relationship giving reasons between sweet potato and potato. **[Average]**
18. Mention the two key concepts of Darwinian theory of evolution. **[Easy]**
19. Genetic drift is a chance while gene flow is constant. Substantiate. **[Average]**
20. Define the terms a) Genetic drift b) Founder effect **[Easy]**
21. What is adaptive radiation? Give an example. **[Easy]**
22. When more than one adaptive radiation occurs in an isolated geographical area it leads to convergent evolution. Give two examples. **[Average]**
23. Give any two examples of natural selection by anthropogenic action. **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. What are homologous organs? Mention one example of homologous organs in plants and one in animals. **[Easy]**
2. What are analogous organs? Give one example of analogous organs in plants and one in animals. **[Easy]**
3. Differentiate between homologous and analogous structures. Give one plant and one animal example for each. **[Average]**
4. Differentiate between divergent evolution and convergent evolution. Give one plant and one animal example for each. **[Average]**
5. Draw a labelled diagrammatic representation of Miller's experiment. **[Average]**
6. a) What were the different gases used by S.L.Miller in his experimental setup?
   b) What conditions of primitive earth was recreated in the flask?
   c) Write the conclusion drawn from this experiment. **[Average]**
7. After industrialization in England, it was observed that white winged moth did not survive.
   a) What is the cause?
   b) What was the change and why it had happened?
   c) Which organism is known as natural indicator to air pollution? **[Average]**
8. "Darwin finches of Galapagos Islands represent on the best example for adaptive radiation. Substantiate". **[Average]**

9. "Australian marsupials and Australian placental mammals explain convergent evolution and adaptive radiation". Justify the statement. **[Average]**
10. How is Darwin's concept of evolution different from that of Hugo deVries concept of mutation? **[Average]**
11. State Hardy-Weinberg principle. List four factors which upset genetic equilibrium. **[Easy]**
12. a) State Hardy-Weinberg principle.
    b) Define the terms: i) Gene pool ii) Genetic equilibrium **[Easy]**
13. Define the terms a) Saltation b) Gene flow c) Natural selection **[Easy]**
14. Arrange the following ancestral forms of man in the order of their evolution:
    a) *Homo habilis* b) *Homo erectus* c) Neanderthal man
    d) *Dryopithecus* e) *Ramapithecus* f) *Australopithecines* **[Average]**
15. A gene locus has two alleles A & a. If the frequency of dominant allele A is 0.4, what will be the frequency of homozygous dominant, heterozygous dominant and recessive individuals in the population? **[Average]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Represent Miller's experiment with the help of a neat labelled diagram.
   Write the conclusion that can be drawn from the experiment. **[Average]**
2. Explain evolution by natural selection by taking an example of white-winged moths and dark-winged moths of England in pre and post-industrialization period. **[Average]**
3. How does comparative anatomy and morphology act as an evidence for evolution?
   Explain with the help of suitable examples. **[Average]**
4. a) State Hardy-Weinberg principle.
   b) Explain any four factors that affect Hardy-Weinberg equilibrium. **[Easy]**
5. Mention the various stages of human evolution along with their with brain capacities. **[Easy]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Classify the following examples as homologous or analogous organs:
   a) Eyes of Octopus & mammals
   b) Wings of Birds & Butterfly
   c) Forelimb of Cheetah & Bat
   d) Flippers of Penguins & Dolphins
   e) Thorns of *Bougainvillea* and tendrils of *Cucurbita* **[Average]**

****

CHAPTER 7
HUMAN HEALTH AND DISEASE

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. Which of the following pairs contains an infectious and a non-infectious disease respectively?
   a) Typhoid and AIDS b) AIDS and cancer
   c) Pneumonia and malaria d) Cancer and malaria **[Easy]**
2. Which of the following is not a bacterial disease in humans?
   a) Pneumonia b) Malaria c) Plague d) Diptheria **[Easy]**
3. One of the following is not the causal organism for ringworm.
   a) *Microsporum* b) *Trichophyton* c) *Epidermophyton* d) *Macrosporum* **[Easy]**
4. An intestinal parasite which causes blockage of the intestinal passage and whose eggs are excreted along with the faeces of infected person is _____.
   a) *Wuchereriabancrofti* b) *Ascaris* c) *Epidermophyton* d) *Microsporum* **[Easy]**
5. Elephantiasis, a chronic inflammation that results in gross deformities is caused by
   a) *Ascaris* b) *E.coli* c) *Wuchereria* d) *Trichophyton* **[Easy]**
6. Which of the following is affected by the infection of *Wuchereria bancrofti*?
   a) Lymphatic vessels b) Respiratory system
   c) Nervous system d) Blood circulation **[Easy]**
7. Which of the following diseases is transmitted by the bite of the female mosquito vector?
   a) Filariasis b) Amoebiasis c) Typhoid d) Pneumonia **[Easy]**
8. Which of the following pairs correctly matches a disease and a pathogen causing it?
   a) Typhoid – *Salmonella typhi* b) Pneumonia – *Haemophilus pneumoniae*
   c) Malaria – *Ascaris lumbricoides* d) Ringworm – *Entamoeba histolytica* **[Average]**
9. Appearance of dry, scaly lesions with itching on various parts of the body are the symptoms of
   a) Elephantiasis b) Ringworm c) Ascariasis d) Amoebiasis **[Average]**
10. Haemozoin is a
    a) Precursor of haemoglobin
    b) Toxin released from *Streptococcus* infected cells
    c) Toxin released from *Plasmodium* infected cells
    d) Toxin released from *Haemophilus* infected cells. **[Average]**

11. A toxic substance, responsible for the chills and high fever recurring every three to four days in malarial fever, is

a) Interferon b) Haemozoin c) Hirudin d) Colostrum **[Average]**

12. During the life cycle of *Plasmodium*, gametocytes develop in

a) Human RBC b) Human liver

c) Salivary glands of mosquito d) Gut of mosquito **[Easy]**

13. Which of the following is a viral disease?

a) Pneumonia b) Typhoid c) Common cold d) Malaria **[Easy]**

14. The human immunodeficiency virus (HIV) primarily attacks and destroys:

a) Red blood cells b) T lymphocytes c) Platelets d) Neurons **[Easy]**

15. Which of the following is an autoimmune disease?

a) Rheumatoid arthritis b) Tuberculosis c) Malaria d) AIDS **[Easy]**

16. The chemical secreted by virus infected cells that can protect other cells from further infection is

a) Serotonin b) Colostrum c) Interferon d) Histamine **[Easy]**

17. Antibodies present in colostrum which protect the new born from certain diseases is

a) Ig G b) Ig A c) Ig D d) Ig E **[Easy]**

18. Which of the following is not a lymphoid tissue?

a) Spleen b) Tonsils c) Pancreas d) Thymus **[Easy]**

19. The site where lymphocytes interact with antigens and proliferate to become effector cells are

a) Thymus and lymph nodes b) Bone marrow and thymus

c) Peyer's patches and tonsils d) Bone marrow and spleen **[Easy]**

20. Antivenom against snake poison contains

a) Antigens b) Antigen-antibody complexes c) Preformed antibodies d) Enzymes **[Easy]**

21. Vaccines work by

a) Stimulating the production of antibodies b) Blocking the entry of pathogens into the body

c) Reducing inflammation d) Killing the pathogen directly **[Easy]**

22. Which form of pathogen is used in vaccination?

a) Activated and strong pathogenic antigens b) Preformed antigens

c) Inactivated and weakened pathogen d) Hyperactive and strong pathogen **[Easy]**

23. Injection of antitoxin in tetanus confers which type of immunisation?

a) Active immunisation b) Passive immunisation

c) Auto-immunisation d) Humoral immunisation **[Average]**

24. Hepatitis B vaccine is produced from

a) Inactivated viruses b) Yeast

c) *Haemophilus influenzae* d) *Salmonella typhimurium.* **[Easy]**

25. The most abundant antibody produced against allergens is

a) IgE b) IgA c) IgG d) IgM. **[Easy]**

26. Which of the following cells actively participate during allergy?

a) B-lymphocytes b) Liver cells c) Mast cells d) Red blood cells **[Easy]**

27. A chemical carcinogen present in tobacco smoke is responsible for

a) Skin cancer b) Pancreatic cancer c) Stomach cancer d) Lung cancer. **[Easy]**

28. Which of the following approaches is not used for the treatment of cancer?

a) Immunotherapy b) Surgery c) Radiotherapy d) Anti Retroviral therapy **[Easy]**

29. Smack' is a drug obtained from the

a) Latex of *Papaver somniferum* b) Leaves of *Cannabis sativa*

c) Flowers of Datura d) Fruits of *Erythroxylum cocao* **[Easy]**

30. Tobacco consumption is known to stimulate secretion of adrenaline and nor-adrenaline. The component causing this could be

a) Nicotine b) Tannic acid c) Curcumin d) Catechin **[Average]**

31. Heroin formed by acetylation of morphine is commonly called as

a) Coke b) Crack c) Smack d) Charas **[Easy]**

32. Marijuana is extracted from.

a) Resin, leaves and flowers of Cannabis plant b) Ergot fungus

c) Roots of hemp plant d) Cocoa plant. **[Easy]**

33. Charas and ganja are the drugs which affect

a) Respiratory system b) Cardiovascular system

c) Digestive system d) Nervous system **[Easy]**

34. Cocaine is obtained from

a) *Erythroxylun coca* b) *Papaver somniferum*

c) *Atropa belladonna* d) *Dature stramonium* **[Easy]**

35. Cocaine is commonly called as

a) Smack b) Coke c) Heroin d) Cola **[Easy]**

36. Cocaine interferes with the transport of the neuro-transmitter _____

a) Acetylcoline b) Glutamate c) Dopamine d) Glycine **[Easy]**

37. Which of the following plants does not possess hallucinogenic properties?

a) *Erythroxylum coca* b) *Atropa belladonna* c) *Datura* d) *Papaver somniferum* **[Easy]**

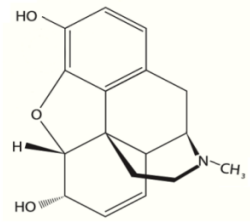
38. The drug that is being abused by some sports persons is

a) Opioids b) Barbiturates c) Cannabinoids d) LSD **[Easy]**

39. Which drug is used as medicine to help patients cope with depression and insomnia?

a) Morphine b) Amphetamines c) Cannabinoids d) Cocaine **[Easy]**

40. A chemical structure is given below. Identify its plant source.



a) *Atropa belladonna* b) *Cannabis sativa*

c) *Papaver somniferum* d) *Erythroxylum coca* **[Difficult]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket.

1 Mark Each

(*Aedes*, Typhoid, Masculination, Smack, Yeast, Metastasis, Lymph nodes, Crack, Memory, Host, Allergens, MALT, Morphine, Houseflies, Pathogens)

1. Disease causing organisms are called _____ **[Easy]**
2. Pathogens adapt to life within the environment of the ____. **[Easy]**
3. Mary Mallon is associated with _____ **[Easy]**
4. _____ act as mechanical carriers and serve to transmit *Entamoeba histolytica* from faeces of infected person to food and food products. **[Easy]**
5. Dengue and chikungunya are spread through _____ **[Easy]**
6. The principle of immunisation is based on the property of _____ of the immune system. **[Easy]**
7. Hepatitis B vaccine is produced from _____ **[Easy]**
8. Mites in dust, pollens, animal dander are common examples of _____ **[Easy]**
9. The _____ are small solid structures located at different points along the lymphatic system. **[Easy]**
10. _____ constitutes about 50 percent of the lymphoid tissue in human body. **[Easy]**
11. _____ is the most feared property of malignant tumors. **[Easy]**
12. _____ is a very effective sedative and painkiller. **[Easy]**
13. _____ when snorted produces a sense of euphoria and increased energy. **[Easy]**
14. The side effects of the use of anabolic steroids in females include _____ **[Easy]**
15. ____ is a white, odourless, bitter crystalline compound which acts as a depressant. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. What are infectious diseases? Give an example. **[Easy]**
2. What are non-infectious diseases? Give an example. **[Easy]**
3. Differentiate infectious diseases from non-infectious diseases. **[Easy]**
4. Mention any four means by which good health can be achieved. **[Easy]**
5. Write the scientific name of the pathogen and any two symptoms of typhoid. **[Easy]**
6. Write the scientific name of the pathogen and any two symptoms of pneumonia. **[Easy]**
7. Write the scientific name of the pathogen and any two symptoms of common cold. **[Easy]**
8. Write the scientific name of the pathogen and any two symptoms of amoebiasis. **[Easy]**
9. Write the scientific name of the pathogen and any two symptoms of ascariasis. **[Easy]**
10. Write the scientific name of the pathogen and any two symptoms of elephantiasis. **[Easy]**
11. Write the scientific name of the pathogen and any two symptoms of ringworm disease. **[Easy]**
12. Name the host and the site where the following occur in the life cycle of a malarial parasite
    (a) Formation of gametocytes. (b) Fusion of gametes. **[Easy]**
13. What is immunity? Mention its types. **[Easy]**
14. Mention the types of barriers of innate immunity. **[Easy]**
15. Differentiate between innate and acquired immunity. **[Easy]**
16. Mention the roles of B and T lymphocytes. **[Easy]**
17. Draw a labelled diagram of structure of an antibody molecule. **[Easy]**
18. Differentiate between active immunity and passive immunity. **[Easy]**
19. Differentiate between primary and secondary immune responses. **[Easy]**
20. What are primary lymphoid organs? Give two examples. **[Easy]**
21. What are secondary lymphoid organs? Give two examples. **[Easy]**
22. Differentiate between primary and secondary lymphoid organs. **[Average]**
23. State the functions of primary and secondary lymphoid organs in humans. **[Easy]**
24. Name the two special types of lymphocytes present in our blood which are responsible for primary and secondary responses. **[Easy]**
25. What is colostrum? Mention its significance. **[Easy]**
26. What is passive immunisation? Give an example. **[Easy]**
27. What is allergy? Mention its symptoms. **[Easy]**
28. What are allergens? Give an example. **[Easy]**
29. Name two chemicals released by the mast cells during an allergic reaction. **[Easy]**

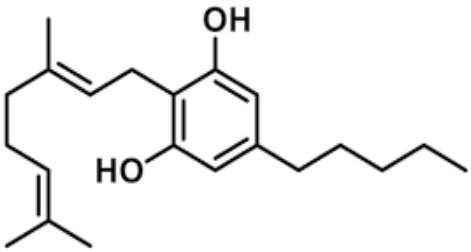
30. What is auto-immune disease? Give an example. **[Easy]**
31. Distinguish between benign and malignant tumors. **[Easy]**
32. Metastasis is the most feared property of malignant tumors. Justify. **[Average]**
33. Mention any two characteristics of cancer cell. **[Easy]**
34. What are carcinogens? Give an example. **[Easy]**
35. Write a note on oncogenic viruses. **[Easy]**
36. Cancer is one of the most dreaded diseases. Explain 'Contact inhibition' and 'Metastasis' with respect to the disease. **[Average]**
37. What is biopsy? Name the disease identified by this. **[Easy]**
38. List the modes of transmission of HIV in humans. **[Easy]**
39. What is drug abuse? Name any three drugs that are commonly abused. **[Easy]**
40. List any four common warning signs of drug and alcohol abuse in youth. **[Easy]**
41. Name an opioid drug and its source plant (scientific name). How does it affect human body? **[Easy]**
42. Name the plant (scientific name) from which cocaine is obtained. What is the effect of cocaine in humans? **[Easy]**
43. Name the plant (scientific name) from which cannabinoid is obtained. What is the effect of cannabinoids in humans? **[Easy]**
44. Name of any two drugs which are commonly abused and write the binomial name of the plants from which these drugs are extracted. **[Easy]**
45. Write the sources and effects of the following drugs on the human body.
    a) Morphine b) Marijuana c) Cocaine **[Easy]**
46. Name a drug used:
    a) As an effective sedative and pain killer
    b) For helping patients cope with mental illness like depression but often misuse. **[Average]**
47. List any four effects of smoking. **[Easy]**
48. Differentiate between addiction and dependence with reference to alcohol and drugs. **[Easy]**
49. List any four withdrawal symptoms exhibited by an alcohol or drug addict. **[Easy]**
50. List the most common warning signs of drug and alcohol abuse among youth. **[Easy]**
51. List any four adverse effects of drug and alcohol abuse. **[Easy]**
52. Mention four side effects of the use of anabolic steroids in males. **[Easy]**
53. Mention four side effects of the use of anabolic steroids in females. **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Define the term 'health'. Mention any four means by which good health can be achieved. **[Easy]**
2. Write the scientific name of the pathogen, mode of transmission and any two symptoms of typhoid. **[Easy]**
3. Write the scientific name of the pathogen, mode of transmission and any two symptoms of pneumonia. **[Easy]**
4. Write the scientific name of the pathogen, mode of transmission and any two symptoms of common cold. **[Easy]**
5. Write the scientific name of the pathogen, mode of transmission and any two symptoms of amoebiasis. **[Easy]**
6. Write the scientific name of the pathogen, mode of transmission and any two symptoms of ascariasis.. **[Easy]**
7. Write the scientific name of the pathogen, mode of transmission and any two symptoms of elephantiasis/filariasis **[Easy]**
8. Write the scientific name of the pathogen, mode of transmission and any two symptoms of ringworm disease. **[Easy]**
9. a) Name the infective stages of *Plasmodium* which *Anopheles* mosquito takes in along the blood meal from an infected person. 1
   b) Why does the infection cause chill and fever in humans? 1
   c) At what stage does *Plasmodium* enter into human body? 1 **[Average]**
10. Differentiate between active immunity and passive immunity. Give an example for each. **[Easy]**
11. Differentiate between primary and secondary lymphoid organs. Give an example for each. **[Easy]**
12. What are allergens? Give examples. Name the type of antibody produced to the allergens. **[Easy]**
13. Mention any three characteristics of cancer cell. **[Easy]**
14. What are carcinogens? Mention any two groups of carcinogens with an example for each. **[Easy]**
15. When the ELISA test was conducted on an immune-suppressed person, he tested positive for a pathogen.
    a) Identify the disease the patient is suffering from. b) Name the causative entity
    c) Mention the cells of the body that are attacked by the pathogen. **[Average]**
16. Mention the names of three drugs which are commonly abused and write the binomial name of the plants from which these drugs are extracted. **[Easy]**

17. Tobacco smoking associated with rise in blood pressure and emphysema (oxygen deficiency in body). Substantiate. **[Difficult]**
18. Briefly explain any three prevention and control measures of drug and alcohol abuse. **[Average]**
19. The skeletal structure of a drug is shown below. Answer the questions given below.



a) Identify the drug

b) Name the plant source of this drug

c) How is this drug generally taken? **[Difficult]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. What is innate immunity? Explain the four types of barriers of innate immunity with suitable examples. **[Average]**
2. Define immunity. Explain active immunity and passive immunity with examples. **[Easy]**
3. What are lymphoid organs? List different lymphoid organs and tissues and explain their role in immunity. **[Easy]**
4. Explain the life cycle of *Plasmodium.* **[Average]**
5. Schematically represent the life cycle of HIV (retroviruses) in human body. **[Average]**
6. Mention any five methods and techniques of cancer detection and diagnosis. **[Easy]**
7. Mention any five common approaches of treatment of cancer. **[Easy]**
8. What is addiction? List four warning signals and four harmful effects caused by drug abuse. **[Easy]**
9. Explain prevention and control measures of alcohol and drug abuse. **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. a) Name the group of genes that have been identified in normal cells that could lead to cancer. 1

   b) Name the techniques that are useful in detecting cancers of internal organs. 3

   c) Why are cancer patients often given α-interferon as part of the treatment? 1 **[Average]**
2. A person has been diagnosed as HIV positive.

   a) Name the diagnostic test which the person underwent. 1

   b) Write full name of pathogen involved. 1

c) Name the genome of the pathogen 1

d) Which particular cells of this person are likely to get destroyed? 1

e) Suggest the partially effective treatment to him. 1 **[Difficult]**

3. Fill the missing data in the table depicting diseases, their causatives and symptoms**:**

Name of the disease		Causative organism	Symptoms
a)	Ascariasis	*Ascaris*	_____
b)	_____	*Trichophyton*	Appearance of scaly lesions on various parts of the body
c)	Typhoid	_____	High fever , weakness, headache, stomach pain , constipation
d)	Pneumonia	*Streptococcus pneumoniae*	_____
e)	_____	Rhino virus	Nasal congestion and discharge, sore throat, cough headache

[Difficult]

4. Mention the scientific names of the pathogens that cause the following diseases:

a) Typhoid b) Common cold c) Pneumonia d) Amoebiasis e) Filariasis

[Easy]

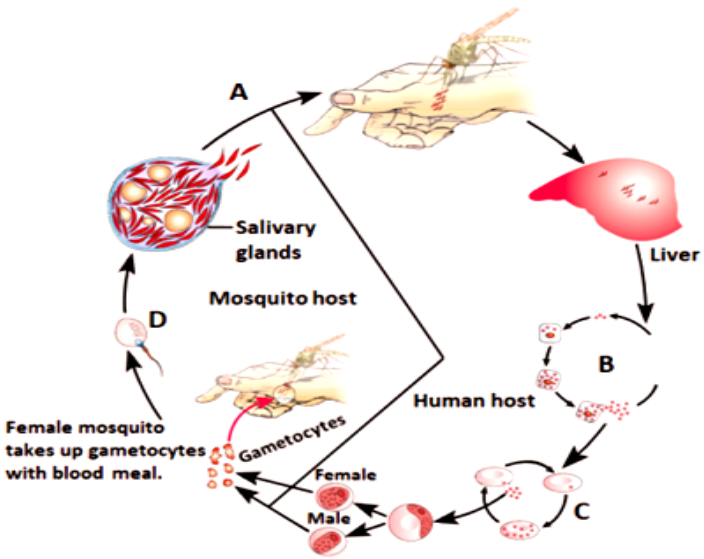
5. Name the diseases caused by the following pathogens:

a) *Wuchereria* b) *Haemophillous influenzae* c) *Entamoeba histolytica*

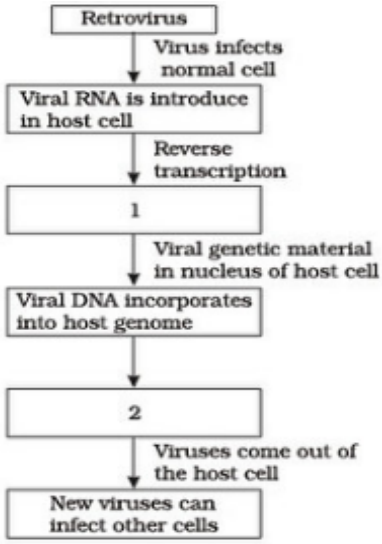
d) *Trichophyton* e) *Samlonella typhi* **[Easy]**

6. a) Stages in the life cycle of *Plasmodium* is given below. Identify A, B, C, D. 4

b) Name the toxin responsible for the appearance of symptoms of malaria in humans. 1 **[Difficult]**



7. The flowchart shows replication of the retrovirus in the host. Answer the questions given below



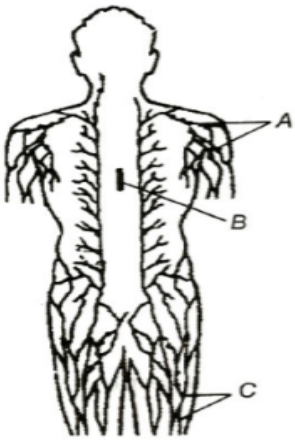
a) Fill in the missing data in boxes labelled 1 & 2. 2

b) Give an example for retrovirus? 1

c) Name the disease caused by this retrovirus. 1

d) Name the diagnostic test for this disease. 1 **[Difficult]**

8. Observe the diagram and answer the questions given below:



a) Identify A and write its function. 2

b) Identify B and write its function. 2

c) Identify C. 1 **[Difficult]**

****

CHAPTER 8
MICROBES IN HUMAN WELFARE

PART-A

I. Select the correct alternative from the choices below: **1 Mark Each**

1. '*Propionibacterium shermanii*' is used in the production of _____ .
   a) Roquefort cheese b) Swiss cheese c) Bamboo shoots d) Ferment fish **[Easy]**
2. *Penicillium notatum* is a _____ .
   a) Bacteria b) Virus c) Blue green algae d) Mould **[Easy]**
3. Alexander Fleming discovered penicillin while working on
   a) *Staphylococci* b) *Streptococcus* c) *E. coli* d) *Salmonella* **[Easy]**
4. Which one of the following alcoholic drinks is produced without distillation?
   a) Wine b) Whisky c) Rum d) Brandy **[Average]**
5. Because of the presence of which vitamin, curd is considered nutritionally richer than milk?
   a) Vitamin B_1 b) Vitamin B_2 c) Vitamin B_9 d) Vitamin B_{12} **[Average]**
6. Bioactive molecules Cyclosporin- A is produced by
   a) Fungus b) Bacteria c) Virus d) Protozoa **[Easy]**
7. Which micro-organism is useful in production of citric acid?
   a) *Acetobacter aceti* b) *Penicillium notatum*
   c) *Aspergillus niger* d) *Clostridium butylicum* **[Easy]**
8. Activated sludge is the sediment deposited in
   a) Settling tank b) Sedimentation tank c) Aerobic tank d) Anaerobic sludge digester **[Average]**
9. Yeast is used in the production of
   a) Bread and beer b) Cheese and butter
   c) Citric acid and lactic acid d) Lipase and pectinase **[Average]**
10. Biogas is the mixture of
    a) CH_4, CO_2, H_2 b) CH_4, SO_2, CO_2 c) CH_4, NO_2, H_2 d) CH_4, SO_2, H_2 **[Average]**
11. Mesh like structure consisting of bacteria associated with fungal filaments is
    a) Primary sludge b) Activated sludge c) Floc d) Anaerobic sludge **[Easy]**
12. The primary sewage treatment involves
    a) Anaerobic bacterial activity b) Sludge digestion
    c) Filtration and sedimentation d) Aerobic bacterial activity **[Easy]**

13. Match column I with column II and select the correct option using the codes given below.

Column -I	Column –II
a. Citric acid	i) *Trichoderma polysporum*
b. Cyclosporin A	ii) *Clostridium butylicum*
c. Statins	iii) *Aspergillus niger*
d. Butyric acid	iv) *Monascus purpureus*

[Difficult]

a) a-(iii), b-(i), c-(ii), d-(iv) b) a-(iii), b-(i), c-(iv), d-(ii)

c) a-(i), b-(iv), c-(ii), d-(iii) d) a-(iii), b-(iv), c-(i), d-(ii)

14. Which of the following in sewage treatment removes suspend solids?

a) Secondary treatment b) Primary treatment

c) Sludge treatment d) Tertiary treatment **[Average]**

15. During sewage treatment, the gas produced in anaerobic sludge digester includes

a) Hydrogen sulphide, nitrogen, methane

b) Methane, oxygen, hydrogen sulphide

c) Methane, hydrogen sulphide, carbon dioxide

d) Hydrogen sulphide, methane, sulphur oxide **[Difficult]**

16. Mycorrhiza is involved in absorption of _____ from soil and passes it to the plant.

a) Nitrogen b) Phosphorus c) Potassium d) Magnesium **[Easy]**

17. Which of the following is not an advantage provided by mycorrhiza to plants?

a) Resistance to root-borne pathogens

b) Increase in yield quality

c) Tolerance to salinity and drought

d) Overall increase in plant growth and development **[Average]**

18. The microbial biocontrol agent for butterfly caterpillar is

a) *Bacillus thuringiensis* b) *Saccharomyces* c) *Cyanobacteria* d) *Staphylococcus* **[Average]**

19. Which of the following is a bacterial biofertilizer?

a) *Anabaena* b) *Nostoc* c) *Rhizobium* d) *Oscillatoria* **[Easy]**

20. Which of the following is not a biofertilizer?

a) *Agrobacterium* b) *Rhizobium* c) *Nostoc* d) *Mycorrhiza* **[Easy]**

21. Which of the following statements regarding flocs is incorrect?

a) They are the masses of bacteria associated with fungal filaments.

b) Produced in aeration tank.

c) They consume major part of organic matter.

d) Formed by useful anaerobic microbes. **[Difficult]**

22. Compare the statements I and II:

Statement I: Mycorrhiza is the symbiotic association of algae and fungi.

Statement II: They absorb nitrogen from soil and pass it to the plant.

a) Statement I is true, statement II is false.

b) Statement I is false, statement II is true.

c) Both the statements I and II are true.

d) Both the statements I and II are false. **[Average]**

23. Select the correct statement from the following.

a) Biogas is produced by the activity of aerobic bacteria on animal waste.

b) *Methanobacterium* is an aerobic bacterium found in rumen of cattle.

c) Biogas, commonly called gobar gas, is pure methane.

d) Activated sludge in settlement tanks of STPs is a rich source of aerobic bacteria. **[Average]**

II. Fill in the blanks by choosing the appropriate word/ words from those given below: 1 Mark Each

(*Saccharomyces cerevisiae*, Immunosuppressive agent, *Clostridium butylicum*, CO_2, Nucleopolyhedrovirus, methanogens, fermentors, Toddy, municipal waste water, clot buster, antibiotics, *Rhizobium*, mycorrhiza, activated sludge, effluent, *Glomus*)

1. Large vessels used in industry for growing microbes are called _____ . **[Easy]**
2. Sediments of flocs in the settling tank are called _____ . **[Easy]**
3. _____ are chemical substances, produced by some microbes and can kill or retard the growth of other microbes. **[Easy]**
4. Baculoviruses used as biocontrol agents are belongs to the genus _____ . **[Easy]**
5. The symbiotic association of fungi in roots of plants are known as _____ . **[Easy]**
6. Many members of the genus _____ form mycorrhiza. **[Easy]**
7. Root nodules in leguminous plants are formed by symbiotic association of ____ **[Average]**
8. _____ are the bacteria found in rumen of cattle. **[Easy]**
9. Cyclosporin A is used as _____ are produced by *Trichoderma polysporum*. **[Easy]**
10. _____ is a traditional drink of some parts of southern India is made by fermenting sap from palms. **[Easy]**
11. The puffed appearance of dough is due to the production of _____ . **[Easy]**
12. Baker's yeast _____ is used for making bread. **[Easy]**
13. Streptokinase is used as _____ for patients with myocardial infarction. **[Easy]**
14. _____ is also called sewage. **[Easy]**

15. _____ is a producer of butyric acid. **[Easy]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Presence of microbes in water is an indication of pollution in water.
   At the same time, microbes also reduce pollution in water. Justify the statement. **[Difficult]**
2. Discuss the importance of aeration tank and anaerobic sludge digester in sewage treatment. **[Average]**
3. Discuss the role of fungi as biofertilizers. **[Easy]**
4. Draw a neat labelled diagram of Biogas plant. **[Easy]**
5. Define BOD. Mention its significance. **[Easy]**
6. Name the microbes responsible for the production of
   a) Large holes in 'Swiss cheese' b) Acetic acid **[Easy]**
7. Mention any two significant roles of LAB. **[Easy]**
8. Mention the source and uses of Cyclosporin A**.** **[Easy]**
9. Mention the source and uses of Statins. **[Easy]**
10. List any two bioactive molecules of fungal origin and explain how those molecules help in restoring good health in humans. **[Average]**
11. Name any two bacteria used in the production of organic acids. **[Easy]**
12. Mention the scientific names of the organisms used in the production of cyclosporine A and statins. **[Easy]**
13. Give any two examples for cyanobacteria which can fix atmospheric nitrogen. **[Average]**
14. How do cyanobacteria acts as biofertilizers? **[Average]**
15. What are flocs? Write their significance in sewage treatment. **[Average]**
16. What is BOD? Write its significance. **[Easy]**
17. Define BOD. BOD is an index of water pollution. Justify. **[Easy]**
18. Name any two gases produced during sludge digestion in sewage treatment. **[Easy]**
19. Name the processes that help in physical removal of particles during sewage treatment. **[Easy]**
20. Mention any two benefits offered by mycorrhiza other than as biofertilizer. **[Average]**
21. Biocontrol of insects and pests is desirable and beneficial compared to the use of insecticides and pesticides. Justify this with two reasons. **[Difficult]**
22. Mention the source and use of penicillin. **[Easy]**
23. Name any two distilled and non-distilled alcoholic beverages. **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Mention the product and uses of the following:
   a) Acid produced by *Aspergillus niger*
   b) Enzyme produced by *Streptococcus*
   c) Bioactive molecule produced by *Trichoderma polysporum* **[Average]**
2. Draw a neat labelled diagram of typical Biogas plant. **[Easy]**
3. Name the microbes that help in the production of the following products commercially.
   a) Cyclosporin A b) Citric acid c) Statins **[Easy]**
4. List any three benefits of mycorrhiza as biofertilizer. **[Easy]**
5. Mention one use each for the following enzymes:
   a) Lipase b) Pectinase c) Streptokinase **[Easy]**
6. Briefly explain the contribution of Alexander Fleming. **[Average]**
7. Mention any three cyanobacteria that are used as biofertilizers. **[Easy]**
8. Mention the name of a bioactive molecule, its microbial sources and uses. **[Average]**
9. Substantiate the following:
   a) Methanogens are found in the rumen of cattle.
   b) *Nucleopolyhedroviruses* (NPV) are species-specific.
   c) BOD is an index of water pollution. **[Difficult]**

9. a) The use of biofertilizers is encouraged in agriculture to reduce the use of chemical fertilizers. Why? 1
   b) Write the role of mycorrhiza as biofertilizer. 2 **[Difficult]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Discuss the role of microbes in house-hold products. **[Difficult]**
2. Explain the role of microbes in 'Industrial Products'. **[Easy]**
3. Explain the process of sewage treatment. **[Easy]**
4. Describe the role of microbes as a) Biocontrol agents b) Biofertilizers **[Easy]**
5. a) Describe the role of microbes in sewage treatment. 3
   b) Draw a neat labelled diagram of typical Biogas plant. 2 **[Easy]**
6. a) Explain the role of three organisms as biocontrol agents. 3
   b) What is the significance of BOD? 1
   c) Give an examples for a fungus found in Mycorrhiza. 1 **[Easy]**

7. a) Name the bioactive molecule for the following: 3

   i) A clot buster agent

   ii) An immunosuppressive agent

   ii) A Blood cholesterol lowering agent

   b) What are biofertilizers? Give one example. 2 **[Average]**

8. a) Draw a neat labelled diagram of biogas plant. 3

   b) What are flocs? Write their significance in sewage treatment. 2 **[Easy]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Complete the below given tabular column with appropriate answers. **[Difficult]**

Microbe	Product	Uses
A	Swiss cheese	B
Penicillum notatum	C	Antibiotic
Monascus purpureus	D	E

2. The biological method used for controlling plant diseases and pests is also a process of natural predation. Justify the statement with appropriate examples. **[Difficult]**

3. The chemical composition of agricultural land can be improved by two methods: The use of chemical fertilizers and the use of microbes. Although both are fertilization methods, The former is considered to increase pollution, while the latter is considered to increase fertility. Discuss. **[Difficult]**

4. With the help of clue given in the brackets, complete the passage with appropriate answers.
   While working on ___A___ (bacteria), ___B___ (scientist) observed the growth of a mould ___C___ (fungus) which does not allowed the bacteria to grow. He found out that it was due to a chemical ___D___ (antibiotic) produced by the mould. This antibiotic was used to treat soldiers of ___E___ (country) wounded in World War II. ___F___ , ___G___ and ___H___ (Scientists) were awarded ___I___ prize in ___J___ (year) for this discovery. **[Difficult]**

5. A person planned to start a biotechnology based industry in the interest of producing some products for human use. Suggest him some names of microorganisms which are commercially useful and guide him regarding the products and their uses produced by those organisms. **[Difficult]**

6. Assign the role of the following microbes:
   a) *Lactobacillus* in milk. b) Methanogens in rumen of cattle.
   c) *Rhizobium* in the nodules of legumes. d) *Glomus* in mycorrhiza.
   e) *Trichoderma* in root ecosystems. **[Difficult]**

****

CHAPTER 9
BIOTECHNOLOGY: PRINCIPLES AND PROCESSES

PART – A

I. Select the correct alternative from the choices given below: **1 Mark Each**

1. Which of the following is not a feature of plasmids?
   a) Extra chromosomal b) Circular structure
   c) Single stranded d) Independent replication **[Difficult]**
2. The procedure through which a piece of DNA is introduced into a host bacterium is
   a) Amplification b) Cloning c) Transformation d) Restriction **[Easy]**
3. Autonomously replicating circular extra-chromosomal DNA is called
   a) Vector b) Restriction enzyme c) Marker d) Plasmid **[Easy]**
4. Plant cells are bombarded with high velocity micro particles of gold or Tungsten coated with rDNA in a method known as
   a) Microinjection d) Biolistics c) Heat shock d) Vector mediated **[Easy]**
5. The commonly used vector for cloning genes in animals is
   a) Rhino viruses b) Disarmed retrovirus
   c) Disarmed Ti plasmid d) *Agrobacterium tumifaciens* **[Easy]**
6. Enzymes which remove nucleotides from the ends of the DNA are
   a) Endonucleases b) Exonucleases c) DNA Ligase d) DNA polymerase **[Easy]**
7. Restriction site of EcoR1 enzyme is
   a) 5'-GAATTC-3' b) 3'-GAATTC-5' c) GAATTC d) 5'-CTTAAG-3' **[Easy]**
8. The technique used to separate DNA fragments in recombinant DNA technology is
   a) Gel electrophoresis b) PCR c) Autoradiography d) Downstream processing **[Easy]**
9. Which of the following is the stain used to visualize DNA fragments in gel electrophoresis?
   a) Cotton blue b) Ethidium bromide c) Acetocarmine d) Saffranine **[Easy]**
10. The extraction of separated bands of DNA by cutting out them from the agarose gel is
    a) Denaturation b) Fragmentation c) Elution d) Separation **[Easy]**
11. The thermostable DNA polymerase enzyme is obtained from
    a) *Agrobacterium tumifaciens* b) *Thermus aquaticus*
    c) *Escherichia coli* d) *Meloidogyne incognitia* **[Easy]**
12. In the naming of restriction enzymes, the first letter of the name is derived from
    a) Species b) Genus c) Strain d) Family **[Easy]**

13. Restriction enzymes belongs to the class

a) Oxidoreductases b) Ligases c) Isomerases d) Nucleases **[Easy]**

14. *Agrobacterium tumifaciens* transforms normal cells into a tumour by delivering a piece of DNA known as

a) Z-DNA b) C-DNA c) T-DNA d) R-DNA **[Average]**

15. Recombinant DNA is directly injected into the nucleus of an animal cell by

a) Microinjection b) Gene gun c) Insertion d) Biolistics **[Easy]**

16. *Agrobacterium tumifaciens* is used as a vector for cloning genes in

a) Plants b) Animals c) Bacteria d) Viruses **[Easy]**

17. Any protein encoding gene is expressed in a heterologous host is called as

a) Recombinant DNA b) Recombinant RNA

c) Recombinant Protein d) Recombinant polysaccharide **[Easy]**

18. Hind II cuts DNA molecule at a specific point by recognizing a specific sequence of

a) 4 base pairs b) 6 base pairs c) 5 base pairs d) 8 base pairs **[Easy]**

19. Which of the following is the correct order of processing of PCR?

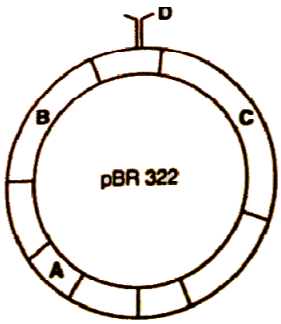
a) Annealing → denaturation → extension

b) Denaturation → extension → annealing

c) Denaturation → annealing → extension

d) Extension → annealing → denaturation **[Average]**

20. In the given diagram of pBR^{322} vector, identify the restriction sites A, B, C and D.



a) A- tet^R, B- *Hind*III, C- Ori, D- amp^R b) A- Ori, B- amp^R, C- tet^R, D- *Hind*III

c) A- *Hind*III, B- tet^R, C- amp^R, D- Ori d) A- amp^R, B- tet^R, C- *Hind*III, D- Ori **[Difficult]**

21. **Statement I:** Sticky ends are resulted due to the action of restriction enzymes.

Statement II: Stickiness facilitates the action of DNA polymerase.

a) Statement I is true, statement II is false. b) Statement I is false, statement II is true.

c) Both the statements I and II are true. d) Both the statements I and II are false. **[Average]**

22. The purpose of stirred tank bioreactor is

a) Providing anaerobic conditions in the culture vessel. b) Purification of the product.

c) Addition and proper mixing of the preservative.

d) Availability of oxygen throughout the process. **[Difficult]**

23. Identify the correct statement.

a) *Hind* II cut the DNA at a specific sequence of 6 base pairs.

b) Microinjection is the technique of introduction of rDNA into plant cells.

c) Biolistics is the technique of introduction of rDNA into animal cells.

d) Retroviruses are commonly used cloning vectors in plants. **[Difficult]**

24. **Statement I:** Plasmid DNA acts as vectors to transfer the piece of DNA attached to it.

Statement II: Vector DNA acts as plasmids to transfer the piece of DNA attached to it.

a) Statement I is true, statement II is false. b) Statement I is false, statement II is true.

c) Both the statements I and II are true. d) Both the statements I and II are false. **[Difficult]**

25. A single strand of nucleic acid tagged with a radioactive molecule is called

a) Vector b) Marker c) Plasmid d) Probe **[Easy]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(*Agrobacterium*, Cancerous cells, Sticky ends, DNA, DNA Ligase, Heat shock, Cellulose, Vector, Restriction endonuclease, Primers, Elution, EcoRI, Tansformation, Polymerase Chain Reaction, Palindrome)

1. The tumor inducing plasmid of _____ has been modified into a cloning vector. **[Easy]**
2. When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of _____ **[Easy]**
3. The plasmid DNA act as _____ to transfer the piece of DNA attached to it. **[Easy]**
4. _____ functions by inspecting the length of a DNA sequence. **[Average]**
5. Small chemically synthesized oligonucleotides that are complementary to the regions of DNA are _____ . **[Easy]**
6. The extraction of DNA bands from agarose gel is known as _____ . **[Easy]**
7. The technique that serves the purpose of early diagnosis is _____ **[Easy]**
8. _____ the enzyme used to break cells to release DNA along with other macromolecules in plants. **[Easy]**
9. Retroviruses in animals have the ability to transform normal cells into _____ . **[Easy]**
10. Stickiness of the cut ends of DNA facilitates the action of the enzyme _____ . **[Easy]**
11. GAATTC is the restriction site of _____ . **[Easy]**

12. _____ is the sequence of base pairs of DNA that reads same on the two strands when orientation of reading is kept same. **[Easy]**
13. _____is a procedure through which a piece of DNA is introduced in a host bacterium. **[Easy]**
14. _____ is a method of making a competent host by placing cells in alternate temperature. **[Easy]**
15. _____ is negatively charged, hence moves towards anode. **[Average]**

PART - B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. What is biotechnology? Mention any two uses of it to mankind. **[Easy]**
2. Mention two core techniques that enabled birth of modern biotechnology. **[Easy]**
3. Name two bacteria that were used for developing recombinant DNA by Stanley Cohen and Herbert Boyer. **[Average]**
4. Mention any four tools used in recombinant DNA technology. **[Easy]**
5. What are restriction enzymes? Mention their types. **[Easy]**
6. Differentiate between exonuclease and endonuclease. **[Average]**
7. Explain the convention for naming restriction endonucleases scientifically. **[Easy]**
8. Mention the roles of restriction endonucleases and DNA ligase in genetic engineering. **[Easy]**
9. Name the technique involved in separation and isolation of DNA fragment. Which dye is used to stain agarose gel to make the DNA visible under UV light? **[Easy]**
10. Draw a neat labeled diagram of a typical agarose gel electrophoresis. **[Easy]**
11. With reference to rDNA technology, what is ‘insertional inactivation’ and write the significance of it. **[Difficult]**
12. What are selectable markers? Mention their significance in genetic engineering? **[Easy]**
13. Draw a neat labeled diagram of stirred tank bioreactor. **[Easy]**
14. Draw a neat labeled diagram of sparged stirred tank bioreactor. **[Easy]**
15. How bacteria like *Escherichia coli* are made competent to take up recombinant DNA? **[Average]**
16. Define polymerase chain reaction. Name the enzyme used in it. **[Easy]**
17. Mention any four techniques available for the introduction of Alien DNA into host cells. **[Easy]**
18. Name any two vector mediated techniques for the introduction of Alien DNA into host cells. **[Easy]**
19. Name any two non vector mediated techniques for the introduction of rDNA into host cells. **[Easy]**
20. What are palindromic nucleotide sequences? Write the restriction site for *EcoRI* enzyme. **[Easy]**
21. Explain the method of preparing a competent host by heat shock method. **[Easy]**
22. Define the terms: a) Microinjection b) Biolistcs **[Easy]**

23. Name the bacterium from which *Taq polymerase* is obtained. What is the unique feature of *Taq polymerase*? **[Average]**

24. Name the vectors for cloning genes in plants and animal cells respectively. **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. What is biotechnology? Mention two core techniques that enabled the birth of modern biotechnology. **[Easy]**
2. Explain three basic steps of gene cloning. **[Easy]**
3. Mention the three basic steps in genetically modifying an organism. **[Easy]**
4. Discuss the mechanism of action of restriction enzymes with a schematic representation. **[Difficult]**
5. Schematically represent the steps in recombinant DNA technology. **[Average]**
6. What is plasmid? Describe any two features of an ideal plasmid. **[Easy]**
7. Mention the significance of the following in plasmid: **[Easy]**
   a) Ori site b) Selectable markers c) Cloning sites
8. Explain the procedure of making bacterial cells competent to take up recombinant DNA. **[Easy]**
9. Draw a labelled diagram of plasmid pBR^{322}. **[Average]**
10. Sketch and label three steps of PCR. **[Average]**
11. What is polymerase chain reaction? List three steps of PCR technique. Mention the polymerase enzyme used in PCR. **[Easy]**
12. Briefly explain the separation and isolation of DNA fragments. **[Easy]**
13. Write the importance of the following in r-DNA technology:
    a) PCR b) Ethidium bromide c) Bioreactor **[Easy]**
14. Explain the feature of cloning vectors. **[Easy]**
15. Define the following with reference to recombinant DNA technology:
    a) Cloning b) Elution c) Transformation **[Easy]**
16. Mention the importance of the following with respect to genetic engineering.
    a) *Agrobacterium tumifaciens* b) *Salmonella typhimurium* c) *Thermus aquaticus* **[Average]**
17. Briefly explain the process of isolation of DNA from a plant cell. **[Easy]**
18. Mention six optimum conditions which are maintained in bioreactor. **[Average]**
19. Mention sequential steps involved in rDNA technology. **[Easy]**
20. What is downstream processing? Write a note on downstream processing. **[Easy]**
21. a) What is recombinant protein?
    b) Name the thermostable enzyme used in polymerase chain reaction.
    c) Name the bacterium from which thermostable *Taq polymerase* is obtained. **[Easy]**

PART – D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. a) Write any four tools used in recombinant DNA technology. 2
   b) Mention any two methods of introducing alien DNA into host cells. 2
   c) Name the stains used to visualize DNA fragments in Gel electrophoresis. 1 **[Easy]**
2. Write the convention for naming the restriction enzymes with suitable example.
3. a) Show diagrammatic representation of recombinant DNA technology. 3
   b) Write note on down streaming processing. 2 **[Easy]**
4. Describe the characteristics that a plasmid should possess to be used as a cloning vector in genetic engineering. **[Average]**
5. Briefly explain the steps involved in recombinant DNA technology. **[Easy]**
6. Explain the process of amplification of gene using PCR. **[Easy]**
7. What is polymerase chain reaction? Name the bacterium from which the polymerase enzyme used in this technique is obtained. Write the schematic representation of this technique. **[Easy]**
8. Describe the process of separation of DNA fragments using gel electrophoresis. **[Easy]**
9. a) Define polymerase chain reaction. Mention its significance in genetic engineering. 2
   b) Write the diagrammatic representation of steps involved in PCR. 3 **[Easy]**
10. What are bioreactors? Mention the control systems of stirred tank bioreactor and write their functions. **[Average]**
11. What is insertional inactivation? Explain with respect to *β-galactosidase* enzyme? How are recombinants and non-recombinants identified? **[Difficult]**
12. Give reason for the following:
    a) Alien DNA is linked with ‘ori’site of a vector in gene cloning.
    b) Restriction enzymes are called ‘molecular scissors’.
    c) DNA ligase is used in genetic engineering.
    d) Gel electrophoresis is considered as a very important technique in recombinant DNA technology.
    e) DNA fragments move towards anode under electric field through a medium in gel electrophoresis. **[Difficult]**
13. Give reasons for the following:
    a) Origin of replication in vector is an essential site required for gene cloning.
    b) Selectable marker in a vector is essential site without which identifying recombinant DNA becomes difficult.
    c) Vectors in rDNA technique needs to have very few, preferably single recognition site.

d) Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure.

e) Chilled ethanol is used in DNA isolation. **[Difficult]**

14. Given below are the steps involved in recombinant DNA technology. Arrange them in a correct order.

i) Transferring the recombinant DNA into the host

ii) Culturing the host cells in a medium at large scale

iii) Extraction of the desired product.

iv) Ligation of the DNA fragment into a vector

v) Isolation of DNA

vi) Fragmentation of DNA by restriction endonucleases

vii) Isolation of desired DNA fragment **[Average]**

15. What are selectable markers? Give any two examples. Mention the significance of selectable markers in genetic engineering. **[Easy]**

16. a) Define plasmid. Give an example. 2

b) Insertion of rDNA into the host is an important step in genetic engineering.

c) An ampicillin resistant gene transferred to *E. coli* acts as selectable marker. Explain. 3 **[Average]**

17. A person working in a laboratory was thinking to use fermentor instead of bioreactor. He was unaware of the systems which one is suitable for aerobic and which one for anaerobic processes. In this respect, guide him by comparing the both. **[Difficult]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Genetic engineering is a complex technology which comprises various techniques. Each technique uses number of chemicals for different purposes.

Complete the following table which is framed in this background.

Gel electrophoresis	A	Visualization of DNA fragments
B	DNA polymerase	C
Gene gun	D	Transfer of recombinant DNA
Isolation of DNA	E	Precipitation of DNA

[Difficult]

2. Mention the importance of the following with respect to genetic engineering:

a) *Agrobacterium tumifaciens*

b) *Thermus aquaticus*

c) *E.coRI* d) DNA Ligase

e) Disarmed retroviruses **[Difficult]**

****

CHAPTER 10
BIOTECHNOLOGY AND ITS APPLICATIONS

PART-A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. In human insulin, the two polypeptide chains are linked together by
   a) Covalent bonds b) Glycosidic bonds
   c) Phosphodiester bonds d) Disulphide bonds **[Easy]**
2. ADA deficiency in humans could be permanently cured by
   a) Bone marrow transplantation b) Introduction of functional ADA through injection
   c) Introduction of gene isolated from marrow cells producing ADA into embryonic cells
   d) Enzyme replacement therapy **[Easy]**
3. A novel strategy adopted to prevent the infestation of tobacco roots by nematode *Meloidegyne incognitia* is
   a) Molecular diagnosis b) RNA interference c) Electrophoresis d) Microinjection **[Easy]**
4. The plants developed by the fusion of protoplasts of two different plants are called
   a) Somaclones b) Somatic hybrids c) Transgenic plants d) GM plants **[Easy]**
5. Which of the following statement is correct with respect to synthesis of insulin?
   a) Synthesized as pro-enzyme and needs to be processed to become functional enzyme.
   b) Synthesized as pro-hormone and needs to be processed to become functional hormone
   c) Synthesized as pro-enzyme and needs to be processed to become functional prohormone.
   d) Synthesized as pro-hormone and needs to be processed to become functional proenzyme. **[Difficult]**
6. Bt- toxin is produced by the bacterium
   a) *Bacillus subtilis* b) *Bacillus thuringiensis*
   c) *Escherichia coli* d) *Salmonella typhi* **[Easy]**
7. Corn borers are controlled by the protein encoded by the gene
   a) *cryIAc* b) *cryIIAb* c) *cryIVAb* d) *cryIAb* **[Easy]**
8. The silencing of mRNA has been used in producing transgenic plants resistant to
   a) White rusts b) Bacterial blights c) Bollworms d) Nematodes **[Easy]**
9. *Meloidogyne incognitia* is a
   a) Protozoa b) Nematode c) Fungi d) Algae **[Easy]**
10. The name given to the first transgenic cow is
    a) Dolly b) Rosie c) Marie d) Lilly **[Easy]**

11. Crystals of BT toxin produced by some bacteria do not kill the bacteria themselves because

a) Bacteria are resistant to the toxin b) Toxin is immature

c) Toxin is inactive d) Bacteria encloses toxin in a special sac **[Average]**

12. Crystalline protein synthesized by *Bacillus thuringiensis* is activated by

a) Acidic pH in insect fore-gut b) Acidic pH in insect mid-gut

c) Alkaline pH in insect fore-gut d) Alkaline pH in insect mid-gut **[Easy]**

13. The use of bioresources by multinational companies and other organizations without proper authorization from the countries and people concerned without compensatory payment is called

a) Bioethics b) Biowar c) Biopiracy d) Bioremedy **[Easy]**

14. The mobile genetic elements are called

a) Transcripts b) Transposons c) Transforms D) Transgenes **[Easy]**

15. Identify the incorrect statement.

a) The transgenic cow Rosie produced milk enriched with α-lactalbumin.

b) Bt toxin genes *cryIAc* controls corn borer.

c) Human protein α-1-antitrypsin is used to treat emphysema.

d) The commercial production of human insulin was made possible by the transgenic species of *E. coli*. **[Average]**

16. Nematode specific genes were introduced into tobacco plants using the plasmid of

a) *Agrobacterium tumifaciens* *b*) *Escherichia coli*

c) *Meloidogyne incognitia* d) *Salmonella typhi* **[Easy]**

17. **Statement I:** The plants genetically identical to the original plants are called somaclones.

Statement II: Somaclones can be obtained through micro-propagation.

a) Statement I is true, statement II is false b) Statement I is false, statement II is true.

c) Both the statements I and II are true d) Both the statements I and II are false. **[Difficult]**

18. Pollen grains of a plant with 2n=48 are cultured by tissue culture to produce a callus. What would be the number chromosomes in the cells of the callus?

a) 48 b) 24 c) 36 d) 12 **[Difficult]**

19. In tissue culture, which part of a diseased plant should be chosen to recover virus free plants?

a) Intercalary meristem b) Apical meristem c) Lateral meristem d) Anther **[Easy]**

20. Identify the correct statement

a) *Bacillus thuringiensis* produces an insecticidal protein that can kill *Meoidogyne incognitia*.

b) The gene *cryIAc* can control corn borer.

c) Using *Agrobacterium* vectors, nematode-specific genes were introduced into the host plant.

d) Bt toxin exists as protein and converts into protoxin in insect gut. **[Average]**

21. Maximum number of existing transgenic animals are

a) Mice b) Rats c) Rabbits d) Cows **[Easy]**

22. The therapy given to cure adenosine deaminase deficiency (ADA) comes under the category of

a) Chemotherapy b) Gene therapy c) Radiation therapy d) Immunotherapy **[Easy]**

23. Which among the following is commonly used in transfer of foreign DNA into plants?

a) *Bacillus thuringiensis* b) *Agrobacterium tumifaciens*

c) *Meloidogyne incognitia* d) *Escherichia coli* **[Easy]**

24. **Statement I:** *Bacillus thuringiensis* produces insecticidal protein crystals called Bt toxin

Statement II: These toxins exist as pro-hormones before entering into insect gut.

a) Statement I is true, statement II is false. b) Statement I is false, statement II is true.

c) Both the statements I and II are true. d) Both the statements I and II are false. **[Average]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Transposons, Corn borer, ELISA, Polypeptide, Meristem, Pro-hormone, *Agrobacterium*, Adenosine deaminase, Cotton bollworm, Somaclones, Totipotency, Eli Lilly, Alpha-lactalbumin, Emphysema)

1. The capacity to generate a whole plant from any cell or explants is called _____ . **[Easy]**
2. Although the plant is infected with a virus, _____ is free of virus. **[Easy]**
3. The first clinical gene therapy was given to a girl with _____ deficiency. **[Easy]**
4. _____ is based on the principle of antigen- antibody interaction. **[Easy]**
5. In mammals insulin is synthesized as a _____ . **[Easy]**
6. The protein encoded by the genes c*ryIAb* controls _____ . **[Easy]**
7. The proteins encoded by the genes *cryIAc* and *cryIIAb* control the _____ . **[Easy]**
8. Using_____ vectors nematode specific genes were introduced into host plants. **[Average]**
9. The artificial human protein α-1-antitrypsin is used to treat _____ . **[Easy]**
10. The transgenic cow Rosie produces milk enriched with _____ protein. **[Easy]**
11. _____is an American company responsible for commercial production of human insulin. **[Easy]**
12. Human insulin consists of two _____ chains. **[Easy]**
13. _____ are the mobile genetic elements. **[Easy]**
14. The plants genetically identical to the original plants from which they were grown are known as _____ . **[Easy]**

PART - B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. What are genetically modified organisms? Give any two examples. **[Easy]**
2. Mention any two benefits of genetically modified plants. **[Easy]**
3. What are *cry* genes? Give two examples. **[Easy]**
4. Mention two genes responsible for producing *cry* protein to control cotton bollworms. **[Easy]**
5. Mention any two problems associated with conventional breeding that were solved through GM crops. Mention one problem associated with its use. **[Easy]**
6. Name the genes of *Bacillus thuringiensis* that produce insecticidal proteins against
   a) Cotton bollworms b) Corn borer **[Easy]**
7. What are somatic hybrids? Give an example. **[Easy]**
8. What are somaclones? Name any two example of somaclones (plants) produced on commercial scale using tissue culture. **[Easy]**
9. Differentiate between somatic hybrids and somaclones. **[Average]**
10. What is meristem culture? Mention its significance. **[Easy]**
11. Differentiate between pro-insulin and a mature insulin. **[Average]**
12. Mention any two biological products produced by transgenic animals with their applications. **[Average]**
13. What is gene therapy? Mention a disease which has been treated using this technique. **[Easy]**
14. Transgenic animals can be used to produce biological products. Justify with two examples. **[Average]**
15. Mention any two biological products obtained from transgenic animals. **[Easy]**
16. Mention any two functions of Genetic Engineering Approval Committee. **[Easy]**
17. Name any two modern diagnosis tools which help in early detection of disease. **[Easy]**
18. Write any two application of PCR as a molecular diagnostic tool. **[Easy]**
19. Give the diagrammatic representation of maturation of pro-insulin into insulin. **[Easy]**
20. PCR and ELISA are the tools for early detection of diseases. Justify the statement by giving the principles behind the techniques. **[Difficult]**
21. How insulin was produced before its production through genetically modified *E. coli*? **[Average]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Mention the three critical areas of biotechnology. **[Easy]**
2. Name any three options that can be thought for increasing food production in agriculture. **[Easy]**
3. In which way genetically modified plants are useful? **[Average]**
4. What is Bt toxin? How does it kill cotton boll worms? **[Easy]**

5. Mention the three groups of insects that are susceptible to Bt toxins. **[Easy]**
6. The development of Bt cotton has reduced the reliance on chemical pesticides. Justify. **[Average]**
7. Write a note on production of human insulin by genetic engineering. **[Easy]**
8. What is RNA interference? Explain how this novel technique is used to develop resistance in tobacco plants against nematode infestation. **[Average]**
9. Briefly explain the steps involved in PCR based molecular diagnosis of mutations in cancer patients. **[Average]**
10. What is gene therapy? Explain the steps involved in curing ADA deficiency by gene therapy. **[Easy]**
11. What is gene therapy? Name a disease against which gene therapy was first employed. **[Easy]**
12. Write a note on gene therapy. **[Easy]**
13. Mention three applications of PCR as a tool of molecular diagnosis. **[Easy]**
14. What is biopiracy? Explain it with reference to Basmati Rice. **[Easy]**
15. Briefly explain the process of somatic hybridization with an example. **[Easy]**
16. Mention the significances of the following in early diagnosis.
    a) PCR b) ELISA 2 + 1 **[Easy]**

PART – D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. "One of the applications of biotechnology is the production of insect resistant crop plant". Justify the statement with reference to Bt cotton. **[Average]**
2. a) What are genetically modified crops? Give example. 2
   b) What is Biopiracy? Comment on the Biopiracy of Basmati rice. 3 **[Easy]**
3. What is Bt cotton. The development of Bt cotton has greatly reduced the reliance on chemical pesticides. Justify. **[Average]**
4. Briefly explain the uses of transgenic animals. **[Easy]**
5. What is RNA interference? How does RNA interference helps to develop resistance in tobacco plant against Nematode infection? **[Difficult]**
6. With reference to tissue culture, define the following terms:
   a) Explants b) Micropropagation c) Totipotency
   d) Somatic hybridization e) Somaclones **[Easy]**
7. Explain any five applications of tissue culture. **[Easy]**
8. Explain the applications of biotechnology in the field of medicine. **[Easy]**
9. In agriculture use of genetically modified crops is more advantageous over conventional method. Justify the statement giving five reasons. **[Average]**

10. In tissue culture, the technique of somatic hybridization is one of the superior thinking of scientists. Substantiate the statement with appropriate example. **[Difficult]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Traditional breeding techniques have failed to keep pace with demand and to provide sufficiently fast and efficient systems for crop improvement. In this context, explain how the tissue culture technique has contributed to crop improvement programmes by mentioning any five of its applications.
2. Biotechnology has its own applications in many ways as it comprises number of techniques. Given below are the few applications created by biotechnology. Name the technique behind the applications.
   a) Detection of pathogen using antigen-antibody interaction.
   b) Identification of mutation in a gene.
   c) Detection of the pathogen before the development of disease symptoms.
   d) Control of *Meloidogyne* infection in tobacco plants.
   e) Production of thousands of plants in short duration using tissue culture. **[Difficult]**
3. Genetic engineering is a technique of altering the chemistry of genetic material to introduce into host organism to create change in phenotype of the host. In light of this, answer the following:
   a) Which is the source organism of cry gene? 1
   b) Name the host plant to which cry gene was inserted? 1
   c) Give any two examples for the cry gene. 1
   d) Explain the mode of action of the protein produced by the cry gene. 2 **[Difficult]**
4. RNA interference is a novel strategy to control *Meloidogyne incognitia* infection in tobacco plants. Given below are the steps involved in this technique. Arrange them in a definite sequence.
   a) Being complementary, the two RNAs forms double stranded.
   b) Nematode specific genes were introduced into the host plants.
   c) DNA produces both sense and anti-sense RNA.
   d) Silences the specific mRNA in the nematode.
   e) Initiates RNAi. **[Difficult]**
5. Give reasons for the following:
   (a) Even though Bt toxin can kill many insects, it will not kill the bacteria that produces it. 2
   (b) Even though Bt toxin is inactive, it kills the insect when enters its midgut. 2
   (c) The gene encoding '*cry*' protein inserted into a crop plant. 1 **[Difficult]**

****

CHAPTER 11
ORGANISIMS AND POPULATIONS

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. The correct order of various levels of biological organization in an ecosystem is:
   a) Organism → Population → Community → Ecosystem → Biome.
   b) Organism → Population → Biome → Ecosystem → Community
   c) Biome → Ecosystem → Population → Organism → Community.
   d) Biome → Community → Ecosystem → Population → Organism. **[Average]**
2. In a pond, there were 20 lotus plants last year and through reproduction 16 new plants were added. Calculate the birth rate:
   a) 0.8 offspring per lotus per year.
   b) 0.4 offspring per lotus per year.
   c) 0.8 offspring per year.
   d) 1.25 offspring per year. **[Average]**
3. In an aquarium, there are 8 fishes and during a week 2 fishes die. Calculate the death rate:
   a) 0.25 individual per fish per week.
   b) 3.0 individual per fish per week.
   c) 0.25 individual fish per week.
   d) 3.0 individual fish per week. **[Average]**
4. The shape of the age pyramid reflects the population's growth status. In this context, the given age pyramid indicates that the population is



   a) Expanding b) Stable c) Declining d) Cannot be depicted **[Difficult]**
5. **Statement 1:** Population size is determined by population density.
   Statement 2: Population density is not the meaningful measure of the population
   a) Both statements 1 & 2 are correct b) Statement 1 is correct & 2 is incorrect.
   c) Statement 1 is incorrect & 2 is correct d) Both the statements are incorrect. **[Difficult]**

6. In a population with a large number of older individuals than younger ones it can be predicted to:
   a) Decline b) Standard. c) Grow d) Does not affect **[Easy]**
7. The change in population size can be used to evaluate :
   a) Impact of predator b) Effect of pesticide.
   c) Introduction of exotic species d) All the above. **[Easy]**
8. **Statement 1:** If the predator is too efficient and over exploits its prey then the predator itself becomes extinct.
   Statement 2: Prey species have evolved various defense s to lessen the impact of the predation.
   a) Both statements 1 & 2 are correct b) Statement 1 is correct & 2 is incorrect.
   c) Statement 1 is incorrect & 2 is correct d) Both the statements are incorrect. **[Difficult]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Exponential, Resource partitioning, 0.0205, Commensalism, Logistic growth)

1. The r value for human population in India in 1981 was _____. **[Easy]**
2. The population growth is called _____ when it finally reaches an asymptote. **[Average]**
3. The population interaction where one species is benefitted and the other is neither benefitted nor harmed is called _____. **[Easy]**
4. Closely related species of Warblers living on the same tree avoid competition by the mechanism called _____ **[Easy]**
5. When resource availability in a habitat is unlimited it leads to _____ growth. **[Average]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Mention the four basic levels of biological organization that ecology is concerned with? **[Easy]**
2. Mention any two population attributes. **[Easy]**
3. Name the four basic processes that fluctuates population density. **[Average]**
4. Mention the two patterns of population growth in organisms. **[Easy]**
5. Differentiate between exponential growth and logistic growth. **[Average]**
6. Give two examples or organisms that breed only once in their life time. **[Easy]**
7. Show diagrammatic representation of exponential and logistic growth curves of population growth in a combined diagram. **[Difficult]**
8. Mention any four types of interspecific interactions in organisms. **[Easy]**
9. Mention two examples for defense of prey against predators in plants. **[Average]**
10. Define predation. Give any two examples. **[Easy]**

11. Mention two adaptations in plants to escape from grazers and browsers. **[Easy]**
12. Give an example for following defense mechanism. **[Average]**
    a) Camouflage b) Distasteful for the predator
13. What is interference competition? Explain. **[Easy]**
14. Explain competitive release with an example. **[Easy]**
15. Write short note on resource partitioning with a suitable example. **[Average]**
16. Mention two adaptations in organisms for parasitic mode of life. **[Average]**
17. What are ectoparasites? Give example **[Easy]**
18. What are endoparasites? Give an example. **[Easy]**
19. Define commensalism. Give examples. **[Easy]**
20. Define mutualism. Give an example. **[Easy]**
21. What is brood parasitism? Give an example. **[Easy]**
22. State Gause's competitive exclusion principle. **[Average]**
23. Female mosquito is not considered as parasite although it feeds on human blood. Why? **[Difficult]**
24. Differentiate between ectoparasite and endoparasite with examples. **[Average]**
25. Cattle egret and grazing cattle is a classical example of commensalism. Comment. **[Easy]**
26. Fig species and pollinator wasp is an example of evolution of mutualists. Justify. **[Difficult]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. If N is population density at time t, B is natality, D is mortality, I is immigration, E is emigration.
   a) Give the equation to determine density at time t + 1.
   b) Which of the above determinants contribute to an increase in population density?
   c) Which of the above determinants contribute to a decrease in population density? **[Average]**
2. List out the three attributes of a population. **[Easy]**
3. Write a note on brood parasitism. **[Easy]**
4. If a plot of N in relation to time (t) results in a sigmoid curve and dN/dt = rN ((K-N) / K)
   a) Name the type of population growth.
   b) What does K represent in the equation?
   c) On what condition does the curve become stationary? **[Difficult]**
5. What is resource partitioning? Describe with an example. **[Easy]**
6. What is parasitism? Mention the types of parasites with an example for each. **[Easy]**
7. What is parasitism? Write a note on brood parasitism. **[Easy]**
8. What is parasitism? Mention two adaptations in organisms for parasitic mode of life. **[Easy]**

9. What is mutualism? Why plant – animal interaction often involves co - evolution of mutualists? **[Average]**
10. Explain how Mediterranean orchid '*Ophrys*' employs sexual deceit to ensure pollination? **[Average]**
11. Name the type of interactions seen in each of the following examples:
    a) A sparrow eating on a seed.
    b) *Abingdon* tortoise and goats in Galapagos Island.
    c) A *Cascuta* growing on a hedge plant. **[Easy]**

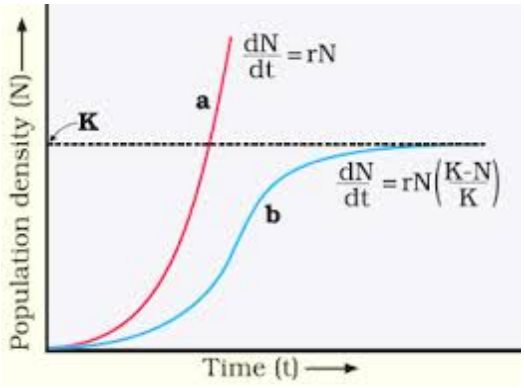
PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Explain Verhulst-Pearl logistic growth with a diagram and write its mathematical expression. **[Difficult]**
2. Graphically represent exponential growth and give its mathematical equation. **[Difficult]**
3. Mention any five population interactions with an example for each. **[Average]**
4. Define competition. Explain interference competition and competitive release with suitable examples. **[Easy]**
5. Define predation. Explain the role of predation in an ecosystem with suitable examples. Add a note on morphological and chemical defensive adaptations in plant and animal preys against their predators? **[Average]**
6. Give an example each for each:
    a) Morphological and chemical defense in animals against their predators
    b) Morphological and chemical defense in plants against grazing animals. **[Average]**
7. What is mutualism? Explain any four examples of mutualism. **[Easy]**
8. a) What is mutualism? Give any two examples. 3
    b) Plant - animal interaction often involves co - evolution of mutualists? Justify your answer with an example. 2 **[Easy]**
9. 'Parasitic mode of life ensures free lodging and free meals'. Justify the statement by listing the special adaptations developed by parasites. **[Average]**
10. Name the type of interactions seen in each of the following examples:
    a) Clown fish living among the tentacles of sea anemone
    b) Koel and cuckoo bird during the breeding season.
    c) Barnacles growing on back of a whale.
    d) Marine fish and copepods.
    e) Sexual deceit in Mediterranean orchid *Ophrys* **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Refer to the graphs given below showing the growth patterns in organisms and answer the questions that follow.



a) Curve 'a' represents which type of growth?

b) Curve 'b' represents which type of growth?

c) Which growth model is considered more realistic one in nature?

d) When bacteria are cultured under controlled optimum environmental conditions. Which curve is obtained?

e) The interactions such as predation, competition and parasitism can be represented by which curve. How? **[Difficult]**

2. Fill in the tabular column with the appropriate terms describing biological interactions among species.

Species A	Species B	Name of interaction
Phytophagous insects	-	Predation
Balanus	*Chathamalus*	-
Marine fish	Copepods	-
-	Orchid	Commensalism
Fig	-	Mutualism

[Average]

****

CHAPTER 12
ECOSYSTEM

PART – A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. The functional unit of nature is called
   a) Community b) Population c) Biome d) Ecosystem **[Easy]**
2. The rate of biomass production is called
   a) Productivity b) Primary productivity
   c) Secondary productivity d) Gross primary productivity **[Easy]**
3. The amount of organic matter or biomass produced per unit area over a time period by producers during photosynthesis is
   a) Productivity b) Primary productivity
   c) Secondary productivity d) Gross primary productivity **[Easy]**
4. The rate of formation of new organic matter by producers is
   a) Net primary productivity b) Primary productivity
   c) Secondary productivity d) Gross primary productivity **[Average]**
5. The rate of production of organic matter during photosynthesis is
   a) Productivity b) Primary productivity
   c) Secondary productivity d) Gross primary productivity **[Easy]**
6. GPP-R is
   a) Primary productivity b) Net primary productivity
   c) Secondary productivity d) Gross primary productivity **[Average]**
7. NPP is represented as
   a) GPP-PP b) GPP-SP c) GPP-R d) GPP-P **[Average]**
8. The annual net primary productivity of the whole biosphere is estimated approximately
   a) 170 billion tons b) 55 billion tons c) 120 billion tons d) 70 billion tons **[Easy]**
9. The annual net primary productivity of oceans is about
   a) 170 billion tons b) 55 billion tons c) 120 billion tons d) 70 billion tons **[Easy]**
10. in GPP-R, the letter ‘R’ represents
    a) Respiratory loss b) Intrinsic rate of natural increase
    c) Respiratory quotient d) Reproductive loss **[Easy]**
11. The productivity available as biomass for the consumption of heterotrophs is
    a) GPP c) NPP c) PP d) SP **[Easy]**

12. The percent of energy that flows to each trophic level in a food chain is

a) 5% b) 10% c) 15% d) 20% **[Average]**

13. In which of the following ecosystem the pyramid of biomass is generally inverted?

a) Grassland b) Tree c) sea d) Forest **[Easy]**

14. Identify the correct sequence of steps in the process of decomposition

a) Fragmentation → Leaching → Catabolism → Humification → Mineralization

b) Leaching → Fragmentation → Catabolism → Humification → Mineralization

c) Humification → Fragmentation → Leaching → Catabolism → Mineralization

d) Catabolism → Fragmentation → Leaching → Humification → Mineralization **[Difficult]**

15. Identify the decomposers among the following in the pond ecosystem.

A) Bacteria B) Fungi C) Flagellates D) Phytoplankton

a) A only b) A & B only c) A,B & C d) A, B, C & D **[Average]**

16. Which of the following statement is not true about ecological pyramids?

a) It does not take into account the same species belonging to two or more trophic levels.

b) It assumes a simple food chain something that almost never exists in nature.

c) Saprophytes are given a place in ecological pyramids as they play a vital role in the ecosystem.

d) It does not accommodate a food web **[Easy]**

17. A food chain with 1000kcal of energy at the first tropic level is expected to transfer what amount of the energy at the third tropic level?

a) 100kcal b) 500kcal c) 10kcal d) 1kcal **[Average]**

18. If the energy produced at the level of the producers is 1000 J, the energy available for the secondary consumers is

a) 1000 J b) 100 J c) 10 J d) 1 J **[Average]**

19. The process of breaking down of complex organic matter into inorganic substances is

a) Fragmentation b) catabolism c) decomposition d) Leaching **[Easy]**

20. Which of the following steps in decomposition does not occur on the detritus?

a) Fragmentation b) Leaching c) Humification d) Catabolism **[Easy]**

21. The rate of decomposition is slower if detritus is rich in

a) Chitin b) Nitrogen c) Water soluble sugars d) Water insoluble sugars **[Easy]**

22. Each trophic level has a certain mass of living material at a particular time and is called

a) Standing state b) standing crop c) Organic matter d) Biomass **[Easy]**

23. **Statement I:** Pyramid of energy maybe upright or inverted.

Statement II: Only 10% of energy goes to next trophic level in a food chain.

a) Both statement I and statement II are correct

b) Both statement I and statement II areincorrect

c) Statement I is correct but statement II is incorrect

d) Statement I is incorrect but statement II is correct **[Difficult]**

24. **Statement I:** The pyramid of biomass in sea is generally inverted.

Statement II: The biomass of fishes far exceeds that of phytoplankton.

a) Both statement I and statement II are correct

b) Both statement I and statement II are incorrect

c) Statement I is correct but statement II is incorrect

d) Statement I is incorrect but statement II iscorrect **[Difficult]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Fragmentation, 10% law, Leaching, Oxygen, Inhibit, Favour, Detritus, GFC, DFC, Food web, Stratification, Earthworm, Catabolism, Soil, Trophic level, Producers, Decline, Terrestrial ecosystem)

1. The vertical distribution of different species occupying different levels is called_____ **[Average]**
2. The dead plant and animal remains including faecal matter constitutes_____ **[Average]**
3. _____is the process of breakdown of detritus into smaller particles. **[Easy]**
4. _____ is an example for detrivore. **[Average]**
5. The process of degradation of detritus into simpler inorganic substances by bacterial and fungal enzymes is_____ **[Easy]**
6. Humification and minimization occur during decomposition in the _____ **[Average]**
7. Decomposition is largely a _____ requiring process. **[Average]**
8. Low temperature and anaerobiosis _____ decomposition. **[Easy]**
9. Warm temperature and moist environments_____ decomposition. **[Easy]**
10. The food chain that begins with dead organic matter is_____ **[Easy]**
11. In aquatic ecosystems_____ is the major conduit for energy flow. **[Average]**
12. The natural interconnections of many food chains are called_____ **[Easy]**
13. The base of each ecological pyramid represents_____ **[Easy]**
14. The specific place of organisms in the food chain based on the source of their nutrition is called_____ **[Average]**
15. Pyramid of biomass shows a sharp_____in biomass at higher trophic levels **[Average]**
16. In _____ greater fraction of energy flows through DFC than GFC. **[Average]**
17. According to the_____only 10% of energy flows to the each trophic level

from the lower trophic level. **[Average]**

PART – B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Define ecosystem. Why is biosphere regarded as global ecosystem by ecologists? **[Easy]**
2. List the basic or fundamental categories of ecosystems on earth with an example for each. **[Easy]**
3. List the four primary functions of an ecosystem. **[Easy]**
4. Define productivity and mention its types. **[Average]**
5. Define a) Primary productivity b) Secondary productivity **[Average]**
6. Distinguish between primary productivity and secondary productivity. **[Average]**
7. Define a) Gross primary productivity b) Net primary productivity **[Average]**
8. Distinguish between gross primary productivity and net primary productivity. **[Average]**
9. Mention two conditions which increase the rate of decomposition in ecosystem. **[Average]**
10. Mention two conditions which decrease the rate of decomposition in ecosystem. **[Average]**
11. Define food chain. Mention the types of food chain. **[Easy]**
12. Distinguish between food chain and food web. **[Average]**
13. Schematically represent a grazing food chain. **[Easy]**
14. 'Ecological pyramids have limitations'. Justify with two reasons. **[Average]**
15. a) Why is the pyramid of biomass in sea generally inverted?

    b) Pyramid of energy is always upright and can never be inverted. Why? **[Average]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Describe the components of an aquatic ecosystem taking pond as an example. **[Easy]**
2. Define the terms a) Detritus. b) Decomposition c) Detrivores **[Easy]**
3. Explain the factors that regulate decomposition. **[Average]**
4. Schematically represent pyramid of numbers with appropriate units. **[Average]**
5. Schematically represent pyramid of biomass with appropriate units. **[Average]**
6. Schematically represent pyramid of energy with appropriate units. **[Average]**
7. 'Ecological pyramids have limitations'. Justify with three reasons. **[Average]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. a) What is productivity? Mention the types of primary productivity. 3

   b) Compare primary productivity and secondary productivity in an ecosystem. 2 **[Difficult]**
2. Define the terms: a) Food chain b) Food web c) Trophic level

d) Stratification e) Standing crop **[Average]**

3. Describe the different steps involved in the process of decomposition. **[Difficult]**

4. 'Ecological pyramids have limitations'. Justify. **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. a) Fill in the blanks with suitable options that correctly identifies A, B and C in the given table: 3

Organism	Trophic level	Food chain
Eagle	A	Grazing
Earthworm	Primary consumer	B
Frog	C	Grazing

.

b) Construct a pyramid of energy representing different trophic levels and their energy content, assuming that 600Kcal at the second trophic level. 2

[Difficult]

****

CHAPTER 13
BIODIVERSITY AND CONSERVATION

PART-A

I. Select the correct alternative from the choices given: **1 Mark Each**

1. The term biodiversity was popularized by
   a) Edward Wilson b) Robert May c) Alexander Von Humboldt d) Paul Ehrlich **[Easy]**
2. Robert May places global species diversity at about
   a) 2 million b) 1.5 million c) 2.5 million d) 7 million **[Easy]**
3. As per species inventories 70% of species recorded are
   a) Plants b) Fungi c) Bryophytes d) Animals **[Easy]**
4. The most species-rich taxonomic group among invertebrates is
   a) Birds b) Insects c) Mammals d) Fishes **[Easy]**
5. The most species-rich taxonomic group among vertebrates is
   a) Birds b) Insects c) Mammals d) Fishes **[Easy]**
6. The most species-rich taxonomic group among plants is
   a) Algae b) Fungi c) Angiosperms d) Ferns **[Easy]**
7. The share of global species diversity of India is
   a) 8.1% b) 8.5% c) 9% d) 2.4% **[Easy]**
8. The species-area relationship was given by
   a) Edward Wilson b) Robert May
   c) Alexander Von Humboldt d) Paul Ehrlich **[Easy]**
9. In the given species-area equation, log S= log C + Z log A, the letter 'Z' represents
   a) Area b) Species richness c) Regression coefficient d)Y-intercept **[Easy]**
10. According to ecologists the value of 'Z' lies in the range of
    a) 0.1-0.5 b) 0.2-0.6 c) 0.6-1.2 d) 0.1-0.2 **[Easy]**
11. The species-area relationship among large areas like the entire content has the 'Z' values in the range of
    a) 0.1-0.5 b) 0.2-0.6 c) 0.6-1.2 d) 0.1-0.2 **[Easy]**
12. The slope of the line (regression coefficient) for fruit-eating birds and mammals in the tropical forests of different continents is found in the range of
    a) 0.1-2.5 b) 1-1.5 c) 0.6-1.2 d) 0.1-0.2 **[Easy]**

13. The rivet popper hypothesis was proposed by

a) Paul Ehrlich b) Alexander Von Humboldt

c) Edward Wilson d) Mac Arthur **[Easy]**

14. Which of the following species introduced for aquaculture purposes is posing serious threats to the indigenous catfishes in our country?

a) *Clarias gariepinus* b) *Eichhirnia* c) *Parthenium* d) *Lantana* **[Easy]**

15. The narrow utilitarian arguments for conserving biodiversity are

a) Cereals & Fruits b) Ecosystem services

c) Intrinsic values d) Aesthetic pleasures **[Easy]**

16. The following are the in-situ conservation approaches except

a) National parks b) Wildlife sanctuaries

c) Biosphere reserves d) Botanical gardens **[Easy]**

17. The following are the ex-situ conservation approaches except

a) Zoological parks b) Seed banks c) Sacred groves d) Botanical gardens **[Easy]**

18. Chanda and Bastar areas known for sacred groves are found in

a) Meghalaya b) Madhya Pradesh c) Karnataka d) Rajasthan **[Easy]**

19. The earth summit was held in Rio de Janeiro in

a) 1991 b) 1992 c) 2010 d) 2002 **[Easy]**

20. The world summit on sustainable development was held in 2002 in

a) Canada b) Johannesburg c) Brazil d) Tokyo **[Easy]**

21. **Statement I:** There are 32 biodiversity hotspots in the world.

Statement II: High level of species richness and high degree of endemism are criteria for declaring a region a biodiversity hotspot.

a) Both statement I and statement II are correct

b) Both statement I and statement II are incorrect

c) Statement I is correct but statementII is incorrect

d) Statement I is incorrect but statement II is correct **[Average]**

22. **Statement I:**Tropical regions have got a long evolutionary time for species diversification as compared to temperate regions.

Statement II: Temperate regions have undergone frequent glaciations in the past whereas tropical regions have remained relatively undisturbed for millions of years.

a) Both statement I and statement II are correct

b) Both statement I and statement II are incorrect

c) Statement I is correct but statement II is incorrect

d) Statement I is incorrect but statement II is correct **[Average]**

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket:

1 Mark Each

(Overexploitation, Decreases, Cryopreservation, Cichlid, 2.4%, Reserpine, Lungs, Increases, Amazon rain forest, In-situ, Ex-situ, 2%, sacred grooves, Biodiversity hotspot, Seed bank, Endemism)

1. The genetic diversity shown by *Rauwolfia vomitoria* growing in Himalayan produces an active chemical called_____ **[Easy]**
2. The extinction of Steller's sea cow and passenger pigeon was due to_____. **[Easy]**
3. The introduction of Nile perch into Lake Victoria in east Africa led to the extinction Of ecologically unique assemblages of_____fish in the lake. **[Easy]**
4. The land area of India is _____ **[Easy]**
5. In general species diversity _____as we move away from the equator towards the poles **[Easy]**
6. _____in south America has the greatest biodiversity on earth **[Easy]**
7. The Amazonian rain forest is called _____of the planet **[Easy]**
8. The protection of endangered or threatened species away from their natural habitat is___ **[Easy]**
9. The protection of the entire forest to save the tiger is an example for_____conservation. **[Easy]**
10. Biodiversity hotspots constitutes about _____ of the earth's land area **[Easy]**
11. Tracts of forests that are set aside and all the trees and wildlife within were venerated and given full protection constitute_____ **[Easy]**
12. The regions with high species richness and a high degree of endemism are called biodiversity_____ **[Easy]**
13. The phenomenon of species confined to a particular region and not found elsewhere is called_____ **[Easy]**
14. Seeds of different genetic strains of commercially important plants can be kept for long periods in_____ **[Easy]**
15. Gametes of threatened species can be preserved in viable and fertile conditions for long periods using_____ **[Easy]**

PART-B

III. Answer the following questions in 3-5 sentences each, wherever applicable: 2 Marks Each

1. Define biodiversity. Who popularized the term 'biodiversity? **[Easy]**
2. Define biodiversity. Mention any two levels of biodiversity. **[Easy]**

3. What is genetic diversity? India has more than 50,000 different strains of rice. Mention the level of biodiversity it represents. **[Easy]**
4. "India is rich in genetic diversity". Justify the statement by giving two examples. **[Average]**
5. Give an example for genetic diversity. Name an active chemical produced by medicinal plant *Rauwolfia vomitoria*. **[Easy]**
6. What is species diversity? Give an example. **[Easy]**
7. What is ecological or ecosystem diversity? Give an example. **[Easy]**
8. Differentiate genetic diversity and species diversity. **[Average]**
9. Differentiate genetic diversity and ecological diversity. **[Average]**
10. Differentiate species diversity and ecological diversity. **[Average]**
11. Name any four recent extinct organisms as per IUCN Red list. **[Easy]**
12. Mention any two effects of biodiversity loss. **[Easy]**
13. Mention David Tilman's observation with respect to species richness in a community. **[Easy]**
14. Mention "The Evil Quartet" of biodiversity losses in a given habitat. **[Easy]**
15. Give any two examples for species which have become extinct due to overexploitation. **[Easy]**
16. "Alien species invasion leads to extinction of indigenous species". Justify the statement by giving two appropriate examples. **[Average]**
17. What is co-extinction? Give an example. **[Easy]**
18. Define the terms: a) Endemism b) Bioprospecting **[Easy]**
19. What are 'biodiversity hotspots'? Name any two biodiversity hotspot in India. **[Easy]**
20. What is *in situ* conservation? Mention any two examples for *in situ* conservation. **[Easy]**
21. What is *ex situ* conservation? Mention any two examples for *ex situ* conservation. **[Easy]**
22. Differentiate *in situ* conservation and *ex situ* conservation. **[Average]**
23. What are sacred grooves? Give an example. **[Easy]**
24. How are sacred grooves important in conserving biodiversity? **[Easy]**
25. How cryopreservation helps in the conservation of biodiversity? **[Easy]**
26. List any two ecosystem services provided by biodiversity. **[Easy]**
27. Write any two advanced techniques used in *ex situ* conservation. **[Easy]**
28. a) What are endemic species?
    b) Who proposed the 'rivet popper" hypotheses? **[Easy]**
29. a) Among vertebrates, which group of animals have the highest number in global biodiversity?
    b) Name the forest which is referred to as the 'lungs of the planet'? **[Easy]**

PART – C

IV. Answer the following questions in 40-80 words each, wherever applicable: 3 Marks Each

1. Briefly explain the three levels of biodiversity with suitable examples. **[Average]**
2. "Tropical rain forests have greater biodiversity compared to temperate forest". Justify with three reasons. **[Average]**
3. Draw a graphical representation of species area relationship. **[Easy]**
4. Explain the species-area relationship of biodiversity as explained by Alexander Von Humboldt. **[Average]**
5. In the species area relationship equation, log S=log C + Z log A what do'S', 'A' and 'Z' represent? **[Easy]**
6. Write the salient features of stable community with reference to biodiversity. **[Easy]**
7. Comment on David Tilman's long term ecosystem experiments and its signicance. **[Easy]**
8. Describe the "rivet popper" hypothesis as explained by Stanford ecologist Paul Ehrlich. Mention its significance. **[Average]**
9. List out the three general effects of loss of biodiversity. **[Easy]**
10. "Introduction of alien species causes biodiversity loss". Justify this statement with any three examples. **[Average]**
11. Why should we conserve biodiversity? Explain. **[Average]**
12. Briefly explain narrowly utilitarian reason for conserving biodiversity. **[Average]**
13. Briefly explain broadly utilitarian reason for conserving biodiversity. **[Average]**
14. "Biodiversity plays a major role in providing many ecosystem services". Justify the statement with three reasons. **[Average]**
15. Name the biodiversity hotspots of India. **[Easy]**
16. a) What is India's share (in percentage) in global species diversity?
    b) How many mega diversity countries have been identified in the world?
    c) Why Western Ghats range is considered as one of the biodiversity hotspots? **[Easy]**

PART- D

V. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks Each

1. Define biodiversity. Explain the major causes of biodiversity loss. **[Average]**
2. What are endemic species? Differentiate *in situ* conservation and *ex situ* conservation with suitable examples. **[Easy]**
3. Explain species area relationship with graphical representation. **[Average]**
4. Describe different ecosystem services provided by biodiversity. **[Average]**

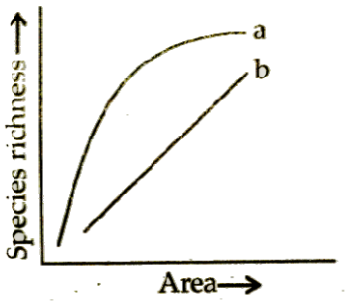
5. Describe different strategies as a measure for conservation of biodiversity. **[Average]**

VI. Answer the following questions in 200-250 words each, wherever applicable: 5 Marks each

1. Refer to the graph showing species-area relationship and answer the questions that follow:

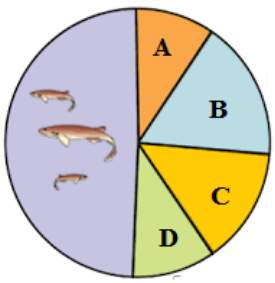
   a) Write the equation of the curve 'a' and 'b'. (2)

   b) Explain how does the value of Z varies. (3)



[Difficult]

2. Refer to the graphical representation depicting the proportionate number of vertebrates species and answer the questions that follow:



   a) Which major taxa of vertebrates do the labelled parts A, B, C, and D represent?

   b) Which plant group species number in the world has greater number than the combined total of fish, amphibian, reptile, and mammal species?

****

- PREVIOUS EXAM QUESTION PAPERS
- MODEL PAPERS
- PRACTICE PAPERS

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 1 (2024)

SUBJECT – BIOLOGY (36)

General Instructions:

1. This question paper consists of four parts A, B, C and D.
2. Part A consists of I & II and Part D consists of two parts, Section I and Section II.
3. All the parts are Compulsory.
4. Draw diagrams wherever necessary. Unlabelled diagrams or illustrations do not attract any marks.
5. For Part A questions, only the first written answers will be considered for evaluation.

PART - A

I. Select the correct alternative from the choices given below: 15 x 1 = 15

1. The function of Tapetum in microsporangium is
   a) It nourishes the developing pollen grains
   b) Helps in dehiscence of anther to release the pollen grains
   c) Helps in protection
   d) Helps in division
2. Which pollination brings genetically different types of pollen grains to the stigma?
   a) Geitonogamy b) Xenogamy c) Autogamy d) Both a) and b)
3. The corpus luteum secretes large amounts of progesterone, which is essential for maintenance of
   a) Perimetrium b) Myometerium c) Endometerium d) Cervical canal
4. The secondary spermatocytes undergo the second meiotic division to produce four equal, haploid
   a) Spermatogonium b) Spermatids
   c) Primary spermatocyte d) Secondary spermatocyte
5. In female sterilisation A is removed or tied. This process is called B .
   a) A - Cervix B - Vasectomy
   b) A - Fallopian tube B - Tubectomy
   c) A - Cervix B - Tubectomy
   d) A - Fallopian tube B - Vasectomy
6. Statement I: Progestogens alone or in combination with estrogen can also be used by females as injections or implants under the skin.
   Statement II: Mode of action of injections or implants is similar to that of pills and their effective periods are much longer.
   a) Both Statement I and II are correct
   b) Statement I is correct and Statement II is incorrect

c) Statement I is incorrect and Statement II is correct

d) Both Statement I and II are incorrect

7. In human males XY type of sex determination is

a) Homogametic b) Heterogametic c) Both a) and b) d) Isogametic

8. In Griffith's experiment, mice infected with, the A die from pneumonia infection, but mice infected with B don't develop pneumonia.

a) A S -Strain; BS -Strain b)A S -Strain; B R -Strain

c) A R -Strain; B S -Strain d) A R -Strain; B R -Strain

9. Which is the correct statement regarding Founder effect?

a) Named after the scientist John founder

b) No large change in frequency

c) The original drifted population become founders

d) Formation of no species

10. Large group of Lymphoid tissue in respiratory, digestive and urogenital tract are collectively called,

a) MALT b) Peyer's Patches c) Lymph nodes d) Alpha globulins

11. Which micro-organism is useful in production of citric acid?

a) Acetobacter aceti b) Penicillium notatum

c) Asoerailus niger d) Clostridium butylicum

12. Which of the following statements are correct for the enzyme Taq polymerase?

I. Taq polymerase is thermally unstable.

II. It requires primers for carrying out the process of polymerisation.

III. Taq polymerase is thermally stable.

Choose the correct option.

a) I and II b) I and III c) II and III d) I, II and III

13. Population density increases as

a) Natality and Emigration increases b) Mortality and Emigration increases

c) Natality and immigration decreases d) Natality and immigration increases

14. Net Primary Productivity (NPP) is equal to

a) GPP b) GPP + R c) d) GPP — R

15. The species confined to a particular region, and not found anywhere else are called as

a) Sacred species b) Endemic c) Extinct d) Vulnerable

II. **Fill in the blanks by choosing the appropriate word/words from those given below: 5 x 1 = 5**

(Oxytocin, Lichen, Ectoparasite, Polyembryony, Honey bee, Endoparasite)

16. If more than one embryo occurs in a seed then it is known as _____
17. The hormone which helps in Parturition is _____
18. Haploid-diploid sex determination system is seen in _____
19. Natural indicator of Industrial Pollution is _____
20. Parasites that feed on the external surface of the host organism are called _____

PART - B

III. Answer any five of the following questions in 3 - 5 sentences wherever applicable: 5 x 2 = 10

21. What is Aminocentesis? Write any one significance.
22. Write the karyotype and symptoms of Klinefelter's syndrome.
23. When does medical termination of pregnancy become essential?
24. List the characteristics of Neanderthal man.
25. Name the two chemicals secreted by Mast cells during allergic reactions in the body.
26. Define BOD. Mention its significance.
27. Differentiate between Exonuclease and Endonuclease.
28. What is meant by Decomposition and Detritus?

PART - C

IV. Answer any five of the following questions in 40 - 80 words each, wherever applicable: 5 x 3 = 15

29. a) Differentiate between false fruit and true fruit. 2
    b) What is meant by Parthenocarpic fruit? 1
30. What is Placenta? Mention any four hormones secreted by Placenta.
31. Write any three criteria, that a molecule has to fulfill to act as a genetic"material.
32. List the difference between Homologous and Analogous organs. Write one example for each in animals.
33. Mention any three uses of genetically modified plants.
34. Write the three critical research areas of Biotechnology.
35. Construct an upright pyramid of biomass.
36. What is Alien species invasion? Give two examples.

PART - D

Section -1

V. Answer any four of the following questions in about 200 - 250 words each, wherever applicable: 4 x 5 = 20

37. Draw a neat labelled diagram of Human female reproductive system.
38. What is Incomplete dominance? Mention one example, write the schematic representation of the same.

39. Write the scientific name of the organism responsible to cause following diseases.
    a) Typhoid b) Pneumonia c) Common cold d) Malaria e) Amoebiasis
40. List any five salient features of Human Genome.
41. a) Draw the schematic structure of a transcription unit. 3
    b) Write the difference between point mutation and frame shift mutation. 2
42. Describe the 'Role of Microbes as Biofertilisers'.
43. Read the following statements and write one appropriate term for each.
    a) Autonomously replicating circular extrachromosomal DNA.
    b) Method to introduce recombinant DNA in animal cells.
    c) Method to introduce recombinant DNA in plant cells.
    d) Specific DNA sequence responsible for initiating replication.
    e) Enzyme used to join the DNA fragments.
44. What is Mutualism? Mention any four examples of Mutualism.

Section - II

VI. Answer any one of the following questions in about 200 - 250 words each, wherever applicable: **1 x 5 = 5**

45. Pollen-Pistil interaction is a dynamic process involving pollen recognition followed by promotion or inhibition. Explain.
46. Name the genetic disorder in which clotting of blood is affected. Write the"features of this genetic disorder.
47. With the help of schematic representation illustrate how an infected"animal cell can survive while viruses are being replicated and released.

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 2 (2024)

SUBJECT – BIOLOGY (36)

General Instructions:

1. This question paper consists of four parts A, B, C and D.
2. Part A consists of I & II and Part D consists of two parts, Section I and Section II.
3. All the parts are compulsory.
4. Draw diagrams wherever necessary. Unlabelled diagrams or illustrations do not attract any marks.
5. For Part A questions, only the first written answers will be considered for evaluation.

PART - A

I. Select the correct alternative from the choices given below: 15 x 1 = 15

1. Occurrence of more than one embryo in a seed is seen in

   a) Grasses a) Asteraceae c) Citrus d) Both A and C

2. Dioecious plants are

   a) Autogamous only

   b) Both autogamous and geitanogamous

   c) Both geitanogamous and xenogamous

   d) Only xenogamous

3. When does the second meiotic division occur in of the secondary oocyte?

   a) At the time of birth b) At the time of fertilization

   c) At the time of puberty d) At the time of ovulation

4. Identify the wrong pair with respect to the menstrual cycle:

   a) Formation of corpus luteum – 15th to 28th day

   b) Secretion of progesterone – 15th to 28th day

   c) Regeneration of endometrium – 5th to 13th day

   d) Regression of endometrium – 2nd to 13th day

5. Which of the following hormones is produced only during pregnancy?

   a) hCG b) LH c) Estrogen d) FSH

6. A reusable contraceptive method that prevents the physical meeting of sperm and ovum is

   a) Condom b) Pills c) Vaults d) Lippes loop

7. The genotypic ratio of incomplete dominance is

   a) 3:1 b) 1:2:1 c) 1:1:1 d) 9:3:3:11:1

8. 47 chromosomes are found in a person with:
   a) Klinefelter's syndrome b) Turner's syndrome
   c) Down's syndrome d) Both a and c
9. Dystrophin gene has
   a) 2.8 million bases b) 2.4 million bases c) 2.6 million bases d) 1.4million bases
10. A prehistoric human with a brain capacity between 652 - 800 cc and who probably did not eat meat
    a) *Homo erectus* b) *Australopithecus* c) *Neanderthal man* d) *Homo habilis*
11. Stagnation of water in and around residential areas may cause an increase in the cases of
    a) Typhoid and Ascaris b) Dengue and Elephantiasis
    c) Amoebiasis and Ringworms d) Pneumonia and Common Cold
12. Chitinase enzyme is used to isolate DNA from
    a) Fungal cell b) Animal cell c) Plant cell d) Bacterial cell
13. An age pyramid shows equal numbers of individuals in pre-reproductive and reproductive age groups. This type of age pyramid _____
    a) Expanding b) Stable c) Declining d) Unstable
14. Which ecological pyramid is always upright?
    a) Pyramid of energy b) Pyramid of biomass
    c) Pyramid of number d) Both a and b
15. Which of the following is an example for in situ conservation?
    a) Zoological parks b) Seed banks c) Sacred grooves d) Botanical gardens

II. Fill in the blanks by choosing the appropriate word/words from those given below: 5 x 1 = 5
(Saltation amensalism, microsporogenesis Glomus coming Commensalism first trimester)

16. The process of formation of microspots from a pollen mother cell through meiosis is called _____
17. MTPs are considered relatively safe during the _____
18. A single step large mutation is called_____
19. Many members of the genus_____ form mycorrhiza.
20. In _____one species is harmed and other is unaffected.

PART - B

III. Answer any five of the following questions in 3 - 5 sentences wherever applicable: 5 x 2 = 10

21. Which assisted reproductive technology can be used to correct infertility when a man has low sperm count in the ejaculation? Write about the technique.
22. Write any four applications which developed later in sexually transmitted infections.
23. Define the term linkage and recombination.
24. Mention any two examples of analogous organs in animals.

25. Name any two types of carcinogens. Give an example for each type.
26. name the microbe responsible for the production of
    a) Large holes in Swiss cheese b) Acetic acid
27. Write the convention for naming restriction endonucleases scientifically with an example.
28. Define productivity mention the types of primary productivity.

PART - C

IV. Answer any five of the following questions in 40 - 80 words each, wherever applicable: 5 x 3 = 15

29. Draw a neat labelled diagram of a mature embryo sac
30. Define the following terms: a) Parturition b) Lactation c) Colostrum
31. Write the steps involved in DNA fingerprinting.
32. Draw a labelled diagram of Miller's experiment.
33. Mention three applications of PCR as a tool for molecular diagnosis.
34. Write any three uses of transmit animals with an example for each.
35. Write any three steps of decomposition process.
36. Give three examples for alien species invasion.

PART - D

Section -1

V. Answer any four of the following questions in about 200 - 250 words each, wherever applicable: 4 x 5 = 20

37. Draw a need labelled diagram of the sectional view of human female reproductive system.
38. Explain one gene inheritance in *Psium sativum* with an example.
39. Write any five characteristic features of genetic code.
40. Describe Griffith's experiment to show transformation in bacteria.
41. What are lymphoid organs mention their types with an example for each.
42. a) Draw a labelled diagram of biogas plant. 3
    b) What are flocs? Write their significance in sewage treatment. 2
43. Give reason for the following statements:
    a) Alien DNA is linked with the Ori site of a vector in a gene cloning.
    b) Restriction enzymes are called molecular scissors.
    c) DNA ligase can be called molecular glue.
    d) Gelelectrophoresis is considered as a very important technique in recombinant DNA technology.
    e) DNA fragments move towards anode under electric field through a medioum in gelelectrophoresis.

44. a) Write the three defence mechanisms in prey species to lessen the impact of predation. 3

    b) Mention the special adaptations in parasites. 2

Section - II

VI. Answer any one of the following questions in about 200 - 250 words each, wherever applicable: **1 x 5 = 5**

45. Continued self pollination results in inbreeding depression. Plants avoid this problem by developing certain devices to discourage self pollination and to encourage cross pollination. Name and explain the concept.

46. A person has severe sneezing, watery eyes, running nose and difficulty in breathing. Name two external agents which could be causing these symptoms and the three drugs which can reduce them.

47. Give an example for the patterns of inheritance mentioned in the cases below:

    a) A single straight is controlled by more than one gene.

    b) The F_1 progeny resembles both the parents.

    c) A single gene exhibits multiple phenotypic expressions.

    d) More than two alleles control the expression of a trait.

    e) More number of mails are affected than females

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 3 (2024)

SUBJECT – BIOLOGY (36)

General Instructions:

1. This question paper consists of four parts A, B, C and D.
2. Part A consists of I & II and Part D consists V and VI
3. For Part A questions, only the first written answers will be considered for evaluation.
4. All the parts are Compulsory.
5. Draw diagrams wherever necessary. Unlabelled diagrams or illustrations do not attract any marks.

PART - A

I. Select the correct alternative from the choices given below: 15 x 1 = 15

1. Which is not a characteristic feature of wind-pollinated flowers?
   a) Feather stigma
   b) Presence of single ovule in ovary
   c) Pollen grains are non-sticky
   d) Flowers are very colourful
2. Gonadotropin-releasing hormone(GnRH) is a hypothalamic hormone which stimulates the pituitary gland to secrete_____
   a) LH and FSH b) FSH and Oxytocin
   c) LH and Oxytocin d) Progesterone
3. Which one of the following is a suitable method of ART advised to a woman if she cannot produce ovum but cannot provide environment for fertilization?
   a) IUD b) GIFT c) ICSI d) ZIFT
4. Which unusual nucleotide is added to the 5' end of hnRNA during capping in eukaryotic transcription?
   a) Methyl guanosine triphosphate b) Adenylate residues
   c) Rho factor d) Sigma factor
5. The pouched mammals of Australia survived due to the following reason:
   a) Continental drift b) Lack of competition from any other mammals
   c) Both a and b d) Convergent evolution
6. With reference to acquired immunity, choose the correct option:
   a) It is pathogen-specific and memory-based b) Produces interferons
   c) Only primary response d) Only secondary response

7. *Propionibacterium shermanii* is used in the production of_____
   a) Roquefort cheese b) Swiss cheese c) Bamboo shoots d) Ferment fish
8. *Penicillium notatum* is a_____
   a) Bacterium b) Virus c) Blue-Green Alga (BGA) d) Mould
9. The following one is used to visualize the separated DNA fragments during gel electrophoresis.
   a) Ampicillin b) Chilled ethanol c) Ethidium bromide d) Enzymes
10. Identify the type of population interaction in the following example:On the rocky sea coasts of Scotland, the larger and superior barnacle *Balanus* dominates the intertidal area and excludes the smaller barnacle *Chthamalus* from that zone.
    a) Predation b) Mutualism c) Commensalism d) Competition
11. Which one of the following basic process will contribute to increase population density?
    a) Natality and Immigration b) Mortality and Emigration
    c) Immigration and Mortality d) Natality and Mortality
12. Each trophic level has a certain mass of living material at a time called as_____
    a) Primary production b) Secondary production
    c) Standing crop d) GPP
13. Which nitrogenous base is present in both DNA and RNA?
    a) Guanine b) Thymine c) Uracil d) Adenine
14. When compared to the last 5 episodes of mass extinction of species, present “Sixth Extinction” episode is different in
    a) Current species extinction rate is to be 100 to 1000 times faster
    b) Human activities c) Both A and B d) Diversification of life
15. Meristem culture in vitro is done to obtain:
    a) Virus-free plants b) Hybrid plants c) Golden rice d) Virus-infected plants

II. Fill in the blanks by choosing the appropriate word/words from those given below: 5 x 1 = 5

16. _____is an example for pleiotropy.
17. “Big Bang theory” explain about_____
18. Pyramid of energy is always_____
19. Rivet popper hypothesis was proposed by_____
20. Gynecomastia is a seen in _____chromosomal disorder.

PART - B

III. Answer any five of the following questions in 3 - 5 sentences wherever applicable: 5 x 2 = 10

21. What is perisperm? Give an example.

22. Define the following: a) Menarche b) Colustrum
23. What are the major tasks of reproductive and child healthcare (RCH) programme?
24. Morgan worked with a tiny flies *Drosophila melanogaster* which were found very suitable to his experiments. Justify with two reasons.
25. Name the scientist who developed the following method or technique
    a) Determination sequences in protein DNA fingerprinting 1
    b) Differentiate between active and passive immunity 1
26. Write the name of a diagnostic test for a) AIDS b) typhoid
27. Differentiate between active and passive immunity.
28. Write a short note on downstream processing.

PART - C

IV. Answer any five of the following questions in 40 - 80 words each, wherever applicable: 5 x 3 = 15

29. Draw a neat, labelled diagram of a mature embryo sac.
30. What is the placenta? Mention any four hormones of placenta.
31. Explain the different natural methods of contraception.
32. Explain the mechanism of sex determination in honeybee.
33. a) Organic farming - the use of biofertilizers encouraged in agriculture to reduce the chemical fertilizers. Why? Give reason. 1
    b) Write the role of mycorrhiza as biofertilizer? 2
34. Name the microbes that help in the production of the following products commercially:
    a) Cyclosporin A 1
    b) Citric acid 1
    c) Statins 1
35. What are the factors affecting primary productivity?
36. When alien species are introduced, they become invasive and cause decline or extinction of indigenous species. Substantiate with three examples.

PART - D

V. Answer any four of the following questions in about 200 - 250 words each, wherever applicable: 4 x 5 = 20

37. Explain the process of double fertilization and triple fusion in angiosperms
38. Draw a neat labelled diagrammatic sectional view of memory gland
39. Write a schematic representation of inheritance of one gene in pea plant

40. a) What is Pedigree analysis? How such pedigree analysis provides a strong tool in human genetics? 3

b) Mention any four symptoms of Down's syndrome. 2

41. Explain the lac operon.

42. List out the silent features of human genome.

43. a) What are the two key concepts of Darwinian Theory of evolution? 2

b) Write the diagrammatic representation of Miller's experiment with labelling. 3

44. a) Briefly explain any three prevention and control measures of drug/alcohol abuse. 3

b) What are lymphoid organs? Give one example for primary lymphoid organs. 2

VI. Answer any one of the following questions in about 200 - 250 words each, wherever applicable: 1 x 5 = 5

45. Write a labelled diagrammatic representation of recombinant DNA technology.

46. What is RNA interference? How does this help in protecting the host plant from nematodal infestation?

47. Assign the plant for the following:

a) Plant which produce highly poisonous glycosides as defence against grazers. 1

b) Plant which employs sexual deceit to get pollination done by a special bee. 1

c) A parasitic plant that is growing on a hedge plant. 1

d) Invasive cactus introduced into Australia in the early 1920. 1

e) Tree has a mutual relationship with wasp and wasp helps in pollination during searching simple for suitable egg lying sites. 1

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 1 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. This question paper contains five Parts – A, B, C, D and E.
2. PART–A consists of I and II and Part–D consists of V and VI.
3. All Parts are compulsory.
4. For PART-A questions, only the first written answers will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.
6. PART-E consists of questions for Visually Challenged students only.

PART – A

I. Select the correct alternative from the choices given below: 15 x 1 = 15

1. Pollen grains are well preserved as fossils" because of the presence of sporopollenin" as

   Statement I: Sporopollenin is the most resistant organic material known.

   Statement II: No enzyme that degrades sporopollenin is so far known.

   a) Both the Statements I and II are correct

   b) Statement I is correct and Statement II is incorrect

   c) Both the Statements I and II are incorrect

   d) Statement I is incorrect and Statement II is correct

2. The process of transformation of spermatids into sperms is called

   a) Spermatogenesis b) Spermiogenesis

   c) Spermiation d) Oogenesis

3. Which one among the following is essential for the maintenance of endometrium and for the implantation of fertilized ovum?

   a) Estrogen b) Progesteron c) LH and FSH d) Androgens

4. Select the one among the given that is a probable reason for explosive growth of population in India.

   a) A rapid increase in MMR

   b) A rapid increase in IMR

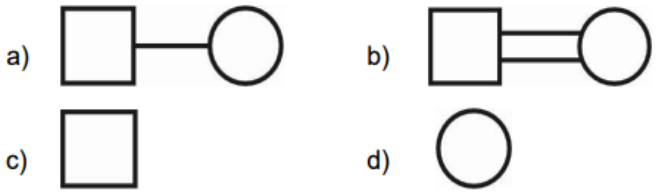
   c) Increase in the number of people in reproducible age

   d) A rapid increase in death rate

5. Which one among the following is a contraceptive method that involves no medicine or device?

   a) Condom b) Saheli c) Cervical caps d) Coitus interrupts

6. Which one among the following ART can be used to rectify if the infertility is due to low sperm count?
   a) GIFT b) IUT c) ZIFT d) IUI 7)
7. Which one among the given is a symbol used in human pedigree analysis for consanguineous mating?



8. Which one among these represents the flow of genetic information?
   a) mRNA → DNA → Protein
   b) Protein → mRNA → DNA
   c) DNA → mRNA → Protein
   d) DNA → Protein → mRNA
9. If the change in gene frequency occurs by chance, it is called
   a) Gene flow b) Mutation c) Genetic recombination d) Genetic drift
10. Which is the test that can be suggested among the given for a patient with symptoms such as - sustained high fever (39°C – 40°C), weakness, stomach pain, constipation, headache and loss of appetite?
    a) Widal test b) ELISA c) CT d) MRI
11. Which among the following is an enzyme modified by genetic engineering and used as “Clot buster”?
    a) Lipase b) Pectinase c) Protease d) Streptokinase
12. In which method the plant cells are bombarded with high velocity microparticles of gold coated with DNA?
    a) Microinjection b) Biolistics c) Heat shock d) Cloning
13. The use of bioresources by multinational companies and other organisations without proper authorisation and compensatory payment is called as
    a) Bioprospecting b) Biofortification c) Biopiracy d) Bioprocessing
14. In which of the processes given, the humus is degraded by some microbes and release of inorganic nutrients occurs?
    a) Leaching b) Mineralisation c) Catabolism d) Humification
15. Which one among these is an example for Ex-situ conservation?
    a) Cryopreservation b) Biosphere reserve
    c) National Park d) Sanctuaries

II. Fill in the blanks by choosing the appropriate word/Words from those given below: 5 x 1 = 5

(*Trichophyton*, Apomixis, Saltation, Oxytocin, Elution, *Glomus*)

16. Few flowering plants have evolved a special mechanism to produce seeds without fertilization called _____.
17. Single step large mutation is known as _____.
18. _____ acts on uterine muscle and cause stronger contractions during parturition.
19. Members of genus _____ form mycorrhiza and absorb phosphorous from soil and supply it to the plant.
20. The step in which bands of DNA are cutout and extracted from gel piece is termed as _____.

PART – B

III. Answer any five of the following questions in 3 to 5 sentences wherever applicable: 5 × 2 = 10

21. Draw a neat labelled diagram of dicot embryo.
22. Differentiate between Linkage and Recombination.
23. "AUG has dual function". Justify the statement.
24. Analyse the table given in relation to the human evolution and write the scientific name for A and B respectively.

Features of Evolutionary man	Scientific Name
First human like being the hominid with brain capacity between 650 cc – 800 cc	A
Large brain capacity around 900 cc and probably ate meat	B

25. Which drug is very useful in patients who have undergone surgery? And why?
26. From which part of, "Virus infected plant" healthy plants can be recovered? Name the technique involved.
27. "Tropical regions have greater biological diversity". Give two hypothesis proposed by ecologists in favour of the above given statement.

PART – C

IV. Answer any five of the following questions in about 40-80 words each, wherever applicable: 5 × 3 = 15

28. List the features required for the flowers pollinated by wind.
29. Name the human male accessory glands.
30. How do IUDs function in bringing about contraception effectively?

31. a) Mention any two examples for evolution by anthropogenic action. 2

    b) State Hardy-Weinberg principle. 1

32. What is a carcinogen? Give an example each for Physical and Biological carcinogens.

33. List the three uses of genetically modified plants.

34. In an ecosystem the number of primary producers is 4 and primary consumers are 21. Construct an ecological pyramid for this ecosystem.

PART – D

V. Answer any four of the following questions in about 200-250 words each, wherever applicable:

4 × 5 = 20

35. Draw a neat labelled diagrammatic sectional view of mammary gland.

36. a) How phenylketonuria is an example for Pleiotropy? 2

    b) Schematically represent the sex determination in honeybee. 3

37. Explain incomplete dominance with an example.

38. Write any five salient features of human genome.

39. a) Name the biocontrol agents of aphids and mosquitoes respectively. 2

    b) How Bacillus thuringiensis can be used as biocontrol agent? 3

40. Diagrammatically represent the recombinant DNA technology.

41. Give suitable terms for the following examples/statements with respect to population interactions.

    a) The method Mediterranean orchid ophrys employs to get pollinated.

    b) Species facing competition might evolve mechanism that promote co-existence rather than exclusion.

    c) The parasitic bird lays eggs in the nest of its host and lets the host bird incubate them.

    d) The interaction between cattle egret and grazing cattle.

    e) The interaction in which one species is harmed while, the other is unaffected.

VI. Answer any one of the following questions in about 200-250 words, wherever applicable:

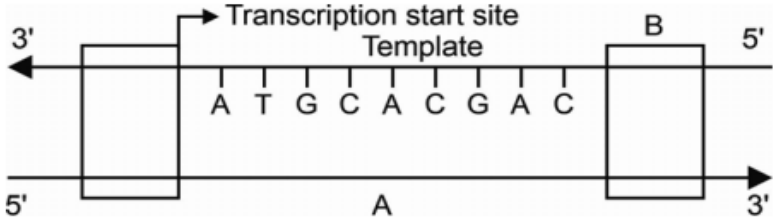
1 × 5 = 5

42. Read the below given statement and answer the given questions. “A breeder is interested in crossing different species of plants to combine desirable characters to produce commercially superior varieties”.

    a) What major approach in “Crop improvement programme”. Would you suggest to the breeder to achieve this goal? 1

    b) Discuss in detail the techniques involved in the process. 4

43. Analyse the below given transcription Unit diagram and answer the given questions.



a) Name the Parts A and B in the diagram. 2

b) Which site of transcription unit provides the binding site for RNA polymerase? 1

c) Write the sequence of bases of the mRNA transcribed from the template given in the diagram. 2

44. "Our body is provided with different types of non-specific immunity barriers at the time of birth". Justify the above statement with suitable example

PART – E

(For Visually Challenged Students only)

7. Failure of segregation of chromatids during cell division cycle results in _____.

a. Haploidy b) Diploidy c) Aneuploidy d) Polyploidy

43. List the salient features of genetic code.

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 2 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. The question paper consists five PARTS – A, B, C, D and E.
2. PART – A consists of I & II and PART – D consists of V and VI.
3. All the parts are compulsory.
4. For PART – A questions, only the first written answers will be considered for evaluation.
5. PART – E consists of questions for Visually-Challenged Students only

PART – A

I. Select the correct alternative from the choices given below: **15 × 1 = 15**

1. Statement I: Apomixis is the production of seeds from unfertilized ovule.

   Statement II: Embryo produced from apomictic seeds are not generally identical to parental plants.

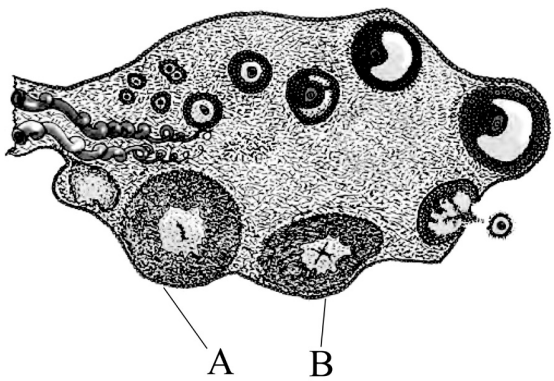
   a) Statement I is true and statement II is false

   b) Statement I is false and statement II is true

   c) Both Statement I and statement II are true

   d) Both Statement I and statement II are false

2. The figure shows a section of human ovary. Select the option which gives correct identification of



   either 'A' or 'B' with function or characteristics.

   a) A – Tertiary follicle – Forms Graffian follicle

   b) B – Corpus luteum – Secretes progesterone

   c) B – Corpus luteum – Secretes estrogen

   d) A – Primary Oocyte – It is prophase I of mitotic division

3. Which of the following hormone is produced only during pregnancy?

   a) L.H b) hCG c) Estrogen d) F.S.H

4. Oral contraceptive mainly contains
   a) Estrogen and L.H. b) Progesterone and Estrogen
   c) Progesterone and L.H d) Progesterone, Estrogen and L.H
5. In a plant, red fruit (R) is dominant over yellow fruit (r). Tallness (T) is dominant over dwarfness (t). If plant RRTt genotype crossed with plant that is rrtt
   a) 25% will be tall with red fruit
   b) 75% will be tall with red fruit
   c) 50% will be tall with red fruit
   d) All the offspring will be tall with red fruit
6. A couple has three daughters. What is the possibility of them having a son next time?
   a) 0% b) 25% c) 50% d) 100%
7. Which of the following characters was not chosen by Mendel?
   a) Pod shape b) Pod colour c) Position of flower d) Position of pod
8. RNA polymerase I transcribes
   a) mRNA b) tRNA c) rRNA d) hnRNA
9. Single step large mutation leading to speciation is called
   a) Natural selection b) Branching descent c) Founder effect d) Saltation
10. Haemozoin is a
    a) Precursor of Haemoglobin
    b) Toxin released from streptococcus-infected cell
    c) Toxin released from plasmodium-infected cell
    d) Toxin released from Haemophilus-infected cell
11. Secondary treatment is mainly
    a) Physical process b) Mechanical process
    c) Biological process d) Chemical process
12. Main Purpose of stirred tank bioreactor is
    a) Purification of the product
    b) Providing anaerobic condition for the culture
    c) Addition of proper mixing preservative
    d) Availability of oxygen throughout the process
13. The capacity to generate a whole plant from any cells or explant is called as
    a) Pluripotent b) Micropropagation c) Somatic hybrids d) Totipotency

14. Identify the decomposer among the following in pond ecosystem.

A. Bacteria B. Fungi C. Flagellates D. Phytoplankton

a) A only b) A & B only c) A & D only d) A, B & C only

15. The most species-rich taxonomic group among invertebrates.

a) Birds b) Insects c) Mammals d) Fishes

II. Fill in the blanks by choosing the appropriate word/words from those given below: 5 × 1 = 5

(Chemical, Mycorhizae, L.H, Endemic, sacred, Recombinant protein)

16. Ovulation is induced by hormone _____

17. Inorganic chemicals are formed by organic molecules which lead to first form of life, this support _____ evolution.

18. The symbiotic association fungi with roots of higher plants _____

19. If any protein encoding gene is expressed in a heterologous host, it is called as _____

20. The species confined to a particular region and not found elsewhere is called as _____

PART – B

III. Answer any five of the following questions in 3–5 sentences each, wherever applicable: 5 × 2 = 10

21. Why apple is called a false fruit? Which part of the flower forms this fruit?

22. List any two preventive methods of Sexually Transmitted Infections (STIs)

23. Differentiate between exons and introns.

24. Comment the similarities and dissimilarities between thorns of Bougainvillea and tendrils of cucurbita with respect to evolution

25. When ELISA test was conducted on an immune-suppressed person, he tested positive for pathogens.

a) Which disease the patient is suffering from? 1

b) Name the causative entity 1

26. How primary treatment is carried out in STP?

27. Mention the 'Evil quartet' which causes biodiversity losses.

PART – C

IV. Answer any five of the following questions in about 40–80 words each wherever applicable:

5 × 3 = 15

28. Draw a neat labelled diagram of embryosac.

29. What is placenta? List the functions of placenta.

30. Categorise the following into different kinds of IUD such as non-medicated Copper releasing and Hormone releasing ones.

a) Lippes loop (b) Cu-T c) Progestosert d) LNG.20 e) Cu-7 f) Multiload-375

31. Darwin's finches of Galapagos Island represent the best example for adaptive radiation. Substantiate.
32. Write the scientific name of microorganism involved in the production of

    a) Penicillin b) Cyclosporin A c) Streptokinase
33. Mention the carcinogens with one example each.
34. Diagrammatically represent pyramid of energy in a grassland ecosystem.

PART – D

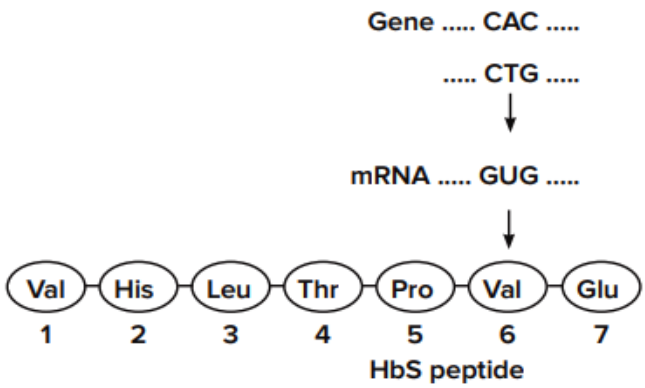
V. Answer any four of the following questions in about 200–250 words each wherever applicable:

4 × 5 = 20

35. Flowering plants have developed many devices to discourage self-pollination and encourage cross-pollination. Justify.
36. Mention the five differences between spermatogenesis and oogenesis.
37. The F_2 progeny of monohybrid cross showed phenotypic ratio 1: 2: 1 unlike that of Mendel's, monohybrid F_2 ratio with the help of suitable example. Work out cross and explain how it is possible?
38. Write a note on prevention & control measures of drug/alcohol abuse.
39. Explain how gene therapy is used to correct adenosine deaminase deficiency disorder.
40. What is mutualism? Mention four examples.
41. Explain how elongated DNA of eukaryotic organism fitted into small-sized nucleus.

VI. Answer any one of the following questions in 200–250 words each. **1 × 5 = 5**

42. Given below is a representation of amino acid composition of relevant translated portion of beta chain of hemoglobin related to shape of human red blood cells.



a) Is this representation indicates a normal or genetically affected person? Give reason in support to your answer. 2

b) Mention the phenotypic difference in a normal and affected person with respect to this gene. 2

c) Who are likely to suffer more from the defect related to this gene among the males, the females or both equally? Why? 1

43. In context to lac-operon, when Jacob and Monad cultured E.coli cells in the presence of lactose medium, E.coli cells metabolized lactose by synthesizing enzymes. Explain how lac-operon switches OFF and switch ON.
44. Describe the characteristics that of plasmid should possess to be as cloning vector in genetic engineering.

PART – E

(For Visually-Challenged Students Only)

2. The human ovary secretes different hormones. Identify correct option with their functions. 1

a) Tertiary follicle – Forms Graffian follicle

b) Corpus luteum – Secretes progesterone hormone

c) Corpus luteum – Secretes estrogen

d) Primary oocyte – It is prophase I of mitotic division

42. Give a brief account on sickle cell anemia. 5

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU QUESTION PAPER EXAM - 3 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. The question paper consists of four Parts A, B, C and D.
2. PART – A consists of I & II and PART – D consists of Section I and Section II.
3. All the parts are compulsory.
4. Draw diagrams wherever necessary, unlabelled diagrams or illustrations do not attract any marks.
5. For PART – A questions, only the first written answers will be considered for valuation.

PART – A

I. Select the correct alternative from the choices given below: **15 × 1 = 15**

1. Statement I: Pollen grains of many species cause chronic respiratory disorders.
   Statement II: Generative cell is bigger than vegetative cell and has abundant food reserve. Choose the correct answer from the options given below.
   a) Both statements I and statement II are correct
   b) Statement I is correct and Statement II is wrong
   c) Statement I is wrong and Statement II is correct
   d) Both Statement I and Statement II are wrong
2. Name the process by which spermatids are transformed into spermatozoa.
   a) Spermiation b) Spermatogenesis c) Spermiogenesis d) Ovulation
3. Oral contraceptive pills mainly contain
   a) Progestogen and LH b) Progestogen, Estrogen and LH
   c) Progestogen, HCG and Estrogen d) Progestogen and Estrogen
4. XO type of sex determination is observed in
   a) Birds (b) Grasshopper (c) Drosophila (d) Honey bees
5. If a double stranded DNA has 30 percent Thymine, then calculate the percentage of Guanine in the DNA.
   a) 20% b) 30% c) 60% d) 70%
6. Identify the following in the order of evolution.
   a) Neanderthal → Homo habilis → Homo erectus → Homo sapiens
   b) Homo habilis → Neanderthal → Homo erectus → Homo sapiens
   c) Homo habilis → Homo erectus → Neanderthal → Homo sapiens
   d) Neanderthal → Homo erectus → Homo habilis → Homo sapiens

7. Which of the following cells actively participate during allergy?
   a) RBC b) B – Lymphocytes c) T – Lymphocytes d) Mast cells
8. Identify the disease when lips and finger nails may turn gray to bluish in colour in severe cases.
   a) Malaria b) Pneumonia c) Hepatitis – B d) AIDS
9. The nutritional quality of curd is improved by which vitamin?
   a) B12 b) B2 c) B6 d) B1
10. The majority of Baculoviruses used as biological control agents are rich in the genus
    a) Nucleopoly hedro virus b) Retro virus c) DNA virus d) Bacterio phage
11. Restriction sequence of ECO–RI is
    a) 5' – C T T A A G – 3' b) 5' – G A A T T C – 3'
    c) 3' – G A A T T C – 5' d) 3' – C T T A A G – 5'
12. Which growth curve is obtained when the resources availability in a habitat is unlimited?
    a) Exponential b) Logistic c) Sigmoid d) Linear
13. With regard to foetus in human beings, some of the following events take place during the development.
    i) Develops limbs and digits
    ii) Major organ systems are formed
    iii) Eye lids separate and eyelashes are formed.
    iv) Heart is formed
    Choose the correct option among the following which shows the correct sequence of development:
    a) i, ii, iii & iv b) i, iii, iv & ii c) ii, iii, iv & i d) iv, i, ii & iii
14. The following are some of the in-situ conservation methods, except
    a) National park b) Wildlife sanctuaries
    c) Biosphere reserves d) Botanical gardens
15. The historic convention "The Earth Summit" held in Rio de Janeiro in 1992 to discuss on
    a) Global warming b) Sustainable developments
    c) Biological diversity d) Ban on plastic

II. Fill in the blanks by choosing the appropriate word/words from those given below: 5 × 1 = 5

(Chitin, Chitinase, *Aedes*, Lobefins, Nematode, *Anopheles*)

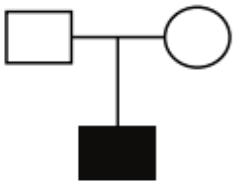
16. The animals called _____ evolved into the first amphibians that lived on both land and water.
17. Dengue and chickengunya diseases are spread through _____ vector.
18. The enzyme used to break open to release DNA in fungal cells is _____.
19. Meloidogyne incognitia is a _____.
20. The rate of decomposition is slower if detritus is rich in _____.

PART – B

III. Answer any FIVE of the following questions in 3–5 sentences each wherever applicable:

5 × 2 = 10

21. What is Perisperm? Give an example of perispermic seed.
22. Mention two uncurable sexually transmitted diseases.
23. Differentiate Exon and Intron.
24. What are fossils? Name the method by which ages of fossils are calculated.
25. Why penicillin and statins drag more attention in the field of medicine?
26. Name any two biodiversity hotspots of India.
27. What is pedigree analysis? Identify the following symbol representing.



PART – C

IV. Answer any FIVE of the following questions in about 40–80 words each wherever applicable:

5 × 3 = 15

28. Draw a neat labelled diagram of typical Anatrophous ovule.
29. What is placenta? List the functions of Human Placenta.
30. What is infertility? How IVF method helps to overcome infertility? Explain.
31. (a) Justify the following statements: 2

    i) Flippers of Penguins and Dolphins are the results of convergent evolution.

    ii) Thorns of Bougainvillea and tendrils of Cucurbita are the result of divergent evolution.

    b) State Hardy-Weinburg principle. 1
32. Name the Plants A, B and C from which the following drugs are extracted.

Drugs	Plants
Smack	A
Crack	B
Marijuana	C

33. Explain briefly biological treatment of sewage water.
34. Define the following terms:
35. a) Standing Crop b) Stratification c) Trophic Level

PART – D

V. Answer any FOUR of the following questions in about 200–250 words each wherever applicable:

4 × 5 = 20

36. a) Some angiospermic plants pollinate only with the help of wind, why? 3

    b) Why apple is called false fruit? Which part of the apple flower forms the fruit? 2

37. Draw a neat labelled diagram of sectional view of mammary gland.

38. Explain Mendel's law of dominance with an example and draw a conclusion.

39. List the goals of Human Genome Project.

40. Schematically represent the life cycle of retrovirus in human body with labels.

41. Briefly explain the steps of recombinant DNA technology.

42. (a) Define Aneuploidy. Give two examples of aneuploidy. 3

    b) Write a note on Phenylketonuria. 2

Section - II

VI. Answer any ONE of the following questions in about 200–250 words wherever applicable: 1 × 5 = 5

43. In the context to Lac operon, when Jacob and Monod cultured E. Coli in a media devoid of Lactose, *E. coli* cells formed the colonies and when Lactose was added, the *E. coli* cells metabolized lactose by synthesizing enzymes. Explain how Lac operon switches off and switches on.

44. a) With respect to tissue culture, define the following terms: 3

    i) Micropropagation ii) Somaclones iii) Somatic hybridization

    b) What is Biopiracy? 1

    c) Expand GEAC with regard to Biotechnology research. 1

45. Name the type of species interactions seen in each of the following example:

    a) Marine fish and Copepods

    b) *Balanus* and *Chathamalus*

    c) Fig tree and wasp

    d) The cattle egrets and grazing cattle

    e) *Ophrys* orchid and Honey bees _____

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU MODEL QUESTION PAPER - 1 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. This question paper consists of four parts A, B, C D and E.
2. Part A consists of I & II and Part D consists of V and VI
3. All the parts are compulsory.
4. The answers for Part A, written in the first two pages of the answer booklet are only considered for evaluation.
5. Part – E consists of questions for visually challenged students only

PART – A

I. Select the correct alternative from the choices given below: **15 x 1 = 15**

1. **Statement I:** Apomixis is the production of seeds from unfertilized ovules

   Statement II: Embryos produced from apomictic seeds are not generally identical to the parent plant.

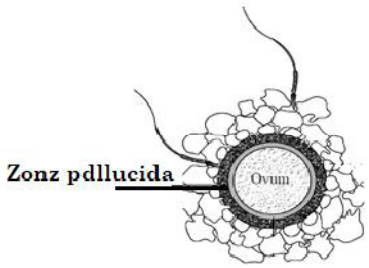
   a) Statement I is true, statement II is false

   b) Statement I is false, statement II is true

   c) Statement I and statement II both are true

   d) Statement I and statement II both are false

2. **Choose the correct option from the table given below for the formation and dissolution of the labelled part in the given diagram.**



Formed by	Dissolved
a) Primary oocyte	Before fertilisation
b) Primary oocyte	After fertilisation
c) Secondary oocyte	Before fertilisation
d) Secondary oocyte	After fertilisation

3. Out of the options given below choose the correct stage for transfer to the fallopian tube for successful IVF results.

   a) Embryo up to 8 blastomeres b) embryo up to 16 blastomeres

   c) Embryo up to 32 blastomeres d) Embryo up to 32 blastomeres

4. 37.2% recombinant *Drosophila* progeny obtained in the T. H. Morgan's dihybrid cross experiment with the phenotypes red eye colour, normal body and white eye colour, miniature body is due to

   a) Loosely linked and shorter distance between genes

   b) Tightly linked and shorter distance between genes

c) Loosely linked and longer distance between genes

d) Tightly linked and longer distance between genes

5. The number of nucleotide pairs present in the DNA of the primary oocyte of a new born in human

a) 3.3×10^9 b) 6.6×10^9 c) 13.2×10^9 d) 3.3×10^9

6. The factors that affect Hardy – Weinberg equilibrium are listed below:

i) Crossing over, Independent assortment ii) Crossing over, Mutation,

iii) Genetic drift, Crossing over iv) Independent assortment, Mutation

Choose the correct options:

a) i, ii and iii b) ii, iii and iv c) i, iii and iv d) i, ii, iii and iv

7. Interferons are most effective in making non-infected cells resistant against the spread of which of the following diseases in humans?

a) Cancer b) Ascariasis c) Ringworm d) Amoebiasis

8. The human host cells in which the gametocytes of malarial parasite develop are

a) Thrombocytes b) Liver cells c) Erythrocytes d) Leucocytes

9. Which of the following water samples in the table given below will have a higher concentration of organic matter?

Water Sample	Level of pollution	Value of BOD
a)	High	High
b)	Low	Low
c)	Low	High
d)	High	Low

10. The steps of Recombinant DNA technology are given below:

i) Insertion of recombinant DNA into the host organism

ii) Amplification of gene of interest using PCR

iii) Cutting of DNA at specific locations

iv) Obtaining the foreign product

v) Downstream processing

vi) Isolation of DNA

Choose the correct option for the sequential steps of Recombinant DNA technology.

a) vi, ii, iii, iv, v, i b) vi, i, iii, iv, v, ii c) vi, ii, iii, iv, v, i d) vi, iii ii, i, iv, v

11. DNA in a clone of cells followed by detection using autoradiography is called

a) Template b) Probe c) Transcript d) Cistron

12. Jeeva was growing a bacterial colony in a culture flask under ideal laboratory conditions where the resources sooner or later become limiting. Which of the following equations will represents the correct growth in this case?
    a) dN/dt = rN
    b) dN/dt = KN
    c) dN/dt = rN (K – N/K)
    d) dN/dt = rN (K +N/K)
13. Which of the following food chains is the major conduit for the energy flow in terrestrial and aquatic ecosystems respectively?

	Terrestrial	Aquatic
a)	Grazing	Grazing
b)	Detritus	Detritus
c)	Detritus	Grazing
d)	Grazing	Detritus

14. Which of the following is not an example of *in-situ* conservation?
    a) National park and seed bank b) National park and Zoological parks
    c) Seed bank and sacred groove d) Seed bank and Botanical gardens
15. Exploration of molecular, genetic and species level diversity for novel products of economic importance is
    a) Biofortification b) Bioprocessing c) Bioprospecting d) Biodiversity

II. Fill in the blanks by choosing the appropriate word/Words from those given below: 5 x 1 = 5

(Competent cells, Bacteria, non-living molecule, vectors, competent cells, Recombinant cells)

16. The interstitial space in seminiferous tubules consists of immunologically ----------
17. The version of biogenesis is accepted by majority, as the first form of life arose slowly through evolutionary forces from ----------
18. Filariasis pathogens are transmitted to a healthy person through ----------
19. Swiss cheese with large holes is produced from ----------
20. The host cells which have the ability to incorporate foreign DNA within them are called ---------

PART - B

III. Answer any FIVE of the following questions in 3 – 5 sentences wherever applicable: 5 x 2 = 10

21. Complete the tabular column given below with respect to the male gametophyte of angiosperms

Cells of the male gametophyte	
Shape of nucleus of the cells	

22. Mention the two medical grounds on which the pregnancies are subjected to termination.
23. Derive the phenotypic and genotypic ratio of a cross between AB blood group parents.

24. Which sequences of bases transcribed from DNA are found both in hnRNA and mRNA?
25. "Potato tubers and Sweet potato tubers are the result of convergent evolution". Justify the statement.
26. What are biofertilizers? Mention its significance.
27. Name any four recent extinct organisms as per IUCN Red list.

PART - C

IV. Answer any FIVE of the following questions in 40 – 80 words each wherever applicable: 5 x 3 = 15

28. Draw a labeled diagram of the fertilised female gametophyte and mention the ploidy of any one of the products of double fertilization.
29. Parturition is induced by complex neuroendocrine mechanism. Comment.
30. The popular and effective contraceptives include IUDs. Mention the types of IUDs with an example of each.
31. Write the salient feature of the following human ancestors;
    i) *Dryopithecus* ii) *Ramapithecus* iii) *Australopithecus*
    iv) *Homo habilis* v) Neanderthal man vi) *Homo erectus*
32. Describe any three properties of Cancerous cells.
33. Complete the below given tabular column with appropriate answers.

Name of the Microbe	Name of the Product	Uses
A	Lactic acid	B
Methanogens	C	D
E	F	Treatment of bacterial diseases

34. Represent diagrammatically the pyramid of number in a Terrestrial ecosystem.

PART – D

V. Answer any FOUR of the following questions in about 200–250 words each wherever applicable: 4 x 5 = 20

35. Flowering plants have developed many devices to discourage self-pollination and encourage cross-pollination. Comment.
36. Draw a labeled diagrammatic sectional view of the human female reproductive system.
37. Few autosome linked recessive gene blood diseases occur in human population. Among them some are related to qualitative and quantitative problem of synthesizing blood proteins. Explain.
38. DNA replication is fast, accurate, energetically expensive, substrate and enzyme dependent, initiated from specific site and cannot uncoil on its entire length. Explain the process of DNA replication considering all these features.

39. Some drug bottles had their name labels missing in a drug store of a hospital.

The staffs need to identify the drugs with their actions still written on them. Analyse their actions listed below and identify the name of each drug and also the source of each one of them.

DRUG	**Effects**
Drug 1	Used by doctors as sedative and pain killer
Drug 2	Help patients to cope with insomnia and depression
Drug 3	Increases blood pressure and heart rate of consumer
Drug 4	Act as depressant and slows down body functions
Drug 5	Affects cardiovascular system of the body

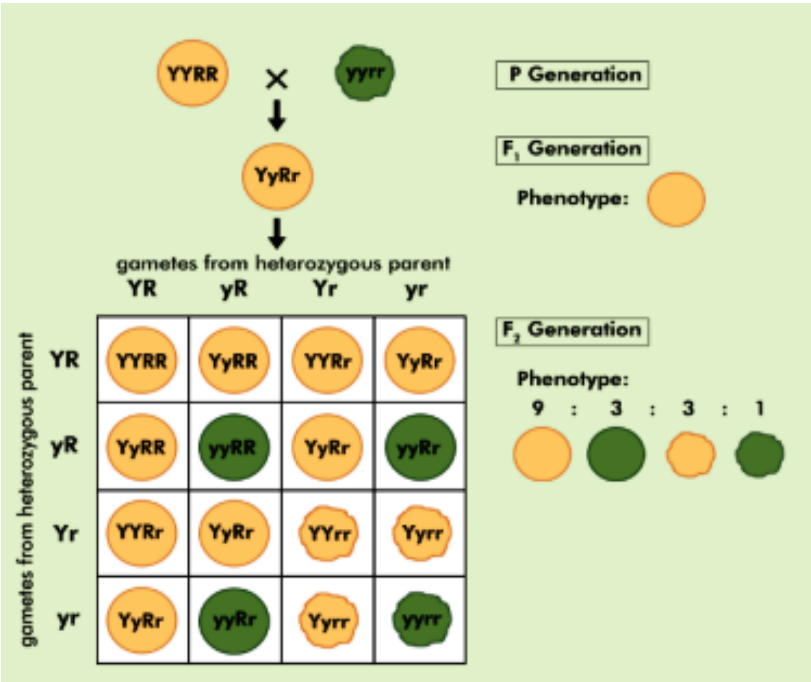
40. Transgenic animals provide innumerable benefits to human beings. Justify the statement with common reasons.

41. Explain the fascinating forms of interactions in;

a) Brood parasitism b) Sexual deceit 3 + 2

VI. Answer any ONE of the following questions in about 200–250 words each wherever applicable: 1 x 5 = 5

42. Results of a Mendelian dihybrid cross are represented in the form of Punnett square.



Answer the following questions with respect to the results of a dihybrid cross.

a) Number of parental types progeny b) Number of recombinant progeny

c) Number of homozygous recessive progeny d) Number of homozygous dominant progeny

e) Number of homozygous progeny for both the traits

f) Number of heterozygous progeny for both the traits

g) Number of pure line progeny h) Number of homozygous progeny for single trait

i) Number of heterozygous progeny for single trait j) Number of recessive progeny for single trait.

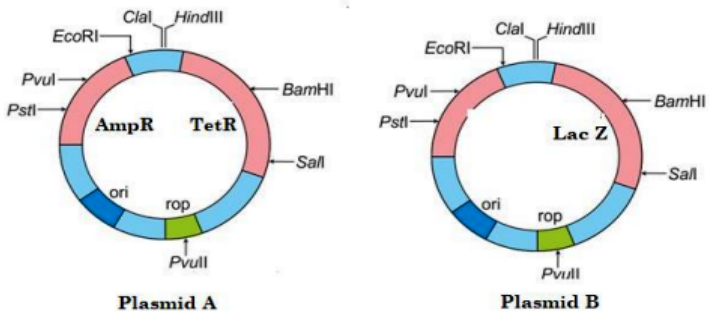
43. Given below are sequences of nucleotides in a particular mRNA and amino acids coded by it;

5'- AUG UUU UUC GAG UUA GUG UAA-3'

- met - phe - phe - glu - leu - val -

Write the properties of genetic code that can be correlated from the above given data

44. Given below are the diagrams of plasmids A and B, observe meticulously and answer the questions that follows:



a) Which plasmid is/are you select for cloning and why? 1

b) What is insertional inactivation? 1

c) Will the number of culture plating you should make is same or different to select recombinant if insertional inactivation is possible.Comment. 3

PART – E

(FOR VISUALLY CHALLENGED STUDENTS ONLY)

2. Choose the correct option for the formation and dissolution of the zona pellucida.

Formed by	Dissolved
a) Primary oocyte	before fertilisation
b) Primary oocyte	after fertilisation
c) Secondary oocyte	before fertilisation
d) Secondary oocyte	after fertilisation

42. Represent schematically the results of incomplete dominance in snapdragon plant taking the flower colour character.

44. a) Mention the tools of genetic engineering. 2

b) What is insertional inactivation? Mention its significance. 2

c) What is a recombinant protein? 1

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD MODEL

II PU MODEL QUESTION PAPER - 2 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. The question paper consists of parts -A, B, C, D and E.
2. Part-A consists of I & II and Part-D consists of V & VI.
3. All the parts are compulsory.
4. For part-A questions, only the first written answers will be considered for evaluation.
5. Part-E consists of questions for visually challenged students only

PART – A

I. Select the correct alternative from the choices given: **15 x 1 = 15**

1. Select the correct sequence of events in microsporogenesis
   a) Sporogenous tissue →Microspore mother cell →Microspore tetrad →Microspores
   b) Microspores →Microspore mother cell →Microspore tetrad → Sporogenous tissue
   c) Sporogenous tissue → Microspore tetrad →Microspores → Microspore mother cell
   d) Microspores → Sporogenous tissue →Microspore tetrad → Microspore mother cell
2. Statement I: The process of release of sperms from the seminiferous tubule is called spermatogenesis.
   Statement II: The spermatids are transformed into spermatozoa by the process called spermiogensis.
   Choose the correct answer from the options given below:
   a) Both statements I and statement II are correct
   b) Both statement I and statement II are incorrect
   c) Statement I is correct but statement II is incorrect
   d) Statement I is incorrect but statement II is correct
3. The function of myometrium layer present in the uterine wall is to
   a) Undergo cyclical changes during menstrual cycle.
   b) Exhibit strong uterine contraction during parturition.
   c) Give protection to the uterus.
   d) Help in the implantation process.
4. The…….hormone is secreted by the ovary in the later phase of pregnancy.
   Androgens b) Estrogens c) Relaxin d) Progestogens

5. Sperms produced by the seminiferous tubules are transported through accessory ducts. Whichduct should be tied and cut for male sterilization?
   a) Vas deferens b) Vasa efferentia c) Rete testis d) Epididymis
6. Reasons for human population explosion are given below:
   i) Rapid decline in maternal mortality rate.
   ii) Rapid decline in infant mortality rate.
   iii) Rapid increase in death rate.
   iv) Increase in the number of people in reproducible age.
   Select the correct answer statements from the options given below:
   a) i), ii) and iii) only b) i) and ii) only
   c) i), ii) and iv) only d) iii) and iv) only
7. In a dihybrid cross in pea plants, Mendel got 9:3:3:1 phenotypic ratio. It denotes that
   a) The alleles of two genes are interacting with each other.
   b) It is a polygenic inheritance.
   c) It is a multiple allelic inheritance.
   d) The alleles of two genes are segregating independently.
8. A DNA segment has a total of 1000 nucleotides, out of which 240 of them are adenine containing nucleotides. How many pyrimidines bases this DNA segement possesses?
   a) 480 b) 500 c) 760 d) 260
9. $(p + q)^2 = p^2 + 2pq + q^2$ represents an equation used in:
   a) Population genetics b) Mendelian genetics
   c) Molecular genetics d) Biometrics
10. A farmer working in a field was bitten by poisonous snake. Doctor gave him an antivenom treatment that contains preformed antibodies. This type of immunization is known as
    a) Autoimmunity b) Passive immunization
    c) Innate immunity d) Active immunization
11. An agriculture labour was spraying some powder mixed with water onto fruit trees to get rid of insect larvae. Which of the following biocontrol agent could have been used here?
    a) *Bacillus thuringiensis* b) *Trichoderma* c) Dragonflies d) Ladybird
12. Choose the correct sequence of polymerase chain reaction steps from the following:
    a) Annealing → Denaturation → Extension
    b) Extension → Annealing →Denaturation
    c) Denaturation → Extension → Annealing

d) Denaturation → Annealing → Extension

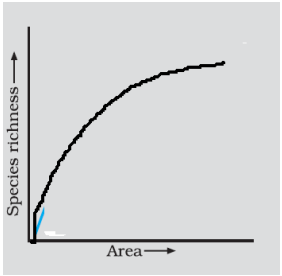
13. Use of bioresources by multinational companies and other organizations without properauthorisation and compensatory payment is referred as

    a) Biopiracy b) Biofortification c) Bioprospecting d) Bioprocessing

14. An example for *ex situ* conservation is

    a) National parks b) Sacred groves c) Biosphere reserves d) Zoological parks

15. The graph given below shows species-area relationships.



Which of the following equation correctly represent the curve?

a) $S = CA^Z$ b) Log S= log C + Z log A c) $A = CS^Z$ d) Log Z = log C + S log A

II. Fill in the blanks by choosing the appropriate word/words from those given in the bracket.

5 x 1 = 5

(Primary productivity, Coelacanth, Secondary productivity, *Glomus,* Amniocentesis, Plasmid)

16. Statutory ban on……..is required to check increasing female foeticides.
17. A fish thought to be extinct and caught in South Africa in 1938 is…….
18. An example for mycorrhiza forming fungi is……..
19. Autonomously replicating circular extra-chromosomal DNA of bacteria is known as…….
20. Rate of formation of new organic matter by consumers is referred as……

PART – B

III. Answer any <u>FIVE</u> of the following questions in 3-5 sentences each, wherever applicable:

5 x 2 = 10

21. Differentiate between geitonogamy and xenogamy.
22. Write a short on sex determination method in birds.
23. Mention the levels where gene expression can be regulated in eukaryotes.
24. Write a short note on Neanderthal man.
25. Draw the structure of an antibody molecule.
26. Write the functions of genes *cryIAb* and *cryIIAb.*
27. Briefly explain the significance of David Tilman's long-term ecosystem experiments using outdoor plots.

PART – C

III. Answer any <u>FIVE</u> of the following questions in 40-80 words each, wherever applicable

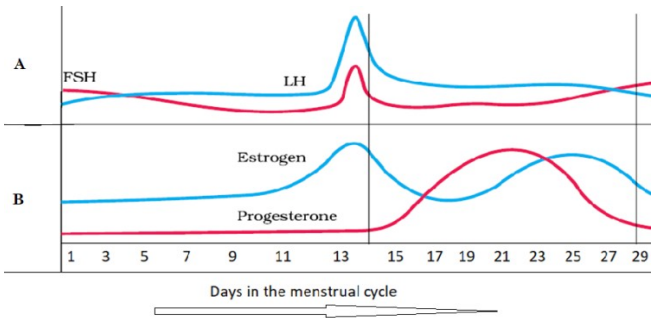
5 x 3 =15

28. Draw a L.S. of grass embryo diagram and label the following parts:

    a) Scutellum b) Coleoptile c) Shoot apex

    d) Radicle e) Root cap f) Coleorrhiza

29. In the figure given below, parts A and B show the level of hormones which influence the menstrual cycle. Study the figure and answer the questions that follow:



    a) Name the organs/glands which secrete the hormones represented in labelled parts A and B. 1

    b) State the role of hormones secreted from part B on the uterus of human female during menstrual cycle. 2

30. Suggest three simple principles through which we can prevent sexually transmitted diseases.

31. What is adaptive radiation? Give any two examples.

32. Name of the drugs, its source and nature is given in the table below. Find *a, b* and *c*.

Name of the drug	**Source plant**	**Nature**
a	*Papaver somniferum*	Depressant
Cannabinoids	***b***	Effects on cardiovascular system
Cocaine	*Erythroxylum cocoa*	***c***

33. How did an American Company Eli Lilly use the knowledge of *r*DNA technology to produce human insulin?

34. An example for grazing food chain is given below:

    Grass → Grasshopper → Birds → Man

    Graphically represent this food chain through pyramid of energy and write different trophic levelswith their energy content.

PART- D

IV. Answer any <u>FOUR</u> of the following questions in 200-250 words each, wherever applicable:

4 x 5 = 20

35. Draw a neat labelled diagram of sectional view of the mammary gland.
36. Schematically represent the inheritance of flower colour in snapdragon and draw conclusions.
37. Give reasons for the following:
    a) A simple cut result in non-stop bleeding in haemophilia affected individuals.
    b) Turner's syndrome affected females are usually sterile.
    c) In Morgan's dihybrid cross experiments on *Drosophila* showed that flies having genes for yellow body and white eyes exhibited less recombination.
    d) Inheritance of skin colour in the humans shows different phenotypes.
    e) Accumulation of phenylalanine in the body of phenylketonuria affected individuals.
38. Describe the steps involved in DNA fingerprinting technique.
39. Name the causative agents of the following diseases:
    a) Malaria b) Filariasis c) Ascariasis d) Amoebiasis e) Pneumonia
40. a) With respect to the microbial products, its source and uses identify the ***a, b*** and ***c*** in the following table: 3

Microbial product	Source	Use
Cyclosporin A	***a***	Immunosuppressant
b	*Monascus purpureus*	Blood cholesterol lowering agent
Streptokinase	*Streptococcus*	***c***

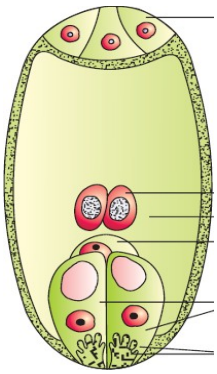
    b) Define the BOD and flocs. 2
41. Mention the population interactions exist among the following:
    a) Abingdon tortoise and goats
    b) Tiger and deer
    c) Sea-anemone and clown fish
    d) Wasp laying eggs in fig fruit
    e) *Cuscuta* growing on hedge plant

VI. Answer any <u>ONE</u> of the following questions in 200-250 words each, wherever applicable:

1 x 5 = 5

42. Picture of a mature angiosperm embryo sac is given below and answer the question that follows.

a) Which cells/nuclei of the embryo sac produce zygote and primary endosperm nucleus?



2

b) What is the ploidy of antipodal cells and primary endosperm nucleus? 2

c) Why the endosperm development precedes embryo development? 1

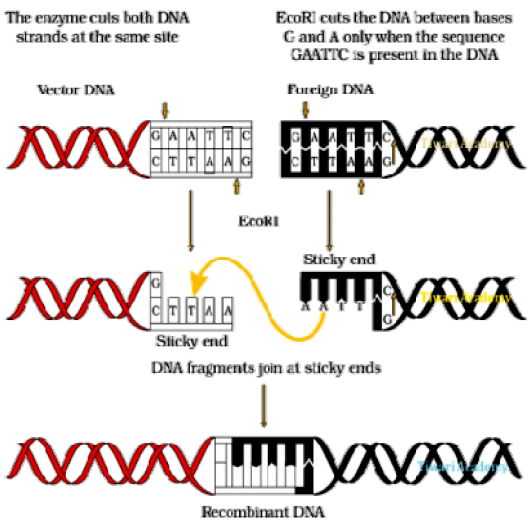
43. Answer the following:

a) Construct a complete transcription unit with promoter and terminator on the basis of the hypothetical template strand given below. 3



b) Write the RNA strand transcribed from the above transcription unit along with polarity. 2

44. Study the diagram given below and answer the questions that follow:



Action of Restriction enzymes

a) What is EcoRI? 1

b) How is the action of exonuclease different from that of endonuclease? 2

c) How are 'sticky ends' formed on a DNA strand? Why are they so called? 2

PART- E

(FOR VISUALLY CHALLENGED STUDENTS ONLY)

15**.** In relation to species – area relationships, what is the expected 'Z' value for frugivorous birds and mammals in the tropical forests of different continents? 1

a) 0.1 b) 0.4 c) 1.15 d) 0.5

29. Define menopause. Mention the different phases of menstrual cycle. 3

42. Answer the following:

a) Draw a neat labeled diagram of typical anatropous ovule. 3

b) What is the ploidy of nucellus? Write the functions of integuments. 2

44. Describe the steps involved in recombinant DNA technology. 5

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU MODEL QUESTION PAPER - 3 (2025)

SUBJECT – BIOLOGY (36)

Time: 3 Hrs. **Max. Marks: 70**

General Instructions:

1. This question paper consists of parts A, B, C, D and E.
2. Part-A consists of I and II and Part D consists of V and VI.
3. All the parts are compulsory.
3. The answers for Part-A, written in the first two pages of the answer booklet are only considered for evaluation.
4. Part-E consists of questions for visually challenged students only.

PART-A

I. Select the correct alternative from the choices given below: **15 x 1 = 15**

1. The ploidy of perisperm present in beet seed is

   a) Haploid b) Diploid c) Tetraploid d) Triploid

2. Some plants such as *Viola* (common pansy), *Oxalis* and *Commelina* produce two types of flowers namely

   a) Chasmogamous and Xenogamous b) Cleistogamous and Geitonogamous

   c) Geitonogamous and Xenogamous d) Chasmogamous and Cleistogamous

3. The correct sequence of embryonic development in angiosperm is

   a) ZygoteGlobular ProembryoHeart shaped Mature embryo

   b) Zygote Heart shaped Proembryo Globular Mature embryo

   c) Zygote ProembryoGlobularHeart shaped Mature embryo

   d) Zygote Globular Heart shaped ProembryoMature embryo

4. During pregnancy the foetus develops limbs and digits

   a) After three weeks b) after eight weeks c) After six weeks d) after nine weeks

5. In spermatogenesis, if FSH hormone is not secreted from the anterior pituitary gland, which stage is affected

   a) Formation of primary spermatocyte b) formation of spermatozoa

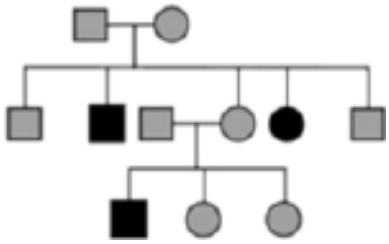
   c) Formation of spermatid d) formation of secondary spermatocyte

6. To produce 2000 sperm and 400 ova, how many spermatogonia and oogonia are required?

   a) 500 spermatogonia and 200 oogonia b) 500 spermatogonia and 400 oogonia

   b) 250 spermatogonia and 250 oogonia d) 250 spermatogonia and 400 oogonia

7. Which among the following sexually transmitted infection (STIs) are not curable
   a) Gonorrhoea, syphilis, hepatitis-B
   b) Chlamydiasis, genital warts, trichomoniasis
   c) Gonorrhea, HIV infection, chlamydiasis
   d) Hepatitis –B, genital herpes and HIV infection
8. A pedigree chart is given below, identify the trait responsible for this inheritance pattern



   a) Autosomal dominant trait b) Autosomal recessive trait
   c) X- linked recessive trait d) X – linked dominant trait
9. In Lac –operon, if mutation occurs in the Z – gene
   a) Transacetylase will not be synthesized
   b) -galactosidase will not be synthesized
   c) Permease will not be synthesized
   d) Lactose digestion will be rapid
10. Match the type of Man with their origin periods and choose the correct answer

Column I	**Column II**
A. Ramapithecus	i) 1.5 mya
B. Australopithecines	ii) 15 mya
C. Homo erectus	iii) 1,00,000–40,000 years back
D. Neanderthal man	iv) 2 mya
	v) 3 mya

   a) A-i, B -iv, C-iii, D-ii b) A-ii, B-iv, C-i, D-iii
   c) A-iii, B-i, C-iv, D-v d) A-iv, B-ii, C-iii, D-v
11. Now a day's disease like dengue and chikungunya are widespread in different parts of India, to prevent the spread of the disease which vector has to be eliminated
   a) *Culex* b) *Anapheles* c) Female anopheles d) *Aedes*
12. If a patient has undergone myocardial infarction leading to heart attack, to remove clots from the blood vessels of patient, which enzyme is used by the doctor to treat the patient

a) Pectinase b) streptokinase c) Protease d) lipase

13. A piece of alien DNA cannot multiply itself in the progeny cells of the organism due to

a) Lack of ori- site b) Denaturation c) Renaturation d) Incompatibility

14. Hind II cuts DNA molecule by recognizing a specific recognition sequence of

a) 3 base pairs b) 6 base pairs c) 8 base pairs d) 10 base pairs

15. Given below is a portion of DNA strand giving the base sequence on the opposite strand, what is so special shown in it?

5'——GAATTC———3'

3'——CTTAAG———5'

a) Replication completed b) Deletion mutation

c) Start codon at the 5' d) Palindromic sequence

II Fill in the blanks by choosing appropriate word/words from those given below: 5 X 1 = 5

(Saccharomyces cerevisiae, Adaptive radiation, Hormone releasing IUDs, RNA interference, sacred groves, Bacteria)

16. _____ make the uterus unsuitable for implantation and the cervix hostile to the sperm.
17. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called _____ .
18. _____ is used for fermenting malted cereals and fruit juices, to produce ethanol.
19. The phenomenon of silencing of a specific mRNA due to a complementary dsRNA molecule that binds to mRNA and prevents translation is _____.
20. _____ are the tracts of forest were set aside and all the trees and wildlife within were venerated and given total protection.

PART-B

III. Answer any FIVE of the following questions 3-5 sentences wherever applicable: 5 × 2 = 10

21. What are emergency contraceptives? Write their hormonal combination.
22. List the measure to check the population growth rate.
23. What is haplo-diploid sex determination mechanism? Mention an animal which exhibits this.
24. Write a note on the role of sigma factor and the rho factor in transcription in prokaryotes.
25. Mention the evolutionary significance of the following organisms.

a) Shrews b) Lobefins

26. Write the role of Cyanobacteria that act as biofertilisers.

27. ELISA is one of the methods of molecular diagnosis, what is the principle of this technique? Name the disease which can be detected by this method.

PART - C

IV Answer any FIVE of the following questions in 40-80 words each wherever applicable: 5 × 3 = 15

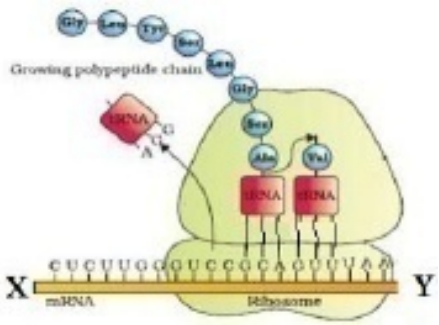
28. Draw a labelled diagram of Transverse section of young anther.
29. During pregnancy the levels of hormones are increase several fold in the maternal blood. Name the hormones and write their importance.
30. Draw a labelled diagram of Miller's experimental set up.
31. The use of biocontrol measures will greatly reduce our dependence on toxic chemicals and pesticides. Justify with an example.
32. What is Gene therapy? Write the steps involved in curing ADA deficiency by gene therapy.
33. Pyramid of energy is always upright, can never be inverted. Give reasons.
34. There are many reasons for conserving biodiversity. Briefly explain the reason for conserving biodiversity from the narrowly utilitarian point of view.

PART - D

V. Answer any FOUR of the following questions in about 200-250 words each wherever applicable.

4 × 5 = 20

35. Draw a neat labelled diagrammatic sectional view of male reproductive system.
36. a) Define aneuploidy. Give two examples of aneuploidy. 2
    b) Mention the karyotype of Klinefelter's syndrome of an affected individual and write its symptoms. 3
37. a) Identify the polarity of X and Y in the above given diagram and how many more amino acids are expected to be added to this elongating polypeptide chain 2



    b) Which sequences of bases are generally found at the X and Y polarity? 1
    c) Mention the anticodon for alanine and valine amino acids based on the diagram. 1

d) What is the composition of the catalyst involved in the peptide bond formation in this process? 1

38. List the salient features of Human Genome.

39. What are lymphoid organs? Write the functions of the following in response to Immune system in our body.

a) Bone marrow b) Thymus c) Spleen d) Lymph nodes.

40. Explain different techniques which help in cancer detection and diagnosis.

41. a) Write three vector free techniques that can be utilized to transfer recombinant DNA into a ready host cell. 3

b) *Agrobacterium tumifaciens* act as natural vector for cloning genes in plants. Substantiate. 2

VI Answer any ONE of the following questions in about 200-250 words each wherever applicable:

1 × 5 = 5

42. Picture related to pollination is given below:



a) Will this pollination confirms fertilization? 2

b) What are the floral rewards provided by the plants to the insects t revisit? 1

c) Mention the characteristics of flower in this pollination other than floral rewards 2

43. Thalassemia is a quantitative problem of synthesizing globin. Explain in detail with respect to the chromosome, number of genes and alleles and the features.

44. a) Predators are 'Prudent and conduits' in Nature. Support your answer by giving reason. 3

b) To lessen the impact of predation, prey species have evolved different defensive methods in plants and animals; support your answer by giving one example each for a plant and an animal. 2

PART- E

(For Visually Challenged Students only)

8. Which of the following is not a recessive gene linked disorder 1

a) Myotonic dystrophy b) Sickle cell anaemia

c) Haemophilia d) Colourblindness

37. Explain the aminoacylation of tRNA and formation of initiation complex in translation. 5

42. What is pollination? Write the characteristics of insect pollinated flowers. 5

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU PRACTICE PAPER - 1 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. The question paper consists of Parts, A, B, C, D and E.
2. Part-A consists of I and II and Part-D consists of V and VI.
3. All the parts are compulsory.
4. For Part-A questions, only the answers written first will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.
6. Part-E consists of questions for visually challenged students only.

PART - A

I. Select the correct alternative from the choices given below: **15 × 1 = 15**

1. Method to discourage self-pollination or promote cross pollination is

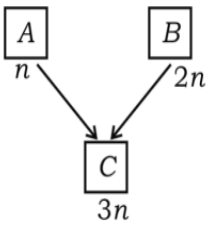
   a) Pollen release and stigma receptivity are not synchronised

   b) Anther and stigma are placed at different positions

   c) Same height of stamen and stigma

   d) Both (a) and (b)

2. Find out A, B and C in the flow chart given below :



   a) A — Female gamete, B — Male gamete, C — Endosperm

   b) A — Endosperm, B — Female gamete, C —Male gamete

   c) A — Male gamete, B — Polar nuclei, C — Endosperm

   d) A — Female gamete, B — Endosperm, C — Male gamete

3. The reason for corpus luteum to degenerate during menstrual cycle is

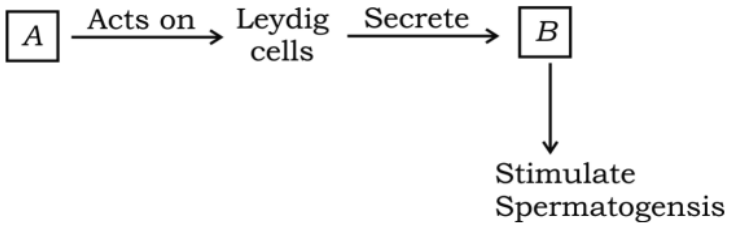
   a) Absence of fertilization

   b) Fertilization

   c) Ovulation

   d) Both (b) and (c)

4. Give the name of two hormones A and B in the sequence given below:



a) FSH and GnRH b) LH and Androgens c) GnRH and LH d) GnRH and FSH

5. Which among the following sexually transmitted infections (STIs) is not curable?

a) Gonorrhoea b) Syphilis c) Hepatitis - B d) Chlamydiasis

6. Thalassemia and sickle cell anaemia are caused due to problem in globin molecule synthesis. Select the correct option given below.

a) Both are due to a qualitative defect in globin chain synthesis

b) Both are due to a quantitative defect in globin chain synthesis

c) Thalassemia is due to the synthesizing too few globin molecules

d) Both (a) and (b)

7. 7. Why both the strands of DNA are not copied during transcription?

a) Because RNA molecule with different sequences will be formed

b) Because RNA molecule with same sequence will be formed

c) Because RNA molecule with identical sequences will be formed

d) Because DNA molecule with different sequences will be formed

8. 8. The correct order of evolution of plant forms through geological period is

a) Palaeozoic → Coenozoic →Mesozoic

b) Coenozoic → Mesozoic → Palaeozoic

c) Palaeozoic → Mesozoic → Coenozoic

d) Mezosoic → Coenozoic → Palaeozoic

9. Which of the following is an autoimmune disease?

a) Rheumatoid arthritis b) Pneumonia c) Malaria d) Ringworms

10. Antibiotic Penicillin is produced from which micro-organisms?

a) *Staphylococci* b) *Trichoderma polysporum*

c) *Penicillium notatum* d) *Monascus purpureus*

11. Agrobacterium tumifaciens transform normal cells into a tumor by delivering a piece of DNA known as

a) Z-DNA b) C-DNA c) T-DNA d) R-DNA

12. Consider the following statements:

Statement I: Plasmid DNA acts as vectors to transfer the piece of DNA attached to it

Statement II: Vector DNA acts as plasmids to transfer the piece of DNA attached to it.

of these:

a) Statement I is true, but statement II is false

b) Statement I is false, but statement II is true

c) Both the statements I and II are true

d) Both the statements I and II are false

13. Identify the Correct statement :

a) The gene cryIAc can control corn borer

b) *Bacillus thuringiensis* produce an insecticidal protein that can kill *Meoidogyne incognitia*

c) Using *Agrobacterium* vectors nematode specific genes were introduced into the host plant

d) Bt. toxin exists as protein that is converted into protoxin in the insect gut

14. During decomposition bacterial and fungal enzymes degrade detritus into simple inorganic substances, the process is known as

a) Humification b) Catabolism c) Leaching d) Fragmentation

15. The following are the in-situ conservation approaches except

a) National parks b) Wildlife sanctuaries c) Biosphere reserves d) Botanical gardens

II. Fill in the blanks by choosing the appropriate word/words from those given below: 5 × 1 = 5

(Adaptive radiation, Stem, Sporopollenin, Immunosuppressive agent, Implant, Intine)

16. Pollen grains are well preserved as fossils because of the presence of _____

17. In blastocyst the inner cell mass contains certain cells called _____cells which have the potency to give rise to all the tissues and organs.

18. Progestogens alone or in combination with estrogen can also be used by females as _____ under the skin.

19. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called_____

20. Cyclosporine – A is used as _____

PART - B

III. Answer any five of the following questions in 3-5 sentences wherever applicable: 5 × 2 = 10

21. What are IUDs? Give two examples for hormone releasing IUDs. How and under which condition is Artificial Insemination (AI) technique used to assist infertility?

22. What is point mutation? Give an example.

23. What are the criteria that a molecule must fulfil to act as genetic material?

24. Write two key concepts of Darwinian Theory of Evolution.

25. What is Allergy? Name the types of antibody produced during allergy.

26. Give reasons for the following:

27. a) Bt. toxin does not kill Bacillus bacteria. 1

b) Milk produced by transgenic cow Rosie is better than natural cow milk. 1

PART – C

IV. Answer any five of the following questions in 40-80 words each, wherever applicable: 5 × 3 = 15

28. Draw a neat labelled diagram of T. S. of young anther.

29. What is Placenta? List two functions of placenta.

30. Draw a neat labelled diagram of Miller's experiment.

31. a) Name any 2 species of malarial parasite. 2

b) Write the cause of periodic (3 - 4 days) recurrence of chill and high fever during malarial attacks in Humans.

32. Write the process of synthesis of insulin by Eli Lilly Company. Name the technique used by the company.

33. Write a note on Primary Productivity.

34. Briefly explain broadly utilitarian reasons for conserving biodiversity.

PART – D

V. Answer any four of the following questions in about 200 – 250 words each, wherever applicable: 4 × 5 = 20

35. How is a female gametophyte developed from functional megaspore? Explain.

36. a) Draw a neat labelled diagram of Human sperm. 4

b) Which part of the sperm helps it to enter into the ovum? 1

37. Write the schematic representation of one gene inheritance with respect to height of Pea plant.

38. a) A woman heterozygous for Blood group A marries a man heterozygous for Blood group B. What are the possible blood groups in their children? 2

b) In some organisms, females produce two types of gametes, with respect to sex chromosomes. What is this phenomenon called? In which type of sex determination is this found? Give an example of an organism exhibiting this type of mechanism. 3

39. Name the process involved in the following:

a) The type of DNA replication, in which the newly synthesized DNA molecule would have one parental and one newly synthesized strand. 1

b) Addition of methyl guanosine triphosphate the 5 / end of hnRNA. 1

c) Introns are removed and exons are joined in a defined order, from hnRNA. 1

d) During transcription Rho factor attaches to RNA polymerase to end the process. 1

e) Polymerisation of amino acids to form a polypeptide. 1

40. a) Cancer is one of the dreaded diseases in human being. What is 'contact inhibition' and 'metastasis' with respect to the disease? 2

b) Name any two techniques that are useful in detecting cancers of internal organs. 2

c) Why is cancer patients often given (Alpha) α-interferons as part of the treatment? 1

41. a) Draw a neat labelled diagram of simple stirred tank bioreactor. 3

b) Alien DNA can be introduced into the host cells. For this purpose which method is used in plant and animal cells respectively? 3

VI. Answer any one of the following questions in about 200 – 250 words each, wherever applicable: 1 × 5 = 5

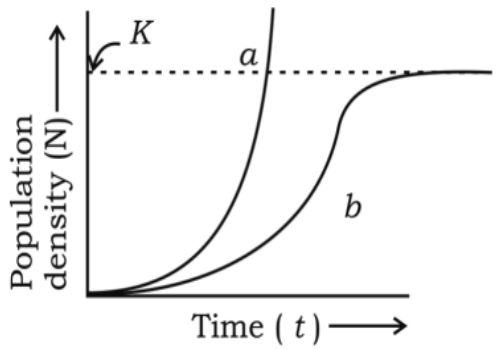
42. a) How does the i-gene regulate the lac-operon in the absence of inducer? 2

b) Draw a schematic labelled diagram of lac-operon in a 'switched on' state. 3

43. a) Some Bacteria, Fungi and Cyanobacteria, increase the soil nutrients and reduce dependence on chemical fertilizers. Substantiate the statement. 3

b) In sewage treatment, primary treatment is a physical process. Comment. 2

44. Study the graph given below and answer the questions that follow :



a) Curves 'a' and ' b' represent which type of growth? 2

b) What is ' K ‘? What is its role in nature? 2

c) Which growth model is considered more realistic one in nature?

PART – E

(For Visually Challenged Students only)

2. Coconut water in a tender coconut is

a) Seed coat b) Nucellus c) Embryo d) Free nuclear endosperm

44. Mention any five population interactions. Give an example for each.

GOVERNMENT OF KARNATAKA

KARNATAKA SCHOOL EXAMINATION AND ASSESEMENT BOARD

II PU PRACTICE PAPER - 2 (2025)

SUBJECT – BIOLOGY (36)

General Instructions:

1. The question paper consists of Parts, A, B, C, D and E.
2. Part-A consists of I and II and Part-D consists of V and VI.
3. All the parts are compulsory.
4. For Part-A questions, only the answers written first will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.
6. Part-E consists of questions for visually challenged students only.

PART – A

I. Select the correct alternative from the choices given below: **15 × 1 = 15**

1. Choose the correct statement from the following.
   a) Cleistogamous flowers always exhibit autogamy.
   b) Chasmogamous flowers always exhibit geitonogamy.
   c) Cleistogamous flowers exhibit both autogamy and geitonogamy.
   d) Chasmogamous flowers never exhibit autogamy.
2. Which is the correct order of menstrual cycle?
   a) Menstrual phase → Secretory phase → Proliferative phase → Ovulatory phase
   b) Menstrual phase → Proliferative phase → Ovulatory phase → Secretory phase
   c) Menstrual phase → Ovulatory phase → Secretory phase → Proliferative phase
   d) Menstrual phase → Ovulatory phase → Proliferative phase → Secretory phase
3. With reference to the Inner cell mass, which of the following statement is correct?
   (a) Embedding of blastocyst in the endometrium
   (b) Cells that would differentiate as embryo
   (c) Protecting the foetus
   (d) Providing nutrition for foetus
4. Cu ions released from copper-releasing Intra Uterine Devices (IUDs)
   a) Make uterus unsuitable for implantation
   b) Increase phagocytosis of sperms
   c) Suppress sperm motility
   d) Prevent ovulation

5. Medical termination of pregnancy is considered safe up to _____ weeks of pregnancy
   a) 8 weeks b) 18 weeks c) 6 weeks d) 12 weeks
6. If Child's Blood group is 'O' then the parents blood group cannot be
   a) A and B b) AB and O c) A and A d) B and O
7. Match the following RNA polymerases with their transcribed product.

1. RNA polymerase I	i) tRNA
2. RNA polymerase II	ii) rRNA
3. RNA polymerase III	iii) hnRNA

   a) 1 – i, 2 – iii, 3 – ii b) 1 – i, 2 – ii, 3 – iii
   c) 1 – ii, 2 – iii, 3 – i d) 1 – iii, 2 – ii, 3 – i
8. When more than one adaptive radiations appeared to have occurred in an isolated geographical area it is _____
   a) Convergent evolution b) Divergent evolution
   c) Natural selection d) Genetic drift
9. Theory of inheritance of acquired characters was given by
   a) Lamarck b) Darwin c) de Vries d) Wallace
10. Transplantation of tissue/organ often fails due to non-acceptance by the patient's body. Which type of immune response is responsible for such rejection?
    a) Humoral-mediated immunity b) Cell-mediated immunity
    c) Hormonal-mediated immunity d) Auto immunity
11. *Monascus purpureus* is a yeast used commercially in the production of
    a) Ethanol b) Streptokinase for removing clots from the blood vessels
    c) Citric acid d) Statins, blood cholesterol lowering agents
12. Statement I: In gel electrophoresis, DNA fragments are separated. Statement II: DNA is negatively charged, so it moves towards anode under electric field.
    Choose the correct option:
    a) Both the statements I and II are correct
    b) Statement I is correct but II is incorrect
    c) Statement I is incorrect but II is correct
    d) Both the statements I and II are incorrect
13. In RNA interference (RNAi), specific mRNA are silenced using
    a) ssDNA b) dsDNA c) dsRNA d) ssRNA

14. Vertical distribution of different species occupying different levels is called

    a) Standing crop b) Stratification c) Trophic levels d) Biological organization

15. The species confined to a region and not found anywhere else is

    a) Alien species b) Species area relationship

    c) Global species d) Endemism

II. Fill in the blanks by choosing the appropriate word from those given in the bracket: 5 × 1 = 5

(Amphibian, Lysozyme, Fungi, Reptiles, Relaxin, Non-Steroidal)

16. Saheli the oral contraception for females contains a _____ preparation.
17. _____ hormone is secreted by ovary in a woman during pregnancy.
18. Roquefort cheese is formed by ripening with _____ for a particular flavour.
19. In bacteria, cell membrane is digested by _____ enzyme.
20. _____ group of animals are more vulnerable to extinction.

PART – B

III. Answer any five of the following questions in 2 to 3 sentences each, wherever applicable: 5 × 2 = 10

21. What are the benefits of dehydration of seeds and dormancy?
22. Two plants (Snapdragon) with red flower and white flower are crossed.

    (a) What phenomenon is shown in F1 generation? 1

    (b) What is the genotypic ratio of F2 generation? 1

23. What was the reason for change in the population of white-winged moth and dark-winged moth in England after industrialization?
24. Mention the two methodologies which were involved in Human Genome Project.
25. A patient has lost his immunity. What disease is associated with it? Why did the patient loose his immunity?
26. What are the main objectives of GEAC?
27. Mention the four major causes of biodiversity losses.

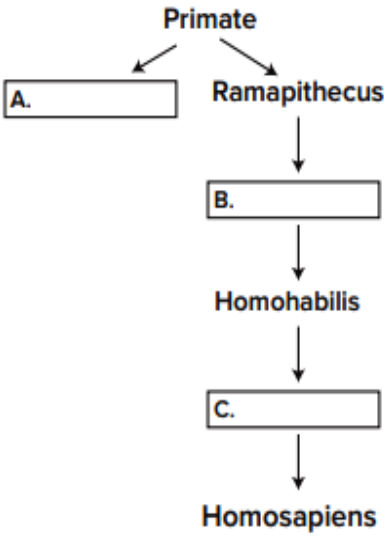
PART – C

IV. Answer any five of the following questions in about 40–80 words each, wherever applicable:

5 × 3 = 15

28. What is double fertilization? Why is it an important character of flowering plants?
29. Draw a neat labelled diagram of human sperm.
30. Mention any three methods to assist infertile couples in having children.

31. Fill up the missing sequence (A, B, C) in the evolution of man.



32. What are carcinogens? Mention any two groups of carcinogens with one example each.

33. Suggest any three possible treatments that can be given to a patient suffering from ADA deficiency.

34. What are the limitations of ecological pyramids?

PART – D

V. Answer any four of the following questions in about 200–250 words each, wherever applicable:

4 × 5 = 20

35. What are outbreeding devices? Explain the various outbreeding devices that have evolved in Angiosperms.

36. Briefly explain the major features of embryonic development at various months of pregnancy.

37. Schematically represent the inheritance of one gene.

38. a) Explain the regulation of Lac-operon in the absence of lactose as an inducer. 3

    b) What are the fulfilling criteria for a molecule to become a genetic material? 2

39. Explain the prevention and control measures of alcohol and drug abuse among adolescents.

40. a) Polymerase Chain Reaction (PCR) is used for amplification of DNA,
    with reference to PCR, answer the following questions.

    i) Write the name of the enzyme required for PCR. 1

    ii) From which organism it is obtained? 1

    iii) What is the significance of this enzyme? 1

    b) Draw a neat labelled diagram of pBR^{322}. 2

41. Name the type of population interaction seen in each of the following:

    a) Herbivores eating plants. 1

    b) Cuscuta and hedge plants. 1

    c) Mycorrhizae and roots of higher plants. 1

d) Orchid and mango tree. 1

e) Balanus and chthamalus 1

VI. Answer any one of the following questions in 200–250 words each, wherever applicable. 1 × 5 = 5

42. a) In humans, males are heterogametic and females are homogametic. Explain. 2

b) Give an example where males are homogametic and females are heterogametic. 1

c) Male and female grasshoppers have different chromosome number. What is the reason for it? 2

43. Meselson and Stahl carried out an experiment to prove a nature of DNA replication. Recall the experiment and answer the following questions.

a) Which two types of Nitrogen were used by them in their experiment and why? 2

b) Why did they take samples of *E.coli* for their experiment? 1

c) State the role of calcium chloride. 1

d) What conclusion did they arrive at the end of the experiment? 1

44. a) For what important characteristic properties, the *Baculoviruses* are considered as the best biological control agents? 2

b) What are the uses of cyanobacteria in paddy fields? 2

c) What role do enzymes play in detergent formulation that we use for washing clothes? 1

**********END**********